

Vague by Design: Performance Evaluation and Learning from Wages^{*}

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Abstract

We study a dynamic principal-agent setting with period-by-period contracting in which both sides learn about the importance of effort. The quality of the agent's output is not observed directly. Instead, the principal jointly designs an evaluation technology and a wage schedule. More precise performance evaluation reduces current agency costs but promotes learning, which can increase future agency costs. As a result, the optimal evaluation technology is coarse. The optimal evaluation features a censorship pattern with a base wage and tailored bonuses/penalties when the threshold is exceeded. With binary output, it is both noisy and tough if effort and ability are strong complements: a bad performance is always sanctioned, but a good one is not always recognized.

1 Introduction

Many firms motivate their workers to exert effort with incentive pay based on objective measures of performance. In the United States, jobs with performance-related pay account for 39 percent of hours worked in the private sector ([Gittleman and Pierce, 2013](#)). Such measures have become richer and easier to obtain. Call centers link repeat contacts, survey responses and retention records to the representative who handled the customer, and language processing extracts the customer's mood from the call itself ([Singer, 2013](#)); software firms estimate in controlled experiments what a shipped feature contributes to engagement; hospitals identify complications from health records. Recent advances in language processing and AI enable data collection and processing across new domains and at unprecedented scale. Using such data, firms can develop better objective measures of workers' contribution to profits.

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Should this additional information be used to set performance pay? What should be measured and how should the information be aggregated? The theory of incentives seems to offer a simple answer to these questions. Providing incentives to workers is costly exactly because the underlying performance measures are only partially informative about effort. Hence, more hard information about the workers' contribution is always helpful and should be used as a basis of performance pay (Holmström, 1979; Grossman and Hart, 1983). In particular, it is not optimal to base incentives on a noisy signal instead of the contribution to output.¹

In this paper, we show that this conclusion changes fundamentally when we also consider how performance evaluation provides information to workers. Often, workers do not directly observe their own contribution to profits – which can be hard to grasp in complex organizations and occurs far from the worker on the consumer-facing side of the firm. Performance evaluations, however, are observed – certainly insofar as they determine their wage and bonus. By basing wages on a precise evaluation of output, the firm therefore reveals information about his output to the worker. Based on this information about performance, the worker can learn about his ability and fit to the job. This learning, in turn, can make it more costly to employ and motivate the agent in the future.

To capture this fundamental trade-off between incentives and information, we develop a model of twice-repeated moral hazard with period-by-period contracting and learning (Section 2). The agent has a persistent match-specific ability that is unknown both to the agent and to the firm. The agent's ability affects not only average output, but also the effectiveness of effort. Output is not observed directly. Instead, in every period the principal flexibly designs not only wages but also the underlying performance evaluation. The agent observes his evaluations and wages. Section 2.3 discusses the employment relationships these assumptions describe, using a customer-service representative as a running example.

Performance evaluation is thus not only the basis for incentives but also a source of workers' learning about their ability and fit to the job. This dual role speaks to several familiar features of incentive systems. First, many bonus plans pay a base wage below a performance threshold, and a substantial bonus upon reaching it that rises with performance above it. In our model, such threshold bonuses arise because the flat region pools the outcomes that would leave the worker most pessimistic.² Since it averages over all outcomes below the threshold, the pooled evaluation is discretely worse evidence of effort than even a performance that barely clears it, so pay jumps there. Second, when performance is measured coarsely and effort and ability are strong complements, as is plausible in law or investment banking, the optimal evaluation is tough: it punishes every failure but only recognizes some successes. Because the trade-off is intertemporal, evaluations are noisier, and here tougher, when future incentives matter more relative to current ones and learning is more important when ability is more uncertain. Both conditions are typical of new hires, in line with the demanding evaluation of junior associates. Finally, evaluations often focus on workers' conduct rather than on available measures of output.

¹That it is not optimal to add noise continues to hold across a wider class of models, including multi-tasking (Holmström and Milgrom, 1991) and linear-Gaussian career concerns models (Holmström, 1999; Hörner and Lambert, 2021). We discuss some settings in which the addition of noise can be beneficial in the related literature.

²Such threshold or hurdle bonus schemes are common in managerial compensation, particularly at the executive level (Murphy and Jensen, 2024), and also arise among nonmanagerial employees, see e.g. Friebe et al. (2017) for retail employees.

In our model, the principal coarsens the evaluation of output to limit what workers learn about their ability; information about conduct does not carry this cost.

We solve the model backward. In the terminal period (Section 3) learning has no further consequences, and a fully informative evaluation is optimal (Holmström, 1979; Grossman and Hart, 1983). In the first period, however, there is our novel trade-off. A more precise evaluation structure reduces agency costs this period, but induces more agent learning. The principal's attitude towards this learning is determined by the shape of profits in the terminal period as a function of the agent's posterior belief. In Section 3.2, we discuss the forces involved – the *filtering* channel, which allows the principal to tailor incentives more precisely to the agent's ability and raises expected profits, and the *agent-learning* channel, which makes incentives more costly because the incentive compatibility constraint must hold at each posterior belief. We derive an elasticity condition comparing how beliefs affect the output distribution (filtering) with how they affect the return to effort (agent-learning). When effort and ability barely interact, filtering dominates and learning is valuable. When the agent-learning effect is sufficiently strong, continuation profits are concave. Under a stronger bound, learning is especially costly where the agent believes his effort matters little: at low beliefs with complements and at high beliefs with substitutes.

We return to the joint design of the evaluation and wage in the first period in Section 4. Our assumptions make this dynamic problem tractable while preserving the trade-off between incentives and learning. Neither party observes output directly, and effort is observable, though not contractible. Both parties therefore interpret the evaluation and wage in the same way, so their beliefs about ability remain common even after a deviation. With period-by-period contracting, the principal then holds the agent to his belief-independent outside option after every history. There is consequently also no ratchet effect: the agent cannot improve his continuation utility by manipulating the principal's beliefs. We show in Section 4.1 that, as a result, the dynamic first-period problem reduces to a static moral-hazard problem in which only the principal's continuation value depends on the posterior belief. We can then write this problem as the joint choice of a distribution of posterior beliefs and a wage at each posterior: an information design problem with the wage function as an additional choice variable and two additional constraints.

We solve for the optimal contract and evaluation structure in Section 4.3 depending on the shape of the continuation value. If the principal prefers the agent to learn, clearly, the optimal evaluation structure is fully informative. If learning is costly, the principal pools the outcomes where it is most costly, and the technology determines which outcomes these are. With complements, these are low outcomes: the evaluation pools output below a threshold and reveals output above it exactly (lower censorship, corresponding to threshold bonuses). With substitutes, the pattern is mirrored (upper censorship): output above a threshold is pooled, which corresponds to a capped wage with deductions for shortfalls. With binary output, the optimal evaluation remains binary, so shaping learning adds no complexity. It is tough when effort and ability are strong complements (log-supermodular technology), and lenient, sometimes rewarding low output, when they are strong substitutes (log-submodular technology).

Section 5 relaxes, in turn, the three assumptions that make the first-period problem static in

the baseline model. First, the principal may privately acquire information about output. We show that under a natural refinement this can only hurt her: any private information that is used is also divulged to the worker through the contract offer, so it carries the learning cost without the incentive benefit it would have had as part of the evaluation. Second, if effort is hidden, a worker who shirks holds a private belief, which brings back the belief-manipulation motive that is absent from the baseline. The incentive constraint becomes dynamic, but with binary output and strong complements the optimal evaluation remains tough. Third, we explore two stronger commitment notions. If the principal can only defer pay but not information, our results extend. Under full commitment instead she could defer information as well, yet in an example she still prefers a partially informative, tough evaluation. Section 6 concludes. The proofs not given in the text are collected in the Appendix.

Related Literature

This paper adds to the large literature on information in moral-hazard models by exploring circumstances under which a principal may favor wages tied to a noisier signal, contrasting with the classical view that greater precision always lowers agency costs (Holmström, 1979; Grossman and Hart, 1983; Kim, 1995).

We show that coarse or noisy evaluation is optimal even though verifiable information about the agent’s true performance would be available. Several strands of the literature show that coarse or noisy evaluation is optimal when such information is not available, for instance with multitasking (Holmström and Milgrom, 1991) or when the agent has private information that would allow him to game a deterministic incentive scheme (Ederer et al., 2018). Similarly, coarse rewards emerge when the evaluation is subjective, i.e. based on unverifiable private information of the principal (MacLeod, 2003; Fuchs, 2007). Fuchs (2007) shows that withholding feedback until dismissal is optimal, with incentives provided through an efficiency wage and the threat of termination. This allows the principal to reuse punishments economizing on the surplus destruction required to provide incentives under subjective evaluation.

That more information about the technology can reduce profits in a moral hazard setting has been noted in the literature in several settings. In general, it is well understood that ex-post incentive compatibility is more demanding than ex-ante incentive compatibility. Lizzeri et al. (2002) show that interim performance evaluation is not optimal when there is no learning.³ Nafziger (2009) demonstrates that it can be optimal to conceal information until after the agent’s effort choice, even though this precludes the principal from adjusting the implemented action. Indeed, such situations occur generically if the problem is sufficiently rich (Jehiel, 2015). In all these papers, the wage is still allowed to depend on the true realization of the signal, even if it is not revealed ex-ante; they study when, not whether information should be revealed. We show that less information about the technology increases profits even if this implies that the wage cannot depend on the state even ex-post.⁴ Ding (2019, Ch. 3) shows that information is detrimental to the principal’s profit in a binary setting with risk-neutral agents and limited

³Ederer (2010) shows that the optimality of interim performance evaluation in a tournament setting with exogenous rewards depends on the shape of the effort cost function.

⁴Under this assumption, Fang and Moscarini (2005) show that information is detrimental if it erodes profitable overconfidence.

liability if and only if the technology is log-supermodular, but does not link this result to performance evaluation.

To our knowledge, this is the first paper to combine the three key features of explicit incentives, learning about a persistent type, and information design. There are several literatures combining each pair of these features.

A growing literature investigates the design of information structures in one-shot moral hazard problems with commitment to a wage scheme. The older literature (Dye, 1986; Feltham and Xie, 1994; Datar et al., 2001) considers the optimal acquisition and aggregation of information within a parametric class;⁵ Demougin and Fluet (2001) study the case with limited liability which results in a binary evaluation. In Georgiadis and Szentes (2020) and Li and Yang (2020) the costs of information acquisition are assumed as part of the monitoring technology. Dai et al. (2019) study the optimal contract when the principal can allocate attention between finding good and bad news. Hoffmann et al. (2021) analyze a setting where the agent takes a single action, but information about his performance arrives over time. Information acquisition requires delayed payments, which creates endogenous costs because of impatience and imperfect risk sharing. Ely et al. (2024) study the joint design of feedback and reward when effort is exerted over time to obtain a breakthrough. Perhaps the closest reference regarding the analyzed trade-off is Orlov (2022), which studies the optimal contract and intensity of monitoring in a dynamic setting with limited liability. The central trade-off is between monitoring to avoid wasteful investment and thereby revealing information about the continuation value which is costly. We analyze not only the intensity but also the shape of the optimal evaluation when monitoring is required for incentives and its costs stem from the agent’s learning about his ability.

Learning about a persistent state and information design are combined in a growing literature. Most closely related to our moral-hazard setting are Smolin (2021) and Ely and Szydlowski (2019) in which the principal uses information, which is valuable for the agent, as an incentive. We analyze the role of information design when – since the principal sets incentives and ability is match-specific – information itself is not valuable. Information and incentive design constrain each other, as the principal reveals at least as much information as is contained in wages. Information and implicit incentives are also linked in models of career concerns (Holmström, 1999), a connection we return to in the conclusion. Waldman (1984) and Ricart i Costa (1988) study job assignments and promotions that also convey information about worker ability to the labor market which affects compensation and can distort assignment. In our setting, evaluations and wages instead shape the worker’s own learning which affects the cost of subsequent explicit incentives even though it does not affect the worker’s outside option.

The literature on learning in moral hazard models (Adrian and Westerfield, 2009; Giat et al., 2010; Prat and Jovanovic, 2014; Demarzo and Sannikov, 2017) studies learning based on output while we study learning based on an information structure that is designed endogenously by the principal. Another important distinction is that we consider learning about the importance of effort as opposed to learning about a state that affects only the level of output, which is often

⁵When restricting attention to linear contracts, it can be optimal to leave information unused (Feltham and Xie, 1994; Datar et al., 2001). This is a consequence of the restricted space of contracts, however.

considerably more tractable (see Bhaskar and Mailath (2019) and Bhaskar and Roketskiy (2023) for notable exceptions).

The formal approach to the analysis in our paper relates to the literature on information design, in particular recent contributions to information design problems with constraints (Boleslavsky and Kim, 2017; Le Treust and Tomala, 2019; Doval and Skreta, 2024) and additional choice variables (Georgiadis and Szentes, 2020) – in our case wages. Once we have reduced our problem to a static joint information and contract design problem, our solution strategy closely follows Georgiadis and Szentes (2020). Jointly designing contractual outcomes and the information they convey has precedents in mechanism design and sequential contracting. Myerson (1982, 1986) studies mediated communication, characterizing implementable outcomes through truthful reporting and obedience to action recommendations. Calzolari and Pavan (2006a,b) analyze how disclosure shapes subsequent contracting and resale. Bester and Strausz (2007) study noisy communication devices that control what the principal learns from the agent’s report under limited commitment. Doval and Skreta (2022) develop a revelation-principle analogue for dynamic mechanism design which results in a joint information and allocation design problem. Tailoring allocations more closely to types requires learning more about types; in our setting, tailoring wages more closely to output results in more learning about ability. Our evaluation conditions on realized output rather than reports of private information. The resulting signals and wages shape both parties’ beliefs about ability, linking current incentive provision to the cost of future contracting.

2 The Model

A principal (she) employs an agent (he) for two periods. The principal is risk neutral, the agent is risk averse with utility index $u(w) = \sqrt{2w}$. The agent can exit at the start of either period and obtain outside utility $U > 0$ in each remaining period. We assume that U is sufficiently large that the nonnegativity constraint that is implicit in square-root utility is slack at every history; we give an explicit condition in Appendix A.1. Both parties use a common discount factor $\delta \in (0, 1]$.

2.1 Technology

Each period, the agent exerts effort $e_t \in \{0, 1\}$ at cost $c \cdot e_t$, with $c > 0$. Effort is observed by both parties but is not contractible. The agent has a time-invariant, match-specific ability $\theta \in \{\theta_L, \theta_H\}$; it does not affect the agent’s outside option U . The principal and the agent share a common prior belief $\mu_0 \in (0, 1)$ that the agent’s ability is high.

Output $y \in Y$ realizes each period in a compact set $Y \subset \mathbb{R}$ according to a distribution $F(\cdot|e, \theta)$. We write $F(\cdot|e, \mu)$ for the expected distribution over output for a belief $\mu \in [0, 1]$. We assume that F can be written as

$$F(\cdot|e, \mu) = F_0(\cdot) + (\Delta_a \mu + b e + \Delta_b \mu e) \Delta_F(\cdot), \quad (1)$$

where F_0 denotes the distribution when both effort and ability are low, the signed measure Δ_F denotes the direction and $(\Delta_a\mu + b e + \Delta_b\mu e)$ denotes the size of the shift in the distribution as effort and ability change. We assume that $\int d\Delta_F(y) = 0$ and $\int y d\Delta_F(y) > 0$. We further assume that effort strictly raises expected output for any level of ability ($b > 0$ and $b + \Delta_b > 0$) and that higher ability weakly raises expected output ($\Delta_a \geq 0$) and strictly so under high effort ($\Delta_a + \Delta_b > 0$). Effort and ability are complements if $\Delta_b > 0$ and substitutes if $\Delta_b < 0$. We also assume that the likelihood ratio $\frac{d\Delta_F(y)}{dF_0(y)}$ is bounded on Y and that it satisfies the monotone likelihood ratio property (MLRP), namely that $\frac{d\Delta_F(y)}{dF_0(y)}$ is increasing in y , which allows us to interpret high output realizations as good news and ensures that optimal wages will be weakly increasing in output.⁶

2.2 Information, Contracts and Commitment

In the main sections, we assume that the principal has full commitment within each period, but no commitment across periods. Within each period t , the timing is as follows: The principal proposes a contract (S_t, p_t) , where S_t is an arbitrary signal space and $p_t : Y \rightarrow \Delta(S_t \times \mathbb{R}_+)$ maps output to a distribution over signals and wages.⁷ The agent observes the contract and either exits or works and chooses effort e_t . Output y realizes. Neither party observes it. The signal and the wage (s_t, w_t) are drawn from $p_t(\cdot|y)$. Both parties observe them, as well as the agent's effort. The signal allows the principal to reveal more about output than the wage; this adds generality and is not essential for our results. We refer to (S_t, p_t) as the contract and to what its outcome reveals about output as the evaluation.

Since both parties know the contract and observe the same outcome, they share a common posterior belief about the agent's ability. After a first-period contract (S, p) and outcome (s, w, e) , this posterior is

$$\mu(s, w, e) = \frac{\mu_0 \int_Y dp(s, w|y) dF(y|e, 1)}{\int_Y dp(s, w|y) dF(y|e, \mu_0)}, \quad (2)$$

including after a deviation to $e = 0$. Since output itself is never observed, the evaluation is both the basis for incentive pay and the agent's only source of information about his ability. Because the principal cannot commit across periods, the terminal contract is chosen at this posterior.

We assume that the principal implements high effort in both periods after every history.⁸ We write the principal's problem backward. After a contract (S_1, p_1) at history (s_1, w_1, e_1) the

⁶To treat the discrete and continuous cases with a uniform notation, throughout the paper we represent distributions as probability measures, use Stieltjes notation for integrals, and interpret likelihood ratios and related expressions such as $\frac{d\Delta_F(y)}{dF_0(y)}$ as Radon–Nikodym derivatives.

⁷The contract can be interpreted as a committed intermediary that conditions on output and implements a joint lottery over public signals and wages. This interpretation connects the contract space to mediated mechanisms (Myerson, 1982, 1986; Bester and Strausz, 2007; Doval and Skreta, 2022). Both parties learn from the outcome, including the realized wage.

⁸Implementing high effort after every history maximizes profit when the gain from effort is large enough; Appendix A.1 makes this precise by deriving sufficient conditions. Focusing on this case sharpens the trade-off between incentives and learning we study: because the implemented effort does not respond to beliefs, information has no instrumental value for the principal.

terminal problem is

$$\Pi_2(s_1, w_1, e_1) = \max_{S_2, p_2} \iint_{S_2 \times \mathbb{R}_+ \times Y} (y - w_2) \, dp_2(s_2, w_2|y) \, dF(y|1, \mu(s_1, w_1, e_1)) \quad (3)$$

$$\text{s.t. } U_2(s_1, w_1, e_1) := \iint_{S_2 \times \mathbb{R}_+ \times Y} u(w_2) \, dp_2(s_2, w_2|y) \, dF(y|1, \mu(s_1, w_1, e_1)) - c \geq U \quad (\text{P-2})$$

$$\begin{aligned} & \iint_{S_2 \times \mathbb{R}_+ \times Y} u(w_2) \, dp_2(s_2, w_2|y) \, dF(y|1, \mu(s_1, w_1, e_1)) - c \\ & \geq \iint_{S_2 \times \mathbb{R}_+ \times Y} u(w_2) \, dp_2(s_2, w_2|y) \, dF(y|0, \mu(s_1, w_1, e_1)) \end{aligned} \quad (\text{IC-2})$$

The principal maximizes her expected profit over contracts, subject to a participation (P-2) and incentive compatibility constraint (IC-2), and $U_2(s_1, w_1, e_1)$ denotes the agent's expected utility under an optimal terminal contract. In the first period, the principal solves

$$\Pi(\mu_0) = \max_{S_1, p_1} \iint_{S_1 \times \mathbb{R}_+ \times Y} (y - w_1 + \delta \Pi_2(s_1, w_1, 1)) \, dp_1(s_1, w_1|y) \, dF(y|1, \mu_0) \quad (4)$$

$$\text{s.t. } \iint_{S_1 \times \mathbb{R}_+ \times Y} (u(w_1) + \delta U_2(s_1, w_1, 1)) \, dp_1(s_1, w_1|y) \, dF(y|1, \mu_0) - c \geq (1 + \delta)U \quad (\text{P-1})$$

$$\begin{aligned} & \iint_{S_1 \times \mathbb{R}_+ \times Y} (u(w_1) + \delta U_2(s_1, w_1, 1)) \, dp_1(s_1, w_1|y) \, dF(y|1, \mu_0) - c \\ & \geq \iint_{S_1 \times \mathbb{R}_+ \times Y} (u(w_1) + \delta U_2(s_1, w_1, 0)) \, dp_1(s_1, w_1|y) \, dF(y|0, \mu_0) \end{aligned} \quad (\text{IC-Dyn})$$

where $\mu(s_1, w_1, e_1)$ is the posterior (2) induced by (S_1, p_1) . The constraint (P-1) requires that working in both periods is at least as good as exiting at the outset. Constraint (IC-Dyn) requires that high effort is optimal, taking into account that effort changes how both parties interpret the evaluation and hence might affect the agent's continuation utility. The principal's problem in every period thus is a standard moral hazard problem with two added features. First, the principal *chooses a wage and evaluation structure* which may use only partial or noisy information about output. Second, the first-period objective contains the *belief-dependent continuation profit* Π_2 . Finally, note that we write the problem as one of profit maximization. Since high effort is implemented after every history, ex-ante expected output does not depend on the contract, and maximizing profit is equivalent to minimizing the expected discounted wage bill.

2.3 Scope and Applicability

A Running Example Consider a customer-service representative in a call center who has recently been assigned to a product line. His contribution to the firm is whether customers' problems stay solved and whether customers stay with the firm. He does not observe this outcome: a problem that seems settled at the end of a call may recur days later, and customers leave or stay long after the call. A separate customer-analytics system measures these outcomes through repeat contacts, post-contact surveys and retention records, attributes them to the representative and condenses them into a score, on which his bonus is based. Neither he nor his team manager sees more than the score: the underlying records stay in the system, and other

information about his customers is never collected or never linked to him. The manager listens to his calls and hears how carefully he diagnoses problems and follows up. How much such care achieves depends on the representative’s fit with the product and its typical problems, which neither of them knows when he is assigned to it. Each quarter, the manager sets the terms on behalf of the call center: how outcomes are scored, and how scores translate into a bonus. The score and the bonus motivate the representative, but they also tell him how well his care works on this product. In setting the terms, the manager therefore also decides how much he learns.

The example illustrates the employment relationships we study. At the heart of our model is the idea that workers learn from their evaluations and wages, and that firms design these instruments. It isolates this idea by modeling an employment relationship with four features: (i) the manager, like the worker, sees output only through the evaluation the firm designs; (ii) effort is visible to the manager, but it cannot be made the basis of an enforceable contract; (iii) how much effort achieves depends on the worker’s fit with the job, which neither side knows at the outset; and (iv) the firm commits to how it evaluates and pays performance in the current period, but not to the terms of future contracts. These features are deliberately stylized. They set aside other forces through which learning and performance information matter in employment, among them private information of the employer (Bénabou and Tirole, 2003), ratchet effects (Bhaskar and Roketskiy, 2023) and career concerns (Holmström, 1999), so that learning from pay can be studied on its own. No job fits them exactly. Workers often know something about their output, managers form impressions beyond the record, effort is never perfectly observed, part of a worker’s ability is valued elsewhere, and firms can often defer some pay. We read each feature as saying that the corresponding force is small in the application, not that it is absent. In the remainder of this section, we take the features in turn and argue that each is a good approximation in the jobs we have in mind. We close with settings in which they are not.

Output is Measured, not Observed Output y is the worker’s contribution to the firm’s profit: for the representative, whether problems stay solved and customers stay with the firm; for a software engineer, whether a shipped feature increases engagement; for a surgeon, whether their patients fare better than their condition predicted. Such contributions materialize away from the worker: at the customer, in another unit of the firm, or later in time. They are also comparisons with what would have happened otherwise, which nobody sees directly. Contributions therefore have to be measured. Repeat contacts and survey responses are collected by customer-relationship systems and survey providers, the effects of product changes are estimated by analytics teams in controlled experiments, and surgical outcomes are adjusted for patient risk using other surgeons’ cases. What is collected, and whether it is linked to the individual worker, is the firm’s choice. In the model, this choice is the contract (S, p) , which maps output to a signal and a wage. The evaluation can depend on output although nobody observes output, because it is the procedure that measures it; leaving an outcome unmeasured, or unlinked to the worker, is one way of designing it.

A worker may see his own work, but not its consequences. The representative hears how a call ends, but not whether the solution lasts; the engineer sees the feature ship, but not its effect on engagement; the surgeon sees whether a patient survives, but not whether a patient of that

risk was likely to. These consequences are what tell the worker whether his effort works in this job, and the model requires that the evaluation is his main source of information about them.⁹

What the Manager Sees The principal in the model is the manager who sets and renews the worker’s terms on behalf of the firm, not every person or database in the organization. What she knows about the worker’s output is what his personnel file contains: his ratings and his pay. She holds no record of his output that he does not also see, for one of two reasons. Some outcomes are never measured. Others are measured, but they are linked to the worker only in the processed form that enters his rating. The manager may well observe aggregates such as profits or revenues, as long as these do not identify the contribution of the individual worker. Organizations routinely divide the work in this way: analytics teams or external survey providers generate the underlying data, while the managers who set pay receive ratings. In most firms this is simply how the work is organized; sometimes it is written into a rule. Pharmaceutical sales representatives are paid on the prescriptions written by the physicians they visit. Under the AMA’s Physician Data Restriction Program, a physician can bar representatives and their first-line managers from seeing his prescribing data, while the company continues to hold the data and to use them for compensation (Hunkler and Musacchio, 2006). Nor does the firm have a reason to give the manager more. In Section 5.1, we let her privately acquire additional information about the worker’s output and show that, under a natural refinement, the firm cannot gain from it (Proposition 6).

The Evaluation is an Announced Rule The evaluation is the rule that leads from the worker’s output to his score and his pay: which outcomes are measured, how they are condensed into a score, and how pay is determined. It is fixed before he exerts effort, and he knows it even though he does not see his output. Firms commit to such rules through compensation plans and HR policies. In the call center studied by Clemens and Sauermann (2025), the thresholds that determine the bonus are set before each evaluation period and communicated to the agents. A rule of this kind need not reflect output finely. There, customers answer on a scale from zero to ten, but the score records only whether an answer falls in the top, the middle or the bottom range, and only a random sample of customers is surveyed. In the model, these are properties of (S, p) : the signal can be coarser than output, and it can be a random function of it.

Effort is Observed but not Contractible Effort is the care the worker puts into his work: the diligence of a diagnosis, the follow-through on an open case. The manager sees this care when she listens to his calls, which is a supervisor’s main task at that call center (Clemens and Sauermann, 2025): she hears what the worker did, though not what it achieved. Care is nevertheless hard to write into an enforceable contract. A recording shows what was said, but whether a diagnosis was careful enough depends on the customer’s problem, and judging it takes the expertise of someone who works on the product, which is difficult to codify. Measures that

⁹The surgeon example illustrates how seeing outcomes and output (contribution over the counterfactual) differ, and that even expert practitioners cannot easily infer one from the other. Before Pennsylvania published risk-adjusted mortality report cards, a cardiac surgeon knew his own mortality rate but not the rate that his patients’ risk predicted. After publication, surgeons improved their quality by more the further the two had been apart, which indicates that this information was news to them (Kolstad, 2013).

can be verified, such as handling times, call counts or compliance with a script, by contrast, capture a different and manipulable object, and paying for them distorts behavior (Holmström and Milgrom, 1991; Baker, 1992). Effort is thus observable but not verifiable, as in the literature on subjective performance measures (Baker et al., 1994; Levin, 2003). Section 5.2 studies the case in which the manager does not observe effort.

Ability is Specific to the Match The worker’s uncertain ability is his fit with a particular job: with the product and customers he supports, the codebase of a team, or the procedures and staff of a hospital. Fit matters for productivity. Star security analysts who move to another bank suffer a lasting drop in performance, unless they move with their team (Groysberg et al., 2008). Fit also shapes how much effort achieves. Additional care may resolve more problems for a representative who knows the product well, so that effort and fit are complements. Or a good fit may secure much of the outcome and care make up for a weaker one, so that they are substitutes. Neither the worker nor the manager knows his fit when the match is new, as when a representative is hired or moved to a new product. For inexperienced workers, how well their talents suit the job is not reflected in the starting wage, but it predicts future outcomes (Fredriksson et al., 2018). Because fit is specific to the match, it does not affect the outside option.

Contracts are Renewed Period by Period Compensation plans often run for a fixed period and are then set anew. In the same call center, for example, an evaluation period lasts three months, and the bonus is paid in the month that follows (Clemens and Sauermann, 2025). Within a period the plan binds the firm, which pays what the announced terms prescribe. Across periods the firm is free to set new terms, and the worker is free to leave. Performance is thus paid for when it is earned, not at the end of the relationship. In the model, this is commitment within a period but not across periods. Section 5.3 studies the case in which the firm can commit beyond the current period.

Where the four features are present, the firm cannot motivate the worker without informing him. It cannot pay for his effort directly, because effort is not contractible, so it must pay for an evaluation of output. It cannot hold that pay back until his fit is known, because terms are set one period at a time. And pay that depends on output tells the worker about his output, which he has no other way to learn. The firm therefore faces the trade-off we study: an evaluation that tracks output more closely gives sharper incentives today, and teaches the worker more about his fit. It also has reason to care about what it teaches him. What he learns tells him how much his effort achieves in this job, and so changes what it takes to motivate him in the next period. Such beliefs about one’s own ability do move effort and performance (Chen and Schildberg-Hörisch, 2019; Bandiera et al., 2015).

Where the Model does not Apply The mechanism is absent or of second order in several settings. When the worker observes his own output, as with piece rates for installation work or commission sales in which each closed deal is visible to the salesperson, the evaluation cannot shape what he learns. When the relevant effort can be verified, as on a machine-paced assembly line, the firm can pay for it directly. When ability is general and publicly visible, as for academics

or fund managers, learning moves the worker's market value, and career concerns and outside offers dominate. When the relationship is long and deep, and performance is rewarded over a career through seniority scales, promotion ladders or deferred pay, pay and information can be shifted and relational aspects dominate. Finally, information can have a high instrumental value. The worker may need to decide whether to stay or which skills to acquire, the firm whom to promote or how to respond to outside offers. The benefits of disclosure that we abstract from can then outweigh the cost we identify.

3 The Contracting Problem in the Terminal Period

We begin our analysis with the second period to determine the properties of the continuation value which influence the design of the performance evaluation in the first period.

3.1 Performance Evaluation in the Terminal Period

In the second period, the principal's problem reduces to a standard moral hazard problem. First, note that the relationship ends after the second period. Absent any reason to manipulate the agent's learning, the only objective in designing the contract is to provide incentives cheaply and there is no reason to leave information about output unused. It is thus optimal to use the most informative signal structure (Grossman and Hart, 1983). Furthermore, the principal wants to use a deterministic wage as a function of output, since the agent is risk averse.

Proposition 1. *In the second period problem, there exists an optimal contract using a fully informative evaluation structure and a deterministic wage, $S^* = Y$, $p^*(y, w(y)|y) = 1$. Furthermore, the participation and incentive compatibility constraints are binding after every history, and the optimal contract, principal value and agent value in the second period depend on the history (s_1, w_1, e_1) only through the posterior belief $\mu(s_1, w_1, e_1)$.*

We can hence rewrite the principal's problem as an optimization over wage schedules given a posterior belief μ .

$$\begin{aligned} \Pi_2(\mu) = \max_{w: Y \rightarrow \mathbb{R}_+} & \int_Y (y - w(y)) \, dF(y|1, \mu) \\ \text{s.t.} & \int_Y u(w(y)) \, dF(y|1, \mu) - c \geq U \\ & \int_Y u(w(y)) \, dF(y|1, \mu) - c \geq \int_Y u(w(y)) \, dF(y|0, \mu) \end{aligned} \quad (5)$$

3.2 The Shape of the Continuation Value

From the perspective of the first period, the profit in the terminal period will induce a continuation value

$$\int_{[0,1]} \Pi_2(\mu) \, dm(\mu), \quad (6)$$

where $m \in \Delta([0, 1])$ is the distribution over posteriors induced by learning from the evaluation in the first period. The shape of this continuation value is crucial in determining the optimal evaluation structure for the first period. If Π_2 is concave, the agent's learning about his ability

imposes a cost on the principal. This cost is larger the more concave the continuation value. In particular, it is more costly to allow agent learning at the top (bottom) if Π_2'' is decreasing (increasing) in μ .

How does information about the agent's ability affect the continuation value? On the one hand, it allows the principal to adapt the contract to the agent's ability. The contract filters out this nuisance parameter more effectively and provides incentives for effort more precisely. As a consequence of this *filtering channel*, the wage can be less risky and it is cheaper to provide incentives. On the other hand, the agent also has more information in the second period when he decides whether to shirk or exert effort. Consequently, the wage has to be more risky on average in order to satisfy incentive compatibility posterior-by-posterior and it is more expensive to provide incentives. In other words, it is easier to satisfy the incentive compatibility constraint in expectation ("ex-ante") rather than for a more informed agent ("interim") and this *agent-learning channel* depresses the principal's profit. The shape of the continuation value and the resulting preferences of the principal over the information structure depend on the balance of these forces.

To gain intuition about this shape and the two channels, consider first the case of binary output.

Binary Output

Suppose that output is either high or low, $y \in \{y_L, y_H\}$. The probability of high output depends on the agent's effort and ability, $\mathbb{P}(y = y_H|e, \mu) = a + \mu\Delta_a + (b + \mu\Delta_b)e$, with our parameter restrictions, where $a > 0$ denotes the probability of high output under low effort and low ability.

In the binary case, the two forces discussed above directly correspond to the parameters Δ_a and Δ_b . The direct effect of learning on the continuation value through the filtering channel depends on the effect of ability on the probability of high output (Δ_a). The effect of the agent-learning channel is stronger the larger the effect of ability on the impact of effort (Δ_b). The agent-learning channel dominates and learning reduces profits whenever the interaction between effort and ability is sufficiently strong. With complementarities ($\Delta_b > 0$), we require that they are sufficiently strong, i.e. that the technology is log-supermodular in effort and ability ($a\Delta_b - b\Delta_a > 0$). With substitutes ($\Delta_b < 0$), we require that the substitutability is sufficiently strong, i.e. that the technology is log-submodular in effort and ability ($b\Delta_a + \Delta_b(1 - a) < 0$).¹⁰ Between these two cases of strong interaction, the agent-learning channel can be dominated by the filtering channel, for example at an additive technology, and learning then reduces the expected cost of incentives.

The technology also determines which kind of information is more costly. To see how, consider the second period IC,

$$u(w_H) - u(w_L) \geq \frac{c}{b + \Delta_b\mu}, \quad (7)$$

where w_L, w_H denote the optimal wage after y_L and y_H . The impact of effort, $b + \Delta_b\mu$, and the bonus (in utility space) required to satisfy incentive compatibility are inversely proportional. As we have seen above, this effect dominates and the continuation value is concave if the interaction

¹⁰We call the technology log-supermodular if the likelihood ratio of high output, $\mathbb{P}(y = y_H|1, \mu)/\mathbb{P}(y = y_H|0, \mu)$, is strictly increasing in μ . This is equivalent to $a\Delta_b - b\Delta_a > 0$. We call it log-submodular if the likelihood ratio of low output, $\mathbb{P}(y = y_L|1, \mu)/\mathbb{P}(y = y_L|0, \mu)$, is strictly increasing in μ . This is equivalent to $b\Delta_a + (1 - a)\Delta_b < 0$.

is sufficiently strong. Furthermore, this effect of learning is stronger when the agent is pessimistic about the impact of his effort, $b + \Delta_b \mu$. Then even a small change in his belief has a large relative effect and causes large changes to the bonus. If effort and ability are strong complements, it is workers with low posterior beliefs who are disproportionately difficult to motivate. They believe that their effort only has a small impact on the likelihood of high output. The principal wants to avoid such beliefs and therefore avoid “information at the bottom”. If effort and ability are strong substitutes, it is instead the workers with high posteriors that are disproportionately difficult to motivate. Such workers are willing to rest on their laurels thinking that their effort is unlikely to matter since their high ability is likely to guarantee success in any case. The principal wants to avoid such beliefs and therefore avoid “information at the top”.

Let us summarize the discussion.

Proposition 2. *Suppose that output is binary. Π_2 is strictly concave if the technology is either log-supermodular ($a\Delta_b - b\Delta_a > 0$) or log-submodular ($b\Delta_a + \Delta_b(1 - a) < 0$). Furthermore, $\Pi_2''' > 0$ if the technology is log-supermodular and $\Pi_2''' < 0$ if the technology is log-submodular.*

The General Case

In the general case, the strength of both effects and thus the shape of the continuation value now depend on the whole distribution and its properties. Let

$$M = [\min_{y \in Y} \mu_0 \frac{dF(y|1, 1)}{dF(y|1, \mu_0)}, \max_{y \in Y} \mu_0 \frac{dF(y|1, 1)}{dF(y|1, \mu_0)}] \subset [0, 1]$$

denote the set of reachable posterior beliefs. Then, define the index (for $\Delta_b \neq 0$)

$$Z(y; \mu) = \frac{\partial_\mu \log dF(y|1, \mu)}{\partial_\mu \log(b + \Delta_b \mu)} = \frac{(\Delta_a + \Delta_b)(b + \Delta_b \mu)}{\Delta_b} \frac{d\Delta_F(y)}{dF(y|1, \mu)}, \quad \mathcal{E} = \max_{\mu \in M} \max_{y \in Y} |Z(y; \mu)|. \quad (8)$$

This index compares the statistical effect of belief on output (the filtering channel) with its effect on the return to effort (the agent-learning channel). A small index means that beliefs primarily change the impact of effort, while the high-effort output distribution changes relatively little. Such changes make the agent-learning channel strong relative to the filtering channel.

Proposition 3 (Elasticity Test). *The following holds throughout M :*

- (i) *If $\Delta_b = 0$, then $\Pi_2'' > 0$.*
- (ii) *If $\mathcal{E} \leq \sqrt{2}$ and $\Delta_b \neq 0$, then $\Pi_2'' < 0$.*
- (iii) *If $\mathcal{E} \leq 4/3$ and $\Delta_b \neq 0$, then $\Pi_2'' < 0$ and $\text{sign } \Pi_2''' = \text{sign } \Delta_b$.*

The bounds need only be checked at the two endpoints of M .

Learning is costly when beliefs have a sufficiently large effect on the return to effort relative to their statistical effect on output, that is when Δ_b is nonzero and $\mathcal{E}$ is not too large. Under a stronger condition on $\mathcal{E}$, we can also determine where it is more costly. In both cases the cost is largest where the agent thinks his effort has little effect: at low beliefs with complements ($\Delta_b > 0$), at high beliefs with substitutes ($\Delta_b < 0$).

The elasticity formula (8) also clarifies several limiting cases. Close to an additive technology $\Delta_b \approx 0$, the filtering channel overwhelms the agent-learning channel as the agent has barely anything to learn about the impact of effort. Direct differentiation gives strict convexity of the continuation value at $\Delta_b = 0$; by continuity, this remains true for sufficiently small interactions of either sign.

For the case of sufficiently strong substitutes, suppose $\Delta_a > 0$ and let Δ_b approach $-\Delta_a$ from above, keeping $b + \Delta_b > 0$. Then

$$\mathcal{E} = \frac{\Delta_a + \Delta_b}{|\Delta_b|} \max_{\mu \in M} \max_{y \in Y} \left| (b + \Delta_b \mu) \frac{d\Delta_F(y)}{dF(y|1, \mu)} \right| \rightarrow 0.$$

Hence sufficiently strong substitutability gives $\Pi_2'' < 0$ and $\Pi_2''' < 0$: ability has almost no effect on output under high effort, but substantially reduces the worker's return to effort. The filtering benefit of learning therefore becomes small relative to its incentive cost.

For the case of sufficiently strong complements, we need an additional bound. Suppose that complements are sufficiently strong, i.e. $\Delta_b \geq 3\Delta_a$, and that the distribution does not shift too much, $\frac{dF(y|0, \mu)}{dF(y|1, \mu)} \leq 2$ for all $\mu \in M$. Then, $\mathcal{E} \leq 4/3$ since $Z(y; \mu) = \left(1 + \frac{\Delta_a}{\Delta_b}\right) \left(1 - \frac{dF(y|0, \mu)}{dF(y|1, \mu)}\right)$. Under these conditions, the continuation value is concave ($\Pi_2'' < 0$), and learning is most costly if it induces low posteriors ($\Pi_2''' > 0$).

A further useful limit instead makes all technological effects small together: write $(b, \Delta_a, \Delta_b) = \varepsilon(\bar{b}, \bar{\Delta}_a, \bar{\Delta}_b)$, with $\bar{\Delta}_b \neq 0$ and the productivity restrictions preserved. To ensure an interior contract for a fixed outside option, also let $c = \varepsilon^2 \bar{c}$. For fixed F_0 and Δ_F , we then have

$$Z = \varepsilon \frac{\bar{\Delta}_a + \bar{\Delta}_b}{\bar{\Delta}_b} \frac{(\bar{b} + \bar{\Delta}_b \mu) d\Delta_F(y)}{dF(y|1, \mu)} = O(\varepsilon)$$

uniformly. Thus, for sufficiently small $\varepsilon > 0$, the continuation value is strictly concave and its third derivative has the sign of $\bar{\Delta}_b$. Beliefs have little effect on the output distribution but retain a substantial proportional effect on the return to effort.

4 The Optimal Evaluation Structure

In this section, we transform and solve the first-period problem. We first show that its constraints (P-1) and (IC-Dyn) are static. Then, we transform the problem of finding the optimal contract (S, p) into the belief-space problem of finding an optimal distribution over posteriors and deterministic wages as a function of the posterior. We then characterize the optimal evaluation structure and derive comparative statics.

Throughout, we suppress time subscripts when no confusion can arise.

4.1 Static Constraints

Consider the first-period participation and incentive compatibility constraints (P-1) and (IC-Dyn). Both constraints are dynamic: they track the continuation utility of the agent U_2 as a function of the history. By Proposition 1, however, we know that the agent is pushed down to his outside option after every history, including after deviations. Hence, $U_2(s_1, w_1, e_1) \equiv U$, and

the constraints simplify to

$$\iint_{S \times \mathbb{R}_+ \times Y} u(w) dp(s, w|y) dF(y|1, \mu_0) - c \geq U \quad (\text{P-Static})$$

$$\iint_{S \times \mathbb{R}_+ \times Y} u(w) dp(s, w|y) dF(y|1, \mu_0) - c \geq \iint_{S \times \mathbb{R}_+ \times Y} u(w) dp(s, w|y) dF(y|0, \mu_0). \quad (\text{IC-Static})$$

The resulting constraints are static. As the continuation payoff of the agent is the same after every history, participation requires a sufficient level of utility in the first period, and incentive compatibility requires sufficient monetary incentives paid out in the first period.

A key assumption to arrive at this simplification is that the principal cannot commit to a long-term contract and that effort is observed after the first period. The first rules out that the principal commits to deferred bonus payments. Together, the two assumptions ensure that the agent cannot endogenously obtain private information about his ability. Even after a deviation to low effort, which changes the interpretation of the signal and hence the agent's posterior, this belief is still shared by the principal (due to the observable effort assumption) who uses it in setting the second period contract (which is made possible by period-by-period contracting). We study ways to relax these two assumptions, the resulting dynamic incentive compatibility constraints and how to deal with them in the analysis in Section 5.

4.2 Transformation to Posterior Space

A first-period contract affects continuation profits only through the distribution over posterior beliefs $\mu(s, w, e)$ it induces. This follows since the principal's continuation profit Π_2 depends on the history only through the posterior and since the constraints of the problem are static by the previous section. We now build from these observations to rewrite the whole first period problem as the choice of a distribution over posterior beliefs and a deterministic wage as a function of this belief.

Let

$$\mu(y) = \frac{\mu_0 dF(y|1, 1)}{dF(y|1, \mu_0)} \quad (9)$$

denote the posterior belief after observing output y and assuming high effort and let $\bar{m} \in \Delta([0, 1])$ denote its distribution. Any contract (S, p) induces a distribution over posterior beliefs $m \in \Delta([0, 1])$ that is a mean-preserving contraction of $\bar{m}$, denoted as $m \leq_{MPS} \bar{m}$. Conversely, for every such distribution, there exists an evaluation structure that induces it.

Define the score as

$$x(y) := \frac{(b + \Delta_b \mu_0) d\Delta_F(y)}{dF(y|1, \mu_0)}$$

The incentive compatibility constraint (**IC-Static**) can then be written as

$$\iint_{S \times \mathbb{R}_+ \times Y} u(w)x(y) dp(s, w|y) dF(y|1, \mu_0) \geq c.$$

By Bayes' rule

$$\mu(y) - \mu_0 = \frac{\mu_0 (dF(y|1, 1) - dF(y|1, \mu_0))}{dF(y|1, \mu_0)} = \mu_0(1 - \mu_0)(\Delta_a + \Delta_b) \frac{d\Delta_F(y)}{dF(y|1, \mu_0)}.$$

Comparing this expression with the definition of the score, it is easy to see that we can write the score as a linear function of the posterior, as

$$x(y) = \chi(\mu(y) - \mu_0), \quad \text{where } \chi = \frac{1}{\mu_0(1 - \mu_0)} \frac{b + \Delta_b \mu_0}{\Delta_a + \Delta_b} > 0. \quad (10)$$

An output realization is indicative of high effort precisely when and to the degree that it is indicative of high ability. This is a result of our linear distribution function specification (1) and a key ingredient in simplifying our problem.

For a fixed contract, let $\mathbb{E}$ denote the expectation under the common prior μ_0 and high first-period effort. Since the evaluation and wage are generated from output, we have that $\mu(s, w, 1) = \mathbb{E}[\mu(y)|s, w, 1]$. Taking the same conditional expectation of the score hence gives

$$\mathbb{E}[x(y)|s, w, 1] = \chi(\mu(s, w, 1) - \mu_0).$$

Using the law of iterated expectations and (10), incentive compatibility becomes

$$\mathbb{E}[u(w)x(y)] = \mathbb{E}[u(w)\chi(\mu(s, w, 1) - \mu_0)] \geq c.$$

Thus, the distribution of output and evaluations enters the incentive constraint only through the joint distribution of wages and on-path posterior beliefs.

It remains to show that random wages conditional on a posterior are unnecessary. For each induced posterior μ , replace the conditional wage distribution by the deterministic wage $w(\mu) := u^{-1}(\mathbb{E}[u(w)|\mu])$. This replacement preserves participation and incentive compatibility. Risk aversion implies that

$$w(\mu) \leq \mathbb{E}[w|\mu],$$

so it weakly reduces wage costs. Pooling histories with the same posterior also preserves the distribution over the posterior, and hence continuation profit is unchanged.

Using this representation, we can thus rewrite the contracting problem (4) as the choice of a distribution over posteriors about ability and a deterministic wage as a function of that posterior.¹¹ Writing $\mathbb{E}_m$ for the expectation with respect to the posterior distribution m , and $x(\mu) = \chi(\mu - \mu_0)$ the reduced problem is

¹¹The problem is written in terms of the posterior belief presuming high effort by the agent. This does not mean that a deviation does not change the continuation belief and continuation contract. It does, and this is essential in the static reduction of the constraints achieved in the previous section. Given this reduction, we can write the problem based on this on-path belief, as we have shown in this section, and indeed we do not have to track the beliefs after a deviation.

Using the on-path belief in the definition of the evaluation structure is convenient, but also arbitrary. We could rewrite the problem in terms of the score, or indeed in terms of the deviation belief.

$$\max_{m \in \Delta([0,1]), w: [0,1] \rightarrow \mathbb{R}_+} \int_Y y \, dF(y|1, \mu_0) + \mathbb{E}_m [\delta \Pi_2(\mu) - w(\mu)] \quad (11)$$

$$\text{s.t. } \mathbb{E}_m [u(w(\mu))] - c \geq U \quad (12)$$

$$\mathbb{E}_m [x(\mu)u(w(\mu))] \geq c \quad (13)$$

$$m \leq_{MPS} \bar{m}$$

Proposition 4. *A contract (S, p) solves the principal's problem (4) if and only if it induces a belief-space contract (m, w) that solves the belief-space problem (11).*

4.3 The Optimal Evaluation Structure

We are now ready to solve the principal's problem. Providing incentives for the agent is cheaper in this period if the evaluation structure is more informative, while the resulting learning can be costly by increasing the expected cost of incentivizing the agent in the next period. How is this trade-off resolved in the optimal contract? We employ the tools of information design to characterize the optimal evaluation structure without imposing any exogenous restrictions.

Definition 1. We call an evaluation structure **generalized upper-censorship** if there exists a cutoff y^* such that it reveals output strictly below y^* , it pools output $(y^*, \max Y]$, and there is a probability $\sigma \in [0, 1]$ such that y^* is revealed with probability $1 - \sigma$ and pooled with the interval $(y^*, \max Y]$ with probability σ . We call an evaluation structure **generalized lower-censorship** if there exists a cutoff y^* such that it reveals output above y^* , it pools output $[\min Y, y^*)$, and there is a probability $\sigma \in [0, 1]$ such that y^* is revealed with probability $1 - \sigma$ and pooled with the interval $[\min Y, y^*)$ with probability σ .

This notion generalizes the censorship policies commonly encountered in linear persuasion problems (see Kolotilin et al., 2022, for a discussion) to account for the possibility of atoms in the underlying distribution. In terms of our application, censorship evaluation structures correspond to commonly encountered bonus schemes. Lower-censorship corresponds to a base wage plus fine-grained performance bonuses that are paid only upon reaching a certain performance threshold. Upper-censorship corresponds to a base wage with fine-grained deductions when the performance requirement is not met.

Theorem 1. *The optimal evaluation structure is*

1. *fully informative if $\Pi_2'' > 0$,*
2. *generalized lower-censorship if $\Pi_2''' > 0$, and*
3. *generalized upper-censorship if $\Pi_2''' < 0$.*

The theorem characterizes the structure of the optimal evaluation for the cases in which the continuation value has an unambiguous shape. If there is no cost from learning ($\Pi_2'' > 0$), a fully informative evaluation is optimal both from the perspective of incentives and from the perspective of learning. If learning is costly ($\Pi_2'' < 0$), the principal prefers a less informative

evaluation. The optimal performance evaluation takes the form of partial pooling as determined by the shape of the continuation value. If pessimistic agents are disproportionately expensive to motivate ($\Pi_2''' > 0$), the principal avoids strong pessimism by pooling at the bottom at a base wage and providing precise evaluation and tailored performance bonuses above a threshold. If by contrast optimistic agents are disproportionately expensive to motivate ($\Pi_2''' < 0$), the principal avoids strong optimism by pooling at the top at a capped high-performance wage and providing precise evaluation and tailored low-performance deductions below a threshold.

In the optimal contract, wages are discontinuous when viewed as a function of the true performance of the agent. Under lower-censorship, for example, the bonus the agent receives when his performance just clears the threshold of the pooling region can be substantial. This discontinuity of the bonus follows naturally from the informational considerations driving our result, as the pooled evaluation is strictly less indicative of high effort than any other evaluation outcome.

The optimal evaluation structure and wage contract are unique up to deviations of measure zero and informationally equivalent modifications of the evaluation structure. For example, in the case of lower censorship with optimal cutoff y^* , the principal could split the pooling interval in an uninformative way, revealing whether output is in $[\min Y, y_1^*) \cup [y_2^*, y^*)$ or in $[y_1^*, y_2^*)$ where y_1^*, y_2^* are chosen such that the posterior after observing the two evaluations is the same as after observing the full pooling set $[\min Y, y^*)$.

We can characterize the optimal evaluation based on the technology by combining Proposition 3 and Theorem 1.

Corollary 1. *Suppose that $\Delta_b = 0$. Then the optimal evaluation is fully informative.*

Suppose that effort and ability are complements ($\Delta_b > 0$) and the agent-learning channel dominates the filtering channel ($\mathcal{E} \leq 4/3$). Then the optimal evaluation is generalized lower-censorship.

Suppose that effort and ability are substitutes ($\Delta_b < 0$) and the agent-learning channel dominates the filtering channel ($\mathcal{E} \leq 4/3$). Then the optimal evaluation is generalized upper-censorship.

Corollary 1 links the form of the evaluation to the technology. With complements, an agent who comes to doubt his ability expects his effort to achieve less and requires a larger bonus, so the evaluation pools weak performance; with substitutes, it is the confident agent who is expensive to motivate, and the evaluation pools strong performance.

In the binary case, we get a different interpretation of upper- and lower-censorship evaluation structures.

Corollary 2. *Suppose that we are in the binary case. Then,*

1. *if the technology is log-supermodular ($a\Delta_b - b\Delta_a > 0$), the optimal evaluation is binary and tough, with $S = \{G, B\}$, wages $w_G > w_B$ and $\sigma \in [0, 1)$ such that*

$$p(G, w_G | y_H) = 1 - \sigma, \quad p(B, w_B | y_H) = \sigma, \quad p(G, w_G | y_L) = 0, \quad p(B, w_B | y_L) = 1; \quad (14)$$

2. if the technology is log-submodular ($b\Delta_a + (1-a)\Delta_b < 0$), the optimal evaluation is binary and lenient, with $S = \{G, B\}$, wages $w_G > w_B$ and $\sigma \in [0, 1)$ such that

$$p(G, w_G | y_H) = 1, \quad p(B, w_B | y_H) = 0, \quad p(G, w_G | y_L) = \sigma, \quad p(B, w_B | y_L) = 1 - \sigma. \quad (15)$$

With strong complementarities, the evaluation appears tough: Even high output does not ensure a good evaluation while low output always results in a bad one. The principal does not engage in “grade inflation”, but instead measures performance against an “unreasonably” high standard. With strong substitutes instead, the evaluation appears lenient: Even low output is sometimes rewarded with a good evaluation. To avoid costly complacency, the principal does not always call out the shortcomings of the agent’s output, which in turn implies that not getting called out is not very informative about high ability.

Note also that the motive to control learning does not increase the complexity of the evaluation structure. While the most informative evaluation is binary, a noisy evaluation can take many forms. The corollary implies that the optimal evaluation remains binary. This result is not an immediate consequence of the assumption of binary output or binary ability. Since the contracting problem has two constraints – participation and incentive compatibility – results from constrained information design suggest that the optimal evaluation structure may involve up to four signals (Le Treust and Tomala, 2019; Doval and Skreta, 2024) even in the binary case. The joint design of wages and information is crucial: this result may not hold when the wage function is fixed exogenously.

Sketch of the Proof of Theorem 1

The proof of Theorem 1 poses the challenge of jointly designing an information structure and a wage scheme. For a fixed wage scheme, the information design problem can in principle be solved by the usual concavification or linear programming approaches (Aumann and Maschler, 1995; Kamenica and Gentzkow, 2011; Kolotilin, 2018; Dworczak and Martini, 2019) taking into account the participation and incentive compatibility constraints (Boleslavsky and Kim, 2017; Le Treust and Tomala, 2019). The constraints make the problem multidimensional so that, although in some cases conceptually tractable, a direct approach is analytically difficult (Doval and Skreta, 2024). We instead exploit the tractability of wage optimization conditional on an information structure. Following Georgiadis and Szentes (2020), we maximize the Lagrangian pointwise over wages, and then characterize the maximizing information structure fixing the Lagrange multipliers. We establish strong duality and dual attainment, ensuring that every optimal contract solves these maximization problems at minimizing multipliers. Georgiadis and Szentes (2020) develop this strategy by casting the duality argument as a zero-sum game and apply it to a static moral-hazard problem with costly information acquisition. Here, evaluation also shapes learning about ability and thereby continuation profits. In the main text, we sketch the main steps of the argument, also suppressing boundary conditions related to the non-negativity constraint on wages. We relegate the explicit proofs and duality arguments to the appendix.

Consider the Lagrangian $\mathcal{L}$ associated to the contracting problem (11), where we retain the condition that $m \leq_{MPS} \bar{m}$ as a constraint and $\lambda = (\lambda_P, \lambda_{IC})$ denotes the Lagrange multipliers

associated to the participation and incentive constraint, respectively,

$$\begin{aligned}\mathcal{L}(m, w; (\lambda_P, \lambda_{IC})) = & \int \left\{ \delta \Pi_2(\mu) - w(\mu) \right. \\ & + \lambda_P (u(w(\mu)) - c - U) \\ & \left. + \lambda_{IC} \left(\frac{1}{\mu_0(1 - \mu_0)} \frac{b + \Delta_b \mu_0}{\Delta_a + \Delta_b} (\mu - \mu_0) u(w(\mu)) - c \right) \right\} dm(\mu).\end{aligned}\tag{16}$$

We have omitted expected output, since it is a constant in the principal's problem.

Wage Setting Fix a feasible distribution $m \in \Delta([0, 1])$ and consider the problem of finding optimal wages subject to the participation and incentive constraint. This is a standard moral hazard problem and the optimal wage schedule follows from pointwise optimization of the Lagrangian.

$$w^*(\lambda, \mu) = \frac{1}{2} \left[\max \left\{ 0, \lambda_P + \lambda_{IC} \frac{1}{\mu_0(1 - \mu_0)} \frac{b + \Delta_b \mu_0}{\Delta_a + \Delta_b} (\mu - \mu_0) \right\} \right]^2 \tag{17}$$

Plugging this function back into the Lagrangian of the problem, it is written as an expectation of a function of the posterior

$$\sup_w \mathcal{L}(m, w; \lambda) = \int \ell^*(\mu; \lambda) dm(\mu) \tag{18}$$

where ℓ^* is the integrand of (16) evaluated at (17).

Information Design Therefore, the information design problem *for a given* λ is of standard form. The principal simply maximizes the expectation of a function of posteriors,

$$\sup_{m \leq_{MPS} \bar{m}} \int \ell^*(\mu; \lambda) dm(\mu), \tag{19}$$

subject to the constraint that the chosen distribution m is a mean-preserving contraction of the distribution $\bar{m}$ which would be induced by the fully revealing performance evaluation. Such an information design problem can be solved via the usual linear programming approaches (Kolotilin, 2018; Dworzak and Martini, 2019; Kolotilin et al., 2022). In order to determine the solution, we need to determine the shape of ℓ^* as a function of μ . Using an envelope argument, it is straightforward to show that

$$\frac{\partial^2}{\partial \mu^2} \ell^*(\mu; \lambda) = \lambda_{IC}^2 \left[\frac{b + \Delta_b \mu_0}{(\Delta_a + \Delta_b) \mu_0 (1 - \mu_0)} \right]^2 + \delta \Pi_2''(\mu). \tag{20}$$

The first term of (20) corresponds to the cost of providing incentives in the first period. It is positive, indicating convexity: the principal prefers the most informative evaluation structure in order to reduce agency costs. The second term corresponds to the impact of beliefs on the continuation value. If the continuation value is also convex, then so is the per-posterior payoff overall, and a fully informative evaluation structure is optimal. If it is concave, there is a

trade-off: the principal wants to keep the agent uninformed in order to reduce agency costs in the next period.

To obtain the optimal evaluation, we then need to further consider the shape of ℓ^* . We have that

$$\frac{\partial^3}{\partial \mu^3} \ell^*(\mu; \lambda) = \delta \Pi_2'''(\mu)$$

With square-root utility, the third derivative of the Lagrangian is fully determined by the continuation value. Hence, we are solving a linear persuasion problem with an S-shaped objective and generalized censorship then follows from Proposition 7 in [Ostrizek \(2026\)](#).

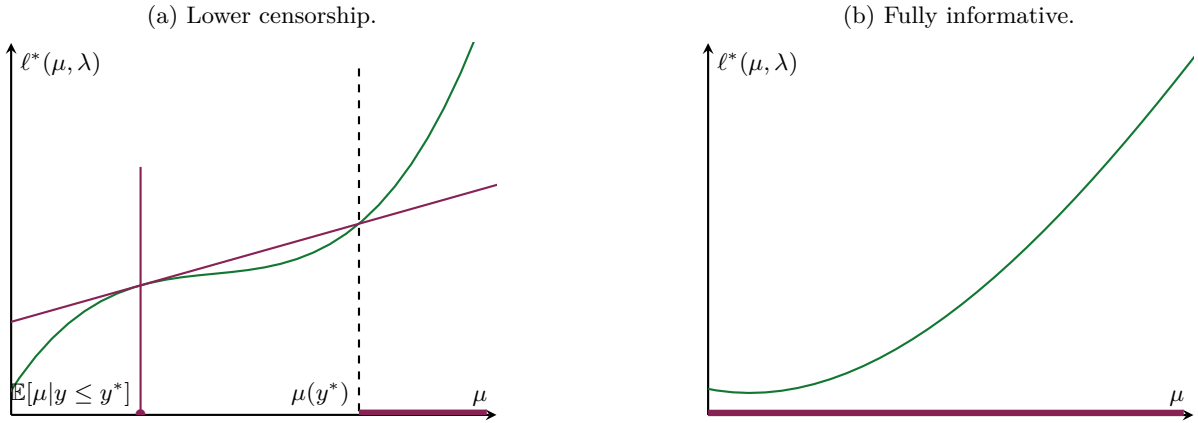


Figure 1: The optimal evaluation structure.

In the relevant case with $\Pi_2'' < 0$ there are three possibilities. If λ_{IC} is sufficiently small, the objective ℓ^* is strictly concave and the optimal information structure is uninformative. Clearly, this cannot be the case in the solution of (11), since the incentive constraint cannot be satisfied without any information. As λ_{IC} increases, we reach a region where, depending on the sign of Π_2''' , ℓ^* is concave for low posteriors and convex for high posteriors or vice versa. The optimal information structure is fully informative in the region where ℓ^* is convex and uses partial pooling at the other side, leading to a point mass in belief space (Fig. 1a). Finally, as λ_{IC} increases further, ℓ^* becomes globally convex as the costs of incentives overwhelm the gains from concealing information. The resulting evaluation structure becomes fully informative (Fig. 1b).

The arguments in the appendix establish through a series of lemmas that a solution to the problem exists and is characterized by the two-step procedure above.

4.4 Analysis of the Solution and Comparative Statics

The properties of an interior solution are pinned down by a tangency condition provided that $\bar{m}$ admits a positive density at the cutoff. The censorship cutoff in belief space, μ^* , and the posterior after the pooling evaluation, $\bar{\mu}$, are determined by

$$\ell^*(\mu^*; \lambda(\mu^*)) = \ell^*(\bar{\mu}; \lambda(\mu^*)) + \frac{\partial \ell^*}{\partial \mu} \Big|_{(\mu; \lambda) = (\bar{\mu}; \lambda(\mu^*))} (\mu^* - \bar{\mu}) \quad (21)$$

(Fig. 1). Note, however, that this condition does not correspond to information design on a given function. Instead, there is an additional dependence on $\lambda(\mu^*)$. This term is present because we design payoffs and information jointly, subject to a participation and incentive compatibility constraint. For the graphical representation of our analysis this implies that, as we vary the cutoff point in Figure 1, not only the tangent line but the whole function ℓ^* shifts.

We can eliminate the multipliers and transform (21) into a more concrete form (assuming upper-censorship, $\Pi_2''' < 0$, for concreteness):

$$\frac{1}{2} \frac{c^2}{\left(\frac{b+\Delta_b\mu_0}{(\Delta_a+\Delta_b)\mu_0(1-\mu_0)}\right)^2} \frac{(\mu^* - \bar{\mu})^2}{\left(\int_0^{\mu^*} (\mu - \mu_0)^2 d\bar{m}(\mu) + (1 - \bar{m}([0, \mu^*])) (\bar{\mu} - \mu_0)^2\right)^2} \quad (22)$$

$$= \delta [\Pi_2(\bar{\mu}) - \Pi_2(\mu^*) + \Pi_2'(\bar{\mu})(\mu^* - \bar{\mu})]$$

The LHS is the benefit from a more informative evaluation structure in period one: revealing additional output realizations decreases the cost of providing incentives today. This effect is larger if agency costs ($\frac{c}{b+\Delta_b\mu_0}$) are already high and if a large dispersion of posteriors is required for a given level of information about output (since output is very informative: $(\Delta_a + \Delta_b)\mu_0(1 - \mu_0)$ large). The RHS is the cost of the additional learning through the continuation value. Indeed, the RHS is a measure of the concavity of the continuation value.

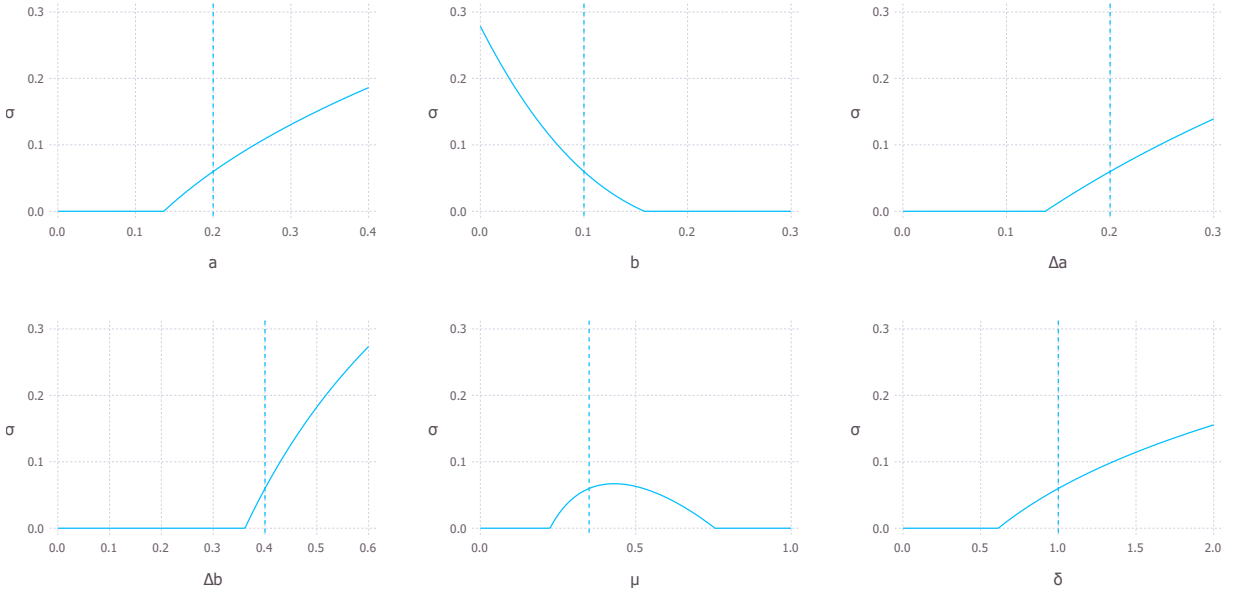


Figure 2: Comparative statics in the binary case (Corollary 2) around $(a, b, \Delta_a, \Delta_b) = (0.2, 0.1, 0.2, 0.4)$, $\mu_0 = 0.35$, $\delta = 1$.

This trade-off yields comparative statics that also apply when output is discrete. To gain further insight into the mechanism, let us distinguish the costs of effort across periods, where c_t denotes the cost in period t .

Proposition 5. *Suppose that learning is costly ($\Pi_2'' < 0$), then:*

1. *The extent of pooling depends on costs and discounting only through $\delta(c_2/c_1)^2$, and is weakly increasing in this ratio.*

2. *In the binary case with strong complementarities ($a\Delta_b > b\Delta_a$) the degree of toughness σ is weakly increasing in Δ_a and Δ_b ; both effects are strict when $\sigma > 0$.*

The first part captures the relative importance of current and future incentives. If first period incentive costs become more important, either because they increase or the principal becomes less patient, the evaluation becomes more informative, and vice versa. This is in line with the intertemporal nature of the effect: the principal chooses a less informative evaluation exactly in order to preserve the worker's motivation (in expectation) in the future.

The second part connects the evaluation policy to the nature of the job. When performance reveals more about the worker's ability, or effort effectiveness depends more strongly on ability, the principal uses a tougher evaluation: more high-output realizations receive the pooled low rating. Concealing this information limits the pessimism that makes workers expensive to motivate in the next period. The other parameters, as well as Δ_a and Δ_b with a general technology, do not have effects that can be signed in general; they jointly affect incentive costs in the first period, the extent of learning, and its cost in the second period; the net effect on the informativeness of the evaluation may go either way. Figure 2 illustrates the comparative statics in an example.

What do these results mean for the customer-service representative of Section 2.3? If care achieves more for a representative who knows the product well, effort and ability are complements. The optimal plan then does not distinguish among weak outcomes: all scores below a target earn the base wage, and the bonus rises with the score above it. The plan pools more when the cost of future incentives weighs more heavily relative to the current one. When outcomes are coarse, like a large case that is resolved or not, the plan is tough: every failure is recorded, but only some successes are certified, and more so in jobs where performance reveals more about fit. For an account manager who retains most clients with ordinary care when he is well matched, effort and ability are substitutes, and the plan is reversed: a fixed amount for meeting a standard, with deductions for shortfalls.

5 Discussion and Extensions

In the previous sections, we made several assumptions to ensure that the agent's posterior belief is the only state variable of the problem and that no party can acquire endogenous private information. In this section, we relax these assumptions. In Section 5.1, the principal can privately acquire information about first-period output, so that her second-period offer may signal what she knows. She cannot gain from this in any equilibrium without implicit commitment (Proposition 6). In Section 5.2, effort is hidden, so that an agent who shirks holds a private belief about his ability. In the binary case, the optimal evaluation remains tough with complements and may add a neutral rating with substitutes (Proposition 8). In Section 5.3, the principal can commit beyond the current period, so that pay, and under full commitment also information, can be deferred. The trade-off between incentives and learning remains at work in both cases.

5.1 Private Information of the Principal

Suppose that the principal can privately acquire more information about first-period output. She may then use it to set the contract of the agent in the second period, who in turn learns from this contract offer. Can the principal improve on the public-information benchmark by acquiring private information? Can she even achieve the benchmark profit despite the endogenous informed principal problem resulting from private information acquisition? In this section, we sketch this model and show that the profit of any equilibrium without implicit commitment is bounded above by the optimal profit in the problem without information acquisition. By using private information, the principal gets the learning cost without the incentive benefit of a more precise evaluation in the first period. Any such equilibrium that attains the bound uses the optimal evaluation of Theorem 1 (on path). The full formal development of this model is relegated to Appendix B.

Model

The setup of the model is identical to the baseline model, except that together with the contract in the first period the principal also chooses an experiment on first period output. Formally, she offers a contract $C_1 = (S_1, p_1)$ as in Section 2. She also chooses an experiment $K = (S_P, p_P)$, where S_P is a private signal space and $p_P : Y \times S_1 \times \mathbb{R}_+ \rightarrow \Delta(S_P)$. Both parties observe s_1 and the wage w_1 ; only the principal observes s_P .

We distinguish two cases. With *observable acquisition*, the agent observes K together with the first-period contract; with *covert acquisition*, the agent observes only C_1 ; both the choice of K and its realization are private. After observing the contract and, when applicable, the experiment, the agent decides whether to participate and chooses effort e_1 , which both parties observe. Output and the two signals then realize.

At the start of the second period, the principal offers a new contract $C_2 = (S_2, p_2)$; the offered contract may depend on her private information. The agent observes the offer, updates his belief about his ability, and decides whether to participate and which effort e_2 to exert.

Let $h = (C_1, s_1, w_1, e_1)$ denote the public history before the second-period offer, including K when acquisition is observable. Write $\mu(h)$ for the public belief at that history which follows from Bayes' rule (2), $\mu_P(h, s_P)$ for the principal's belief after observing her private signal

$$\mu_P(h, s_P) = \frac{\mu_0 \int_Y dp_P(s_P|y, s_1, w_1) dp_1(s_1, w_1|y) dF(y|e_1, 1)}{\int_Y dp_P(s_P|y, s_1, w_1) dp_1(s_1, w_1|y) dF(y|e_1, \mu_0)}, \quad (23)$$

and $\mu_A(h, C_2)$ for the agent's belief after observing the offer. The principal uses μ_P to evaluate the cost and revenue of a contract, while the agent uses μ_A to evaluate participation and effort. These beliefs can differ even after the agent has observed the offer. On the equilibrium path, Bayes' rule implies $\mu_A(h, C_2) = \mathbb{E}[\mu_P(h, s_P)|h, C_2]$.

A perfect Bayesian equilibrium specifies the principal's choice of contract and experiment, her second-period offer after each public and private history, the agent's participation and effort decisions in both periods, and the agent's beliefs about ability and the principal's private information, including her acquisition choice when it is unobserved. Strategies must be

sequentially rational and beliefs must satisfy Bayes' rule wherever it applies. Appendix B.1 gives the formal definition and all updating formulas.

Definition 2. An equilibrium is *no-holdup* if, at almost every on-path terminal information set, the agent accepts, chooses high effort, and receives net utility U :

$$\iint_{S_2 \times \mathbb{R}_+ \times Y} u(w_2) \, dp_2(s_2, w_2|y) \, dF(y|1, \mu_A(h, C_2)) - c = U. \quad (24)$$

An equilibrium in the observable-acquisition game satisfies *no-signaling-what-you-don't-know* if $\mu_A(h, C_2)$ lies in the closed convex hull of the support of $\mu_P(h, \cdot)$ under the announced experiment.

No-holdup restricts the class of equilibria we compare with the baseline. Without it, the threat of unfavorable beliefs could deter the principal from holding the agent to his outside option. The resulting continuation rents would then act as deferred compensation. This could give the principal implicit commitment power with effects akin to commitment to continuation values studied in Section 5.3, a channel that can make the principal better off that is not directly related to private information. No-signaling-what-you-don't-know requires that an offer cannot convince the agent of something the principal could not have learned given what he observed.

The Value of Private Information Acquisition

Let $w_2(y; \mu(h))$ denote the optimal second-period contract with public belief $\mu(h)$ from Proposition 1.

Proposition 6. *In either acquisition game, the principal's expected profit in any no-holdup perfect Bayesian equilibrium implementing high effort in both periods is at most $\Pi(\mu_0)$. If it equals $\Pi(\mu_0)$, then on the equilibrium path:*

1. *the first-period contract is optimal in the public-information problem (4); in the cases covered by Theorem 1, it is the evaluation characterized there;*
2. *the terminal offer leaves the agent's belief unchanged, $\mu_A(h, C_2) = \mu(h)$;*
3. *the terminal contract pays $w_2(\cdot; \mu(h))$, the optimal wage at the common belief $\mu(h)$.*

The intuition behind this result is simple. Take any private information of the principal. Private information is valuable only if the principal uses it, and using it reveals it: any information that shapes the terminal offer reaches the agent through that offer. Revealed one period earlier, as part of the evaluation, the same information would have produced the same learning. It would also have made first-period incentives cheaper, because pay could have been tailored to it. Revealed through the offer, it carries the learning cost without the incentive benefit. The optimal evaluation is therefore what the principal would choose if she could commit to what she learns.

The proposition shows that private information acquisition can only hurt the principal. She would like to commit not to acquire any private information and not to use any private information that was acquired. In organizations, this can be achieved through a separation of

concerns: a data analytics or quality assurance team gathers data about output and only the resulting evaluation is communicated to the manager setting the bonus scheme and negotiating the contract with the worker. When no formal commitment is available, when can the principal achieve the bound in an equilibrium?

Attaining the Bound

When the agent observes the experiment, the principal can publicly forgo private information. After the null experiment, no-signaling-what-you-don't-know keeps the agent at the public belief after every offer. The terminal stage is then the common-information problem of Section 3, and the agent obtains U after either observed effort. The first-period problem then coincides with the baseline problem.

Remark 1. In the observable-acquisition game, the principal obtains $\Pi(\mu_0)$ in every no-holdup perfect Bayesian equilibrium that implements high effort and satisfies no-signaling-what-you-don't-know. By Proposition 6, every such equilibrium implements the optimal evaluation.

Note that the remark does not guarantee that such an equilibrium exists. An equilibrium must specify credible play after every announced experiment, where the terminal stage is an informed-principal game with a continuum of types and contracts. In particular, to construct such an equilibrium it suffices that, after every announced first-period contract and experiment, the continuation game has an equilibrium that implements high effort, satisfies no-holdup, and satisfies no-signaling-what-you-don't-know. By Proposition 6, no such continuation can give the principal more than $\Pi(\mu_0)$, so she has no incentive to announce an informative experiment. Proposition 7 below provides such a construction for the binary case with complements.

With *covert acquisition*, the principal can observe output in secret and, if an offer tailored to what she saw is cheaper, make it. The bound can be attained only if the agent's responses make such deviations unprofitable. Call a belief attainable at h if the principal can hold it at h after observing first-period output.

Proposition 7. *Suppose that output is binary and that effort and ability are weak complements, $\Delta_b \geq 0$. Then the covert-acquisition game has a no-holdup perfect Bayesian equilibrium in which the principal obtains $\Pi(\mu_0)$. In this equilibrium, the principal observes first-period output and offers $w_2(\cdot; \mu(h))$ whatever she observes. The agent holds belief $\mu(h)$ after this offer. After any other offer, his beliefs are skeptical: he holds the lowest attainable belief if the offer pays weakly higher expected utility after high output than after low output, and the highest attainable belief otherwise.*

The same construction yields an equilibrium of the observable-acquisition game that satisfies no-signaling-what-you-don't-know, with the attainable beliefs being those the principal can hold under the announced experiment.

The construction rests on the agent's skepticism about unexpected offers: after a deviation, he adopts the attainable belief at which accepting the proposed contract is least attractive. Under complementarity and our maintained assumption that high effort is optimal, skeptical beliefs make any departure from $w_2(\cdot; \mu(h))$ unprofitable, whatever the principal privately learns. Private information then has no value and no cost. The principal offers the same contract after

every private observation, and the public-information benchmark $\Pi(\mu_0)$ is attained even though she can acquire information covertly.¹²

5.2 Unobservable Effort

In the main sections, we assume that effort is observed but not contractible. This ensures that even after a deviation, the principal and the agent share a common belief over the agent's ability. Assume instead that effort is not observed by the principal. This does not affect beliefs on the equilibrium path, since the conjectured high effort is correct. After a deviation to $e_1 = 0$, however, the agent updates using the effort he actually chose, while the principal continues to use the on-path updating rule. Depending on the evaluation, the agent is then more or less optimistic than the principal. The second-period contract may leave him a rent, or he may prefer the outside option. As a result, a deviation to low effort in the first period is more profitable because of this belief-manipulation effect.¹³

We now sketch this model and solve for the optimal evaluation in the binary case. We can extend our results: if the technology is log-supermodular, the optimal evaluation is binary and tough; if it is log-submodular, the evaluation is often binary and lenient, though for more extreme parameters the optimal information structure may have an additional uninformative signal.

Model

Again, we solve the problem backwards. Note that the problem in the second period is unchanged. The principal offers the optimal contract for the public posterior whether the agent privately deviated in the first period or not. Let $w_2(y; \mu)$ denote this optimal wage in the second period for a public posterior μ .

Let $\tilde{\mu}(\mu)$ denote the agent's posterior if he shirked, i.e. $\mu(s_1, w_1, 0)$, as a function of the public posterior, i.e. $\mu(s_1, w_1, 1)$. Bayes' rule gives

$$\tilde{\mu}(\mu) = \mu_0 + \frac{\Delta_a}{\Delta_a + \Delta_b} \frac{\mu - \mu_0}{1 - x(\mu)}. \quad (25)$$

When $\Delta_a = 0$, a shirker learns nothing about ability and $\tilde{\mu}(\mu) = \mu_0$. Whenever $\Delta_a > 0$, shirking leads to learning. The deviation posterior is increasing in μ , strictly convex and coincides with the public belief μ at the prior μ_0 . It intersects from above when $\Delta_b > 0$ and from below when $\Delta_b < 0$. There may also be a second crossing.

¹²This construction depends on the skeptical beliefs of the agent off-path. One possible deviation of the principal involves a small increase in the base wage, combined with a reduction in the bonus, which can be chosen to increase profits only when the principal's posterior exceeds a cutoff above the public belief. An agent who interprets this offer as evidence of favorable private information becomes more willing to participate and exert effort, making the deviation profitable. Psychologically, convincing a worker that he is high ability by cutting his bonus may not be the most plausible. The skeptical beliefs supporting our equilibrium prevent this inference.

¹³This effect is central in the analysis of many models of moral hazard with learning, e.g. [Prat and Jovanovic \(2014\)](#); [Demarzo and Sannikov \(2017\)](#). [Bhaskar and Mailath \(2019\)](#) show that this motive implies that the costs of providing incentives using spot contracts grow unboundedly with the length of the time horizon in a model similar to ours, but with learning from output. It is unknown whether the design of the information structure can reverse this conclusion.

In the first period, we can still rewrite the problem in the space of the (public) posterior belief, but we can no longer reduce to a static incentive-compatibility constraint because of the belief-manipulation effect. The first period IC then reads

$$\begin{aligned} & \int x(\mu) u(w_1(\mu)) dm(\mu) - c + \delta U \\ & \geq \delta \int \max\left\{ \underbrace{U}_{(i)}, \underbrace{\int u(w_2(y; \mu)) dF(y|1, \tilde{\mu}(\mu)) - c}_{(ii)}, \underbrace{\int u(w_2(y; \mu)) dF(y|0, \tilde{\mu}(\mu))}_{(iii)} \right\} (1 - x(\mu)) dm(\mu). \end{aligned} \quad (26)$$

The LHS integral denotes the incentive effect of the wage in the first period, which is unchanged. If the agent does not deviate, he will pay effort cost c and obtain his reservation utility δU in the final period. If effort were observable, he would also obtain the reservation utility after a deviation, so this term would cancel and we obtain our familiar constraint. Since effort is not observable, the agent has private information about his ability after a deviation and has a nontrivial choice in the second period between: (i) not participating, (ii) exerting effort, and (iii) shirking. Not participating is optimal if he is more pessimistic after the deviation ($\tilde{\mu} < \mu$): The wage $w_2(y; \mu)$ is monotonic in output and just calibrated to ensure that an agent with belief μ is willing to participate. If the agent is more optimistic after a deviation ($\tilde{\mu} > \mu$), he participates. He exerts effort in the second period and IC is over-satisfied if effort and ability are complements ($\Delta_b > 0$) and shirks if they are substitutes ($\Delta_b < 0$). Finally, we integrate this continuation value according to $(1 - x(\mu)) dm(\mu)$, which is the distribution over public posterior beliefs conditional on exerting low effort.

The Optimal Evaluation with Hidden Effort

To proceed with the analysis, we now specialize the model to binary output. We can translate this dynamic IC into belief space and plug in the optimal wage contract to obtain

$$\begin{aligned} & \int \frac{(b + \mu_0 \Delta_b)}{\mu_0(1 - \mu_0)(\Delta_a + \Delta_b)} (\mu - \mu_0) u(w_1(\mu)) \\ & - \delta(1 - x(\mu)) \max\left\{0, c \frac{\Delta_a + \Delta_b}{b + \mu \Delta_b} (\tilde{\mu}(\mu) - \mu), c \frac{\Delta_a}{b + \mu \Delta_b} (\tilde{\mu}(\mu) - \mu)\right\} dm(\mu) \geq c \end{aligned} \quad (27)$$

We then proceed with the analysis just as before, using (27) as the IC constraint, and arrive at a Lagrangian $\tilde{\ell}$ that can be written as the Lagrangian with observable effort, $\ell^*(\mu; \lambda)$, plus a term adjusting for the rent after a deviation

$$\tilde{\ell}(\mu; \lambda) = \ell^*(\mu; \lambda) - \lambda_{IC} \delta(1 - x(\mu)) \max\left\{0, c \frac{\Delta_a + \Delta_b}{b + \mu \Delta_b} (\tilde{\mu}(\mu) - \mu), c \frac{\Delta_a}{b + \mu \Delta_b} (\tilde{\mu}(\mu) - \mu)\right\}.$$

The rent term introduces a kink at the prior and it affects the shape of $\tilde{\ell}$ in the interior. There is no second kink, as there is no second crossing between $\tilde{\mu}(\mu)$ and μ in the region of feasible posteriors when the interaction is sufficiently strong (log-super or submodular). We show in the appendix that hidden effort preserves increasing curvature with strong complements and decreasing curvature with strong substitutes; at the prior, however, the slope jumps downward in both cases. This creates a concave kink which may introduce a third, uninformative evaluation, see Figure 3.

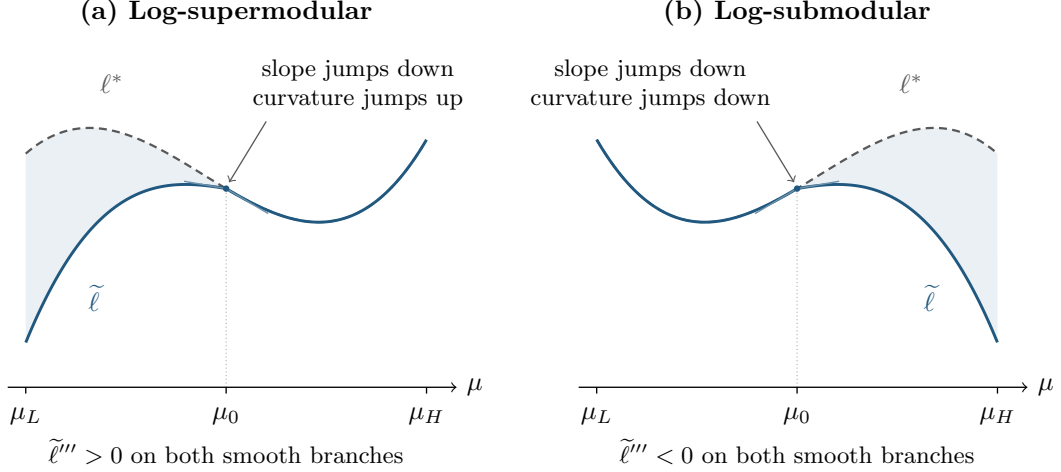


Figure 3: The effect of hidden effort on the Lagrangian (illustration).

Proposition 8. *Suppose that output is binary and that effort is hidden. Then,*

1. *if the technology is log-supermodular ($a\Delta_b - b\Delta_a > 0$), the optimal evaluation is binary and tough, with $S = \{G, B\}$, wages $w_G > w_B$ and $\sigma \in [0, 1)$ such that*

$$p(G, w_G | y_H) = 1 - \sigma, \quad p(B, w_B | y_H) = \sigma, \quad p(G, w_G | y_L) = 0, \quad p(B, w_B | y_L) = 1;$$

2. *if the technology is log-submodular ($b\Delta_a + (1 - a)\Delta_b < 0$), the optimal evaluation uses $S = \{G, B, N\}$ and wages $w_G > w_N > w_B$, sending*

(a) *with probability $p_N \in [0, 1)$ an uninformative signal N and wage w_N with $p(N, w_N | y_H) = p(N, w_N | y_L) = p_N$, which induces posterior $\mu = \mu_0$, and*

(b) *with probability $1 - p_N$ a signal in $\{G, B\}$ (binary and lenient), with $\sigma \in [0, 1)$, and*

$$\begin{aligned} p(G, w_G | y_H) &= 1 - p_N, \quad p(B, w_B | y_H) = 0, \\ p(G, w_G | y_L) &= \sigma(1 - p_N), \quad p(B, w_B | y_L) = (1 - \sigma)(1 - p_N). \end{aligned}$$

The proposition establishes that in the binary case the optimal evaluation retains its structure even though the effort choice is hidden. The proof first allows unrestricted evaluations and obtains three possible posteriors, including the prior. With substitutes, this reduction is sharp. The optimal evaluation structure sends an output-independent neutral evaluation with some probability, possibly zero. This neutral evaluation is interpreted the same way independently of effort and thus helps to reduce learning off-path and thereby reduces the cost of deterring double deviations. Conditional on receiving an informative evaluation, the optimum remains binary and lenient. We provide an example that shows that the neutral evaluation can indeed be part of the optimal signal structure in the Appendix (Example 2).¹⁴ With complements, we can rule out the neutral evaluation; the evaluation is always binary and tough.

¹⁴It features a very high prior and a very low impact of effort if ability is high. Both features seem necessary, based on numerical exploration.

The asymmetry between the two cases comes from the nature of the double deviation. With complements, the optimistic deviator works, so his gain is governed by the ability effect under high effort, $\Delta_a + \Delta_b$, just like on path. With substitutes, he shirks, and his gain is governed by Δ_a , which can be substantially larger than $\Delta_a + \Delta_b$ and importantly differs from the on-path case. This can create a substantial private rent even when learning is modest. The neutral rating limits this rent.

5.3 Long-Run Commitment

In the main analysis, we assumed period-by-period contracting. This assumption plays two roles. First, it forces the principal to pay for first-period effort before the second period begins. Rewards reveal the performance they are based on, so the agent learns about his ability before he chooses his second-period effort: incentives and learning cannot be separated in time. Second, because effort is observed, the principal sets the second-period contract at the belief the agent actually holds, also after a deviation. The agent never has private information about his ability, the first-period constraints are static, and the problem splits into two static problems linked by the public belief.

This section asks which of these roles our results depend on, by considering two levels of commitment. With *commitment to continuation values*, the first-period contract also promises the agent a level of second-period utility, but the second-period incentive scheme is still set sequentially at the public posterior. The principal can defer pay, but not information. Our analysis goes through in that case, and the optimal evaluation pools weakly more: deferring pay makes first-period incentives cheaper but leaves the cost of learning unchanged. With *full commitment*, the principal fixes the second-period wage schedule at the outset and may condition it on a record of first-period output that the agent does not see. She can then defer information as well as pay, and could avoid learning altogether. But a committed schedule cannot respond to effort, and the problem loses its recursive structure. We show that the trade-off between incentives and learning is nevertheless still at work. First, when ability is known, full revelation is optimal even with full commitment (Proposition 10). Second, with learning, avoiding it by deferring information altogether can be prohibitively costly. In an example, we show that the principal prefers not to defer all information and instead chooses a partially informative tough binary evaluation. Limited commitment is not the source of the trade-off in our main sections, which arises whenever it is costly to separate incentive pay from information about performance; rather, limited commitment sharpens the trade-off. It makes this separation infeasible rather than merely costly, and it keeps the problem recursive and the optimal evaluation tractable.

Commitment to Continuation Values

Consider first a principal who can only postpone payments. Together with a signal and a wage, the first-period contract now assigns a continuation value $U_2 \geq 0$, so that $p : Y \rightarrow \Delta(S \times \mathbb{R}_+ \times \mathbb{R}_+)$. The continuation value is an enforceable right to a flat second-period wage $U_2^2/2$, which gives the agent utility U_2 at low effort. We assume that U_2 also replaces the outside option in the second period; the agent can only exit in the first period to obtain $(1 + \delta)U$. At the beginning of the second period, the principal observes first-period effort and may offer a replacement contract;

the agent chooses between the flat wage and the replacement contract. The principal can thus reward and punish first-period performance through future utility, but the contract on path will still be adapted to the public posterior. The continuation value is part of the outcome that the agent observes, and it enters his posterior $\mu(s, w, U_2, e)$, defined analogously to (2). Commitment to continuation values therefore defers payment, but not information.

By the argument of Proposition 1, the contract holds the agent to his promised continuation value U_2 after every level of effort. The dynamic constraints (P-1) and (IC-Dyn) therefore reduce to static constraints on the total promised utility $u(w) + \delta U_2$. The principal delivers the total promised utility most cheaply by equating the current utility payment with the expected second-period utility payment:

$$u(w(\mu)) = U_2(\mu) + c. \quad (28)$$

Proposition 9. *Under commitment to continuation values, the optimal evaluation coincides with the optimal evaluation of the period-by-period problem in which the first-period effort cost is $c/\sqrt{1+\delta}$ instead of c . In particular, it is characterized by Theorem 1. Under the conditions of Proposition 5, it pools weakly more than the evaluation under period-by-period contracting.*

Smoothing payments across both periods lowers the cost of first-period incentives by the factor $1/(1+\delta)$. The cost of learning, which works through the incentive costs of the next period, is still present. Commitment to continuation values does not resolve the trade-off between incentives and learning; it pushes toward concealing information.¹⁵

Full Commitment

Suppose now that the principal commits at the outset to the first-period evaluation and wage and to a second-period wage schedule, without renegotiation. The agent can still leave before the second period, so the schedule must respect an interim participation constraint. The second period evaluation and wage can also depend on first-period output, which the principal records but does not disclose before the agent's second-period effort. A contract therefore is a tuple $C = (S_1, S_2, p_1, p_2)$ comprising signal spaces S_1, S_2 for the first and second period evaluation, a first period evaluation and wage $p_1 : Y \rightarrow \Delta(S_1 \times \mathbb{R}_+)$, and a second period evaluation and wage $p_2 : S_1 \times \mathbb{R}_+ \times Y \times Y \rightarrow \Delta(S_2 \times \mathbb{R}_+)$. Before his second-period effort, the agent observes the first-period signal and the wage, and hence knows which schedule $p_2(s_1, w_1, \cdot, \cdot)$ applies, but he does not learn first-period output. He values the schedule using his posterior about y_1 given the signal and his own first-period effort. We give a formal statement of the contracting problem in Appendix E.

Full commitment changes the contracting problem in two ways. First, observing effort no longer helps the principal. Since effort is unverifiable, a committed schedule cannot respond to it. Conditional on an evaluation and wage (s_1, w_1) , a shirking agent faces the same continuation

¹⁵If the agent can exit and obtain U in the second period, the continuation value must satisfy $U_2 \geq U$. Initial participation and (28) imply that this floor binds after unfavorable evaluations and is slack after favorable ones: favorable evaluations are rewarded partly through higher future pay, while unfavorable ones are penalized only through current pay. The floor creates an upward jump in the curvature of the principal's Lagrangian over posteriors, at the posterior where it starts to bind. When $\Pi_2''' > 0$, the jump reinforces the concave-then-convex shape, and the optimal evaluation remains generalized lower censorship. When $\Pi_2''' < 0$, the jump works against upper censorship and the optimal evaluation may no longer be generalized upper censorship.

contract as an obedient one and values it at his post-deviation belief $\tilde{\mu}(\mu)$ from Section 5.2. He may work, shirk, or exit, and the first-period incentive constraint must again rule out these double deviations. Observing effort made the first-period constraint static under period-by-period contracting only because the principal could act on it when offering the next contract. Second, the second-period incentive constraint need not bind. A terminal bonus larger than second-period incentives require changes how a first-period shirker values the continuation, and it can relax the first-period incentive constraint. The first-period wage and the second-period schedule must therefore both be designed jointly with the evaluation. Consequently, the full commitment problem is much less tractable. We will illustrate how our mechanism extends to the full commitment case by first showing that when ability is known to both parties a fully informative performance evaluation is optimal. We then show by way of a numerical example that when there is learning, the principal may want to use a partially informative evaluation in the first period even with full commitment.

Consider first the benchmark in which ability is known to both parties, $\theta_L = \theta_H$. The principal can still postpone rewards for first-period output by conditioning the second-period wage on a retained record, but there is no learning about the technology by either party. Nevertheless, full revelation is optimal.

Proposition 10. *Suppose that ability is commonly known, $\theta_L = \theta_H$. Under full commitment, a fully informative first-period evaluation is optimal.*

Thus, in the absence of learning, the principal has no reason to conceal information about the first period output. Risk sharing and interim participation do not by themselves make concealment profitable.¹⁶

The following example shows that concealing information about output can be profitable with full commitment when there is learning about the agent’s ability.

Example 1. Output is binary with $y_L = 0$ and $y_H = 15$ and the technology is log-supermodular with $(a, \Delta_a, b, \Delta_b) = (1/10, 1/10, 1/10, 3/5)$, $\mu_0 = 3/10$, $c = 1/2$, $\delta = 1$, and $U = 1$. We give the optimal evaluation and the expected discounted wage (“cost”) for this example across different commitment assumptions in Table 1. Expected output is the same in every row, so the table reports expected discounted compensation cost.

Four observations follow. First, commitment to continuation values lowers compensation cost by 7.9%. The saving comes from smoothing: the continuation value carries part of the bonus after the good evaluation and part of the penalty after the bad one. The optimal evaluation remains tough, but pools more. Second, commitment can be detrimental for the principal. Fixing the evaluation at full revelation in the first period, full commitment costs more than period-by-period contracting, because the commitment contract cannot adapt to the belief that follows observed shirking. Third, a retained record makes learning avoidable but doing so is not always optimal. Withholding all information is feasible but costs more than the optimal period-by-period contract. The *optimal* evaluation with full commitment and retained records is still tough: it discloses a success with probability 0.683 and otherwise pools it with failures.

¹⁶Note that square root utility is essential for this result. With a general utility function, the cost of providing incentives is different at different levels of utility, and concealing information about output in the first period can ensure that the incentives in the second period are provided in a region that is favorable for the principal.

Table 1: Commitment under common primitives.

Commitment Assumption	First-Period Evaluation	Expected Wage	Rent
period-by-period	full revelation	3.462	0
period-by-period	tough; pools 14.1% of successes (optimal)	3.389	0
continuation values	full revelation	3.269	0
continuation values	tough; pools 24.8% of successes (optimal)	3.120	0
full commitment	full revelation	4.201	0.432
full commitment	no information	3.599	0.500
full commitment	tough; pools 31.7% of successes (optimal)	3.224	0.257

Rent is the agent’s expected second-period utility, net of effort cost, in excess of U . Wages are optimized for every evaluation. Rows marked optimal also optimize over evaluations.

Fourth, after its favorable evaluation, the optimal full commitment contract commits to a second-period bonus in excess of what second-period incentive compatibility requires. This promised rent helps satisfy the dynamic first period incentive compatibility constraint.

Other primitives produce different patterns. The appendix gives an example in which full information in the first period is optimal with full commitment (Example 3), and another in which no information in the first period is optimal with full commitment (Example 4).

6 Concluding Remarks

Our model demonstrates why it can be in a principal’s interest to base incentives on a coarse or noisy evaluation of the agent’s performance, even when the principal could measure true output and commit to contingent wages. The underlying insight is that output contains information both about effort, which she wants to reward, and the agent’s match-specific ability, which she may prefer to keep shrouded. Richer measurement opportunities therefore do not mean that evaluations and wages should use them all: whatever enters the evaluation and the wage is also revealed to the agent.

The optimal resolution of the trade-off between incentives and learning pools outcomes to avoid creating agents who are expensive to motivate. When effort and ability are substitutes, this involves a fixed wage with tailored deductions for low performance, to avoid revealing top performance, which would cause complacency in the future. When they are complements, it involves a threshold contract: a base wage below a performance threshold and a discontinuous bonus that rises with performance above it, to avoid revealing detailed information about low performance, which is demotivating. When the underlying measure of output is coarse, the optimal evaluation randomizes. When effort and ability are strong complements, it is tough: Good performance is not always recognized, but bad performance is always punished. Again, the reason is learning, as a tough evaluation ensures that even after a bad evaluation, the agent is not too pessimistic about his ability.

Our analysis has two parts that can be read separately. Section 3.2 shows when common learning about the technology is costly for the principal in a standard moral hazard problem. Section 4 shows how incentives are designed when the learning they create is costly; the characterization there (Theorem 1) depends on the second period only through the shape of the continuation value. Methodologically, this paper shows how a dynamic contracting problem

with learning and designed information can be kept tractable. Period-by-period contracting and observable, yet non-contractible, effort together thread a needle: neither party ever acquires private information, the common posterior belief is the only state variable, and the first-period problem becomes a static moral-hazard problem with a belief-dependent continuation value. With a technology that is linear in effort and ability, what an outcome reveals about effort is proportional to what it reveals about ability. The problem is then one of persuasion over posterior beliefs with the wage as an additional choice, and its solution takes a simple censorship form despite two constraints. Once this benchmark is understood, the assumptions behind it can be relaxed, and we do so one at a time: for private information of the principal, hidden effort and long-term commitment.

Two lessons concern commitment. First, the principal must be able to commit to the evaluation within a period. After effort is exerted, she would otherwise prefer to pay low wages and to convince the agent that his effort is essential for his success, whatever his performance. That the principal commits to the wage scheme is standard in moral hazard problems; that she commits to the evaluation is the assumption our analysis shares with information design. Firms have several means to this end. Performance indicators can be written into the employment contract. More broadly, the firm designs its organizational structure and evaluation guidelines, and persistent practices and internal rules determine how data collected in the course of business are aggregated and linked back to an individual worker. The principal can also leverage reputation across time (Best and Quigley, 2024) and across the workers in the organization (Lin and Liu, 2024). A further way a firm can commit to the style of evaluation is through the selection and training of evaluators. Interpreted from this angle, our results suggest, for example, that in environments with strong complementarities between effort and ability unreasonably strict evaluators and drill-sergeant mentality may be part of the optimal organization design. Second, the principal gains from committing not to know more than the evaluation reveals. Finer information that shaped the next contract would reach the agent through her offer, with the cost of learning but without the incentive benefit it would have had as part of the evaluation. Organizations achieve this by separating those who measure output from those who set pay.

Our results inform not only the optimal evaluation of employee performance, but also the selection of information sources. Among measures that combine information about effort and ability, the principal prefers measures that are less sensitive to ability. This may motivate secrecy about salary differences between employees in the same job as opposed to differences in bonuses, to the extent that base pay reflects the principal’s estimate of ability while bonuses more directly reward effort. Ideally, incentives would rest on a signal that is informative about effort but independent of ability, such as a verifiable measure of conduct, which rewards effort without teaching the agent about his ability. Both patterns are in sharp contrast with models of implicit incentives through career concerns (Holmström, 1999). In such models, the fact that a signal combines information about effort and ability is not a friction but the source of incentives, as the agent exerts effort to avoid being perceived as low-ability. To achieve the right mixture of information about effort and ability the principal might add noise (Dewatripont et al., 1999) or combine information from different sources or vintages (Hörner and Lambert, 2021). In our setting with explicit incentives, by contrast, such entangled information is the source of the

friction. The analysis of evaluation design when both explicit and implicit incentives are present is an interesting avenue for future research. So are longer horizons, where the two-period logic suggests coarser evaluations early in a relationship, more than two levels of ability, where the posterior is no longer one-dimensional, and a general characterization under full commitment.

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A Proof Appendix

This appendix provides bounds and the proofs for the results in Sections 3 and 4. The proofs for the extensions in Section 5 together with the full formal definition and treatment of these extensions are deferred to the Online Appendix.

A.1 Interiority Bounds

Before we proceed to prove the results of the main text, let us deliver the promised sufficient conditions on primitives that ensure that the principal actually wants to implement high effort and that wages are interior.

Inducing High Effort is Optimal after every History

We now provide a lower bound on the dispersion of output that ensures that the principal wants to implement high effort and wants to employ the agent after every history. To this purpose, let us write $Y = \kappa[0, 1]$, for a scale of output $\kappa > 0$, and we keep F_0 and Δ_F fixed with support in $[0, 1]$. We can write any compact Y like this, perhaps after shifting it up and taking the closed convex hull. Note that the information content of output is unchanged by this rescaling.

In the second period, the principal wants to implement high effort for every common belief about ability if

$$\kappa > \frac{(U + c)^2}{2 \min\{b, b + \Delta_b\} \int_0^1 y d\Delta_F(y)} + \frac{c^2}{8 \left[\min\{b, b + \Delta_b\} \int_0^1 y d\Delta_F(y) \right]^3} \quad (29)$$

In the first period, a sufficient condition is

$$\kappa > \frac{(U + c)^2}{2 \min\{b, b + \Delta_b\} \int_0^1 y d\Delta_F(y)} + \frac{(1 + \delta)c^2}{8 \left[\min\{b, b + \Delta_b\} \int_0^1 y d\Delta_F(y) \right]^3} \quad (30)$$

The second condition also implies the first. It compares a fully informative high-effort contract with any contract inducing low effort, allowing for the extra cost of future incentives caused by learning. Both bounds are finite under the assumed strictly positive effects of effort. Thus a sufficiently large κ justifies high effort in both periods at every common belief, including beliefs following observed low effort.

For binary nonnegative output, these conditions simplify to in the second and first period, respectively

$$y_H - y_L > \frac{(U + c)^2}{2 \min\{b, b + \Delta_b\}} + \frac{c^2}{8 \min\{b, b + \Delta_b\}^3}, \quad y_H - y_L > \frac{(U + c)^2}{2 \min\{b, b + \Delta_b\}} + \frac{(1 + \delta)c^2}{8 \min\{b, b + \Delta_b\}^3}.$$

Hence it suffices to raise y_H while holding y_L and the success probabilities fixed.

In the extensions, there can be histories where the principal's and the agent's belief about ability differ. This is the case with private principal information acquisition; with hidden effort, this only happens off-path and does not affect the analysis of the principal problem; it can also happen implicitly after a deviation from a full commitment contract. The following

condition ensures that the principal with any belief $\mu_P \in [0, 1]$ prefers to induce high effort and participation from an agent with belief $\mu_A \in [0, 1]$ in the second period, making no joint restrictions on the beliefs.

$$\kappa > \frac{\left[U + \frac{c}{\min\{b, b + \Delta_b\} \int_0^1 y d\Delta_F(y)} \right]^2}{2 \min\{b, b + \Delta_b\} \int_0^1 y d\Delta_F(y)}. \quad (31)$$

With hidden effort, the first period cost of providing incentives is increased because of the possibility of double deviations (Section 5.2). The following condition ensures that the principal prefers inducing high effort and participation despite this additional cost.

$$\begin{aligned} \kappa &> \frac{1}{2 \min\{b, b + \Delta_b\} \int_0^1 y d\Delta_F(y)} \left[U + \frac{c \left(1 + \delta \frac{\max\{\Delta_a, \Delta_a + \Delta_b\}}{\min\{b, b + \Delta_b\}} \right)}{\min\{b, b + \Delta_b\} \int_0^1 y d\Delta_F(y)} \right]^2 \\ &\quad + \frac{\delta c^2}{8 \left[\min\{b, b + \Delta_b\} \int_0^1 y d\Delta_F(y) \right]^3}. \end{aligned}$$

Derivation of the bounds. We only describe the key steps to deriving the conditions above; in the interest of conserving space, the detailed computations for each condition are omitted.

The conditions follow by bounding the cost of a feasible high-effort contract and comparing it with the expected output gain from effort. With square-root utility, expected wages satisfy

$$\mathbb{E}[w] = \frac{\mathbb{E}[u(w)]^2}{2} + \frac{\mathbb{V}(u(w))}{2}.$$

For common-belief contracts, binding participation fixes expected utility payments at $U + c$, and the additional cost of incentives is bounded using Cauchy–Schwarz and $\mathbb{V}(y) \leq \frac{1}{4}$ for output normalized to $[0, 1]$. For comparisons involving continuation payoffs, we bound the wage cost with full information in the first period. With asymmetric beliefs, we construct an output-increasing utility payment

$$u(w(y)) = U + \frac{cy}{\min\{b, b + \Delta_b\} \int_0^1 t d\Delta_F(t)}$$

that induces acceptance and high effort at every agent belief, and bound its cost at every principal belief. With hidden effort, the initial bonus additionally covers the largest continuation gain from a private deviation. These arguments provide cost bounds that are uniform over the relevant histories and beliefs. Scaling output increases the revenue gain from effort while leaving these bounds unchanged, yielding the stated sufficient conditions. $\square$

Wages are Positive after every History

We now provide a lower bound on the outside utility U that ensures that the nonnegativity constraint on wages, which is implicit in assuming square-root utility, is slack at the optimal contract after every history. We keep the normalization of the previous subsection, $Y = \kappa[0, 1]$

with F_0 and Δ_F supported in $[0, 1]$, and denote by

$$\Lambda = \max_{y \in Y} \frac{d\Delta_F(y)}{dF_0(y)}$$

the largest likelihood ratio of the shift in the output distribution, which is finite by assumption. Throughout, a history is attainable if it can be reached from the prior μ_0 , including histories following observed low effort. All conditions are sufficient rather than necessary.

In the second period, the optimal wage is strictly positive after every attainable history if

$$U + c > \frac{c \max\{b, b + \Delta_b\} (1 + \max\{\Delta_a, \Delta_a + \Delta_b\} \Lambda)}{4(1 - \mu_0)(\Delta_a + \Delta_b) \left[\min\{b, b + \Delta_b\} \int_0^1 y d\Delta_F(y) \right]^2}.$$

Given this condition, a sufficient condition for strictly positive wages in the first period is

$$U + c > \frac{(1 + \delta)c \max\{b, b + \Delta_b\} (1 + \max\{\Delta_a, \Delta_a + \Delta_b\} \Lambda)}{4(1 - \mu_0)(\Delta_a + \Delta_b) \left[\min\{b, b + \Delta_b\} \int_0^1 y d\Delta_F(y) \right]^2}.$$

Both conditions are implied by

$$U > \frac{(1 + \delta)c \max\{b, b + \Delta_b\} (1 + \max\{\Delta_a, \Delta_a + \Delta_b\} \Lambda)}{4(1 - \mu_0)(\Delta_a + \Delta_b) \left[\min\{b, b + \Delta_b\} \int_0^1 y d\Delta_F(y) \right]^2}. \quad (32)$$

The bound depends only on the primitives, not on the optimal evaluation, and it is finite under the maintained assumptions $\Delta_a + \Delta_b > 0$, $\min\{b, b + \Delta_b\} > 0$, and the bounded likelihood ratio. It is also independent of the scale of output κ . Since the bounds on κ in the previous subsection increase in U , all conditions hold simultaneously if we first choose U to satisfy (32) and then choose κ sufficiently large.

For binary output with a log-supermodular technology, $a\Delta_b - b\Delta_a > 0$, the condition simplifies. Then, $U > \frac{ca}{b}$ ensures strictly positive wages in both periods.

In the private-information acquisition game, the equilibria that attain $\Pi(\mu_0)$ offer the second-period contracts of the main model at the public belief $\mu(h)$, so (32) also ensures positive wages there. With hidden effort, the first-period wage must in addition deter double deviations (Section 5.2). With commitment to continuation values (Section 5.3), the promised continuation value U_2 must be positive. Moreover, a low promise can trigger the nonnegativity constraint in the second period and leave the agent a participation rent. The following single condition ensures strictly positive wages in both extensions and strictly positive promised continuation values, and it ensures that contracts leaving such rents are not optimal:

$$U > \frac{5c}{4 \left[\min\{b, b + \Delta_b\} \int_0^1 y d\Delta_F(y) \right]^2} \left(1 + \frac{\max\{\Delta_a, \Delta_a + \Delta_b\}}{\min\{b, b + \Delta_b\}} \right)^2 \times \left[1 + \frac{\max\{b, b + \Delta_b\} (1 + \max\{\Delta_a, \Delta_a + \Delta_b\} \Lambda)}{(1 - \mu_0)(\Delta_a + \Delta_b)} \right]. \quad (33)$$

For commitment to continuation values alone, the squared factor can be omitted.

Under full commitment, the second-period incentive constraint need not bind, and we derive

the bound separately. When ability is known, $\theta_L = \theta_H$, the output distribution does not depend on the belief, and we suppress this argument. The optimal contract conditional on full first-period revelation in Proposition 10 has strictly positive wages if

$$U + c > \frac{c \max_{y \in Y} \frac{dF(y|0)}{dF(y|1)}}{\int_Y \left(1 - \frac{dF(y|0)}{dF(y|1)}\right)^2 dF(y|1)}.$$

Here, the likelihood ratio of low to high effort must be bounded and effort must affect the output distribution. This is a separate requirement, which is not implied by the finiteness of Λ . For binary output with success probabilities a and $a + b$ in $(0, 1)$, a simpler sufficient condition is

$$U > \frac{ca}{b} + \frac{c(a+b)(1-a-b)}{2b^2}.$$

With learning, we verify the nonnegativity constraints of Example 1 directly. All seven contracts in Table 1 have strictly positive wages whenever $U > 0.966072$, and the two contracts with commitment to continuation values have strictly positive promised values. The example's $U = 1$ therefore satisfies all nonnegativity constraints.

Derivation of the bounds. We only describe the key steps to deriving the conditions above; in the interest of conserving space, the detailed computations for each condition are omitted.

We drop the nonnegativity constraints and show that the relaxed optimum pays strictly positive utility after every attainable history, so that it also solves the original problem. With square-root utility and binding constraints, the relaxed terminal contract at belief μ pays

$$u(w(y)) = U + c + \frac{cx(y; \mu)}{\int_Y x(t; \mu)^2 dF(t|1, \mu)},$$

with the score x evaluated at μ in place of μ_0 . It therefore suffices to bound the second moment of the score away from zero and the score from below. The first bound follows from Cauchy–Schwarz and $\mathbb{V}(y) \leq 1/4$, as for the output-scale bounds. For the second, $dF(y|1, 1) = dF(y|1, \mu) + (1 - \mu)(\Delta_a + \Delta_b) d\Delta_F(y) \geq 0$ gives $x(y; \mu) \geq -\max\{b, b + \Delta_b\}/[(1 - \mu)(\Delta_a + \Delta_b)]$. By Bayes' rule and the bounded likelihood ratio, after any first-period evaluation, including one that follows a deviation,

$$1 - \mu \geq (1 - \mu_0) [1 + \max\{\Delta_a, \Delta_a + \Delta_b\} \Lambda]^{-1}.$$

In the first period, we again drop the nonnegativity constraint. For an evaluation with posterior distribution m , write $\mathbb{V}_m[\mu] := \mathbb{E}_m[(\mu - \mu_0)^2]$ for the variance of the posterior. The relaxed optimal contract pays

$$u(w(\mu)) = U + c + \frac{c}{\chi \mathbb{V}_m[\mu]} (\mu - \mu_0)$$

at posterior μ , with incentive cost $c^2/(2\chi^2 \mathbb{V}_m[\mu])$. As in the terminal period, U enters wage costs only through the additive constant $(U + c)^2/2$, so the relaxed optimal evaluation does not depend on U . It must be weakly better than full revelation. Its first-period incentive cost can

therefore exceed the cost under full revelation by at most the discounted reduction in expected terminal incentive costs. Both the cost under full revelation and this reduction are bounded by the uniform bound on terminal incentive costs, so $c/(\chi^2 \mathbb{V}_m[\mu])$ is at most $(1 + \delta)c$ divided by the lower bound on the second moment of the score. The slope of the payment is thus bounded by primitives at the relaxed optimal evaluation, even though it is unbounded across arbitrary evaluations. Since $\chi(\mu - \mu_0)$ is a conditional expectation of the score at μ_0 , it satisfies the same lower bound, and the payment is strictly positive on M under (32). The relaxed optimum is therefore feasible in (11) and solves it. Conversely, every solution of (11) attains the relaxed value and is a relaxed optimum.

In the binary case with a log-supermodular technology, the lowest payment at belief μ is $U - c \frac{a + \Delta_a \mu}{b + \Delta_b \mu}$, after low output in the terminal period and after the bad evaluation in the first. It is smallest at $\mu = 0$, which gives $U > ca/b$.

The remaining conditions follow from the same bounds. With hidden effort, the first-period bonus must also cover the largest continuation gain from a private deviation, which accounts for the squared factor in (33). With commitment to continuation values, contracts that leave terminal participation rents are ruled out by comparing their quadratic wage cost with their incentive benefit. In the known-ability benchmark, the score of the fully revealing contract is bounded using the maximum of $\frac{dF(y|0)}{dF(y|1)}$, and the numerical bound follows by checking each displayed contract. $\square$

A.2 Proofs for Sections 3 and 4

Proof of Proposition 1: Fix a history (s_1, w_1, e_1) . For any feasible contract, revealing output and paying the deterministic wage satisfying

$$u(w(y)) = \int_{S_2 \times \mathbb{R}_+} u(w_2) dp_2(s_2, w_2 | y)$$

preserves expected utility under either effort choice and weakly reduces expected payments by Jensen's inequality. Full revelation has no further consequences in the terminal period, so this replacement is weakly profitable.

Both constraints bind. Since nonnegativity is assumed slack, slack participation would permit a uniform reduction in utility payments, leaving incentives unchanged. If incentive compatibility were slack, contracting utility payments toward their mean under high effort would preserve participation and strictly reduce expected wage costs while maintaining incentives.

The resulting problem depends on the history only through $\mu(s_1, w_1, e_1)$. Strict convexity of u^{-1} gives a unique optimal utility schedule, up to null sets, so the optimal wage and both parties' values depend only on this posterior. In particular, $U_2(s_1, w_1, e_1) = U$ after every history, including deviations, because first-period effort is observed. $\square$

Proof of Proposition 2: For the binary case, we denote the probability of high output as $P_\mu := a + b + (\Delta_a + \Delta_b) \mu$. By standard arguments, both the participation and the incentive constraint are binding. Hence, solving the problem in utility space, we have

$$\Pi_2(\mu) = y_L + P_\mu(y_H - y_L) - P_\mu \frac{1}{2} u_H^2 - (1 - P_\mu) \frac{1}{2} u_L^2,$$

where $u_L = U - \frac{a+\mu\Delta_a}{b+\mu\Delta_b}c$ and $u_H = U + \frac{1-a-\mu\Delta_a}{b+\mu\Delta_b}c$. Note that we require $U - \frac{a+\mu\Delta_a}{b+\mu\Delta_b}c > 0$ to satisfy the implicit nonnegativity constraint in the agent's utility function. Then, at an interior solution

$$\begin{aligned} \Pi_2''(\mu) = & -\frac{c^2}{(b+\mu\Delta_b)^4} \left[P_\mu (b\Delta_a + (1-a)\Delta_b)^2 + (1-P_\mu) (b\Delta_a - a\Delta_b)^2 \right. \\ & \left. - 2(b\Delta_a - a\Delta_b)(b\Delta_a + (1-a)\Delta_b) \right] < 0, \end{aligned}$$

where the final term is weakly negative since $b\Delta_a - a\Delta_b \leq 0$ or $b\Delta_a + \Delta_b(1-a) \leq 0$.

Only P_μ depends on μ inside the bracket, with slope $\Delta_a + \Delta_b$. The two factors differ by $(b\Delta_a + (1-a)\Delta_b) - (b\Delta_a - a\Delta_b) = \Delta_b$. Hence, differentiation of Π_2'' gives

$$\Pi_2'''(\mu) = -\frac{3\Delta_b}{b+\mu\Delta_b} \left[\Pi_2''(\mu) + \frac{c^2 (b\Delta_a - a\Delta_b)(b\Delta_a + (1-a)\Delta_b)}{(b+\mu\Delta_b)^4} \right].$$

If the technology is log-supermodular or log-submodular, then $(b\Delta_a - a\Delta_b)(b\Delta_a + (1-a)\Delta_b) \leq 0$ and $\Pi_2'' < 0$. The bracket is therefore negative, and $\text{sign } \Pi_2''' = \text{sign } \Delta_b$. $\square$

Proof of Proposition 3: Fix $\mu \in M$. Write $\mathbb{E}_\nu$ and $\mathbb{V}_\nu$ for expectation and variance under probability weights proportional to

$$\left(\frac{d\Delta_F(y)}{dF(y|1, \mu)} \right)^2 dF(y|1, \mu),$$

and suppress the arguments of Z . Differentiating the minimized terminal wage cost,

$$\frac{(U+c)^2}{2} + \frac{c^2}{2(b+\Delta_b\mu)^2 \int_Y \left(\frac{d\Delta_F(y)}{dF(y|1, \mu)} \right)^2 dF(y|1, \mu)},$$

gives

$$\Pi_2''(\mu) = \frac{c^2}{(b+\Delta_b\mu)^4 \int_Y \left(\frac{d\Delta_F(y)}{dF(y|1, \mu)} \right)^2 dF(y|1, \mu)} \left[\mathbb{V}_\nu[\Delta_b Z] + 2\Delta_b \mathbb{E}_\nu[\Delta_b Z] - 3\Delta_b^2 \right].$$

Here $\Delta_b Z(y; \mu) = (\Delta_a + \Delta_b)(b + \Delta_b\mu) \frac{d\Delta_F(y)}{dF(y|1, \mu)}$, which is also well defined for $\Delta_b = 0$. It is then easy to see that $\Pi_2''(\mu) > 0$ if $\Delta_b = 0$, as all terms in the parenthesis are zero except

$$\mathbb{V}_\nu[\Delta_b Z] = (b + \Delta_b\mu)^2 (\Delta_a + \Delta_b)^2 \mathbb{V}_\nu \left[\frac{d\Delta_F(y)}{dF(y|1, \mu)} \right] > 0.$$

For $\Delta_b \neq 0$, we can write

$$\begin{aligned} \text{sign}(-\Pi_2'') &= \text{sign}(3 - 2\mathbb{E}_\nu Z - \mathbb{V}_\nu Z), \\ \text{sign}(\Delta_b \Pi_2''') &= \text{sign}\left(4 - 3\mathbb{E}_\nu Z - 2\mathbb{V}_\nu Z - \mathbb{E}_\nu[Z(Z - \mathbb{E}_\nu Z)^2]\right). \end{aligned}$$

Since $|Z| \leq \mathcal{E}$,

$$3 - 2\mathbb{E}_\nu Z - \mathbb{V}_\nu Z \geq 2 - \mathcal{E}^2 + (\mathbb{E}_\nu Z - 1)^2.$$

This is positive when $\mathcal{E} < \sqrt{2}$. At $\mathcal{E} = \sqrt{2}$, equality would require $Z \in \{-\sqrt{2}, \sqrt{2}\}$ almost surely and $\mathbb{E}_\nu Z = 1$. But then $\mathbb{E}_\nu[1/Z] = 1/2$, contradicting $\mathbb{E}_\nu[1/Z] = 0$, which follows from $\int_Y d\Delta_F(y) = 0$. Thus $\Pi_2'' < 0$ also at the bound.

If $\mathcal{E} \leq 4/3$, concavity follows from the first claim. For the third derivative, Cauchy–Schwarz applied to the nonnegative variable $4/3 - Z$, followed by completion of the square, gives

$$\begin{aligned} & 4 - 3\mathbb{E}_\nu Z - 2\mathbb{V}_\nu Z - \mathbb{E}_\nu[Z(Z - \mathbb{E}_\nu Z)^2] \\ & \geq \frac{2}{9} \left(\frac{4}{3} - \mathbb{E}_\nu Z \right) + \frac{\left[\mathbb{V}_\nu Z - \frac{5}{3} \left(\frac{4}{3} - \mathbb{E}_\nu Z \right) \right]^2}{\frac{4}{3} - \mathbb{E}_\nu Z} > 0. \end{aligned}$$

The inequality is strict as Z takes both signs under ν since Δ_F is nonzero and has zero total mass. Hence $\text{sign } \Pi_2''' = \text{sign } \Delta_b$.

Finally,

$$\partial_\mu Z(y; \mu) = \frac{\Delta_b}{b + \Delta_b \mu} Z(y; \mu)(1 - Z(y; \mu)).$$

For each output, Z hence either equals 0 or 1 throughout, or cannot cross either value by uniqueness of solutions to this differential equation. The derivative of Z therefore has a constant sign, so it is monotone and $|Z|$ is largest at an endpoint of M . $\square$

Proof of Proposition 4: By Proposition 1, the agent receives continuation utility U after every history, including deviations. The first-period constraints are therefore static. Moreover, Bayes' rule and (10) imply

$$\mathbb{E}[u(w_1)x(y)] = \chi \mathbb{E}[u(w_1)(\mu(s, w, 1) - \mu_0)].$$

Thus incentives, participation and expected profits depend only on the joint distribution of wages and posteriors.

Pool histories inducing the same posterior μ and replace their wages by the conditional certainty equivalent,

$$w(\mu) = u^{-1}(\mathbb{E}[u(w_1)|\mu]).$$

Pooling preserves the posterior, and the replacement preserves both constraints and continuation profits. Jensen's inequality gives

$$w(\mu) \leq \mathbb{E}[w_1|\mu],$$

strictly whenever wages are nonconstant conditional on μ . Hence every optimal contract pays a deterministic wage conditional on the posterior, up to null sets. Its objective and constraints then coincide with those in (11).

Finally, an evaluation is a garbling of output, so its posterior distribution satisfies $m \leq_{MPS} \bar{m}$. Conversely, every such distribution can be induced by a garbling of the full-information posterior $\mu(y)$. Reporting the resulting posterior and paying $w(\mu)$ implements any feasible pair (m, w) without revealing further information. This implementation has exactly the constraints and

value in (11). The two problems therefore have the same value, and their optimal contracts correspond as claimed. $\square$

Proof of Theorem 1: Throughout, we omit expected output, which is a constant in the contracting problem. The contracting problem (11) is equivalent to

$$\sup_{w, m \leq_{MPS} \bar{m}} \inf_{\lambda \geq 0} \mathcal{L}(m, w; (\lambda_P, \lambda_{IC})).$$

To see this, note that since

$$\inf_{\lambda \geq 0} \mathcal{L}(m, w; \lambda) = \begin{cases} \int (\delta \Pi_2(\mu) - w(\mu)) \, dm(\mu) & \text{if (P)\&(IC) are satisfied} \\ -\infty & \text{else} \end{cases}, \quad (34)$$

the infimum simply wraps the constraints into the objective function. It is always the case that $\inf \sup \mathcal{L} \geq \sup \inf \mathcal{L}$, where the supremum is taken over the choice variables and the infimum over the multipliers. If this condition holds with equality, i.e. if we can exchange sup and inf, we say that the optimization problem satisfies *strong duality*.

Fix a distribution $m \leq_{MPS} \bar{m}$ and consider the problem of finding optimal wages subject to the participation and incentive constraint. It can be solved by point-wise optimization, arriving at (17).

Lemma 1. *The wage setting problem satisfies strong duality, i.e.*

$$\sup_w \inf_{\lambda \geq 0} \mathcal{L}(m, w; \lambda) = \inf_{\lambda \geq 0} \sup_w \mathcal{L}(m, w; \lambda),$$

and, if m is nondegenerate, both outer extrema are attained.

Proof of Lemma: If m is degenerate (a point mass on μ_0), the problem is infeasible and both the primal and dual value are $-\infty$. If m is nondegenerate, it is easy to see that the problem can be written in utility space where the objective is a concave functional and the constraints are linear. Furthermore, a strictly feasible utility promise exists (e.g., provide utility U after every posterior with a suitable large bonus if and only if $\mu > \mu_0$). Therefore, by Luenberger (1969, Section 8.6, Theorem 1), the problem satisfies strong duality and attains the dual (RHS). The primal (LHS) is attained since it is a standard moral hazard problem with strictly convex cost of wages and bounded likelihood ratios for any m . $\triangle$

Pointwise maximization yields (17) and

$$\sup_w \mathcal{L}(m, w; \lambda) = \int \ell^*(\mu; \lambda) \, dm(\mu),$$

where

$$\ell^*(\mu; \lambda) = \delta \Pi_2(\mu) + \frac{1}{2} [\max\{0, \lambda_P + \lambda_{IC} \chi(\mu - \mu_0)\}]^2 - \lambda_P(U + c) - \lambda_{IC} c.$$

Consider next the choice of m . The information design problem given λ reads

$$\sup_{m \leq_{MPS} \bar{m}} \int \ell^*(\mu; \lambda) \, dm(\mu), \quad (35)$$

Lemma 2. *The information design problem satisfies strong duality, i.e.*

$$\sup_{m \leq_{MPS} \bar{m}} \inf_{\lambda \geq 0} \int \ell^*(\mu; \lambda) d\mu = \inf_{\lambda \geq 0} \sup_{m \leq_{MPS} \bar{m}} \int \ell^*(\mu; \lambda) d\mu, \quad (36)$$

and both outer extrema are attained.

Proof of Lemma: The space of posterior distributions satisfying $m \leq_{MPS} \bar{m}$ is convex and compact in the weak* topology, and, as ℓ^* is jointly continuous and bounded for any λ , the objective is continuous and affine in m and continuous and convex in λ : convexity follows directly because it is a supremum of affine functions of the multipliers.

Therefore, the problem satisfies the conditions of Sion's Minimax Theorem and

$$\sup_{m \leq_{MPS} \bar{m}} \inf_{\lambda \geq 0} \int \ell^*(\mu; \lambda) d\mu = \inf_{\lambda \geq 0} \sup_{m \leq_{MPS} \bar{m}} \int \ell^*(\mu; \lambda) d\mu$$

Both extrema are attained: On the LHS, the infimum of continuous functions of m is upper semicontinuous, so compactness gives a maximizer. Full revelation admits a feasible contract with finite wage cost, whereas the degenerate law cannot provide incentives and has value $-\infty$. The maximizing law is therefore nondegenerate. On the RHS, fix a full-revelation contract that satisfies both participation and incentive compatibility strictly. Its Lagrangian equals its profit plus each multiplier times the corresponding strictly positive constraint slack. It therefore tends to infinity whenever either nonnegative multiplier tends to infinity. The dual objective, which maximizes the Lagrangian over contracts, is at least this large. Consequently, minimizing multipliers can be restricted to a bounded closed set. The dual objective is continuous, so it attains its minimum at some λ^* . $\triangle$

Using Lemma 1 and 2, we have

$$\sup_{w, m \leq_{MPS} \bar{m}} \inf_{\lambda \geq 0} \mathcal{L}(m, w; \lambda) = \sup_{m \leq_{MPS} \bar{m}} \inf_{\lambda \geq 0} \sup_w \mathcal{L}(m, w; \lambda) = \inf_{\lambda \geq 0} \sup_{w, m \leq_{MPS} \bar{m}} \mathcal{L}(m, w; \lambda),$$

i.e. the original problem satisfies strong duality.

Fix minimizing multipliers λ^* . Strong duality and attainment imply that every optimal contract maximizes the Lagrangian at λ^* . Its wage schedule is therefore given by (17), and its posterior distribution solves

$$\max_{m \leq_{MPS} \bar{m}} \int \ell^*(\mu; \lambda^*) d\mu. \quad (37)$$

Moreover, $\lambda_{IC}^* > 0$, since otherwise pointwise optimization would give a constant wage, which cannot provide incentives.

Lemma 3. *The optimal evaluation structure in (37) is essentially unique and*

- (i) *fully informative if $\Pi_2'' > 0$,*
- (ii) *generalized lower-censorship if $\Pi_2''' > 0$, and*
- (iii) *generalized upper-censorship if $\Pi_2''' < 0$.*

Proof of Lemma: Where wages are positive,

$$\ell^{*''}(\mu; \lambda^*) = \delta \Pi_2''(\mu) + \left(\lambda_{IC}^* \frac{b + \Delta_b \mu_0}{(\Delta_a + \Delta_b) \mu_0 (1 - \mu_0)} \right)^2,$$

whereas where wages are zero, $\ell^{*''}(\mu; \lambda^*) = \delta \Pi_2''(\mu)$. Furthermore, $\ell^{*'''}(\mu; \lambda^*) = \delta \Pi_2'''(\mu)$. The first derivative is continuous at the wage threshold, while the second derivative jumps upward by $\left(\lambda_{IC}^* \frac{b + \Delta_b \mu_0}{(\Delta_a + \Delta_b) \mu_0 (1 - \mu_0)} \right)^2$ as wages become positive.

If $\Pi_2'' > 0$, the objective is strictly convex, so full revelation, $m = \bar{m}$, is uniquely optimal.

If $\Pi_2''' > 0$, curvature is strictly increasing on either side of the wage threshold and jumps upward at the threshold. Thus ℓ^* is strictly concave followed by strictly convex. Proposition 7 of [Ostrizek \(2026\)](#) applied to the underlying distribution $\bar{m}$, implies that the unique optimal posterior law is generalized lower censorship. If $\Pi_2''' < 0$ and the supporting wage schedule is interior throughout M , there is no wage threshold on this interval. Curvature is strictly decreasing, so ℓ^* is convex followed by concave. The same proposition then gives generalized upper censorship and uniqueness of the optimal posterior law. Condition (32) ensures that there is no wage threshold on this interval. $\triangle$

In either case, one curvature region may be empty. A globally strictly convex objective gives full revelation, which is a boundary case of censorship. A globally strictly concave objective would give complete pooling and is therefore incompatible with incentive compatibility at the optimum. Since $\mu(y)$ is increasing in y , the censorship characterization also applies to the underlying output realizations.

Finally, every optimal contract maximizes the same Lagrangian at λ^* , so uniqueness of its distribution over posteriors implies uniqueness in the original contracting problem. Pointwise wage optimization is strictly concave in promised utility and hence determines wages uniquely, up to null sets. The contract is therefore unique up to informationally equivalent signal splittings. $\square$

Proof of Proposition 5: For this proof, we distinguish effort cost in the first and second period as c_1 and c_2 . Write the terminal wage cost as

$$\frac{(U + c_2)^2}{2} + \frac{c_2^2}{2} C(\mu), \quad C(\mu) = \left[\int_Y \left(1 - \frac{dF(y|0, \mu)}{dF(y|1, \mu)} \right)^2 dF(y|1, \mu) \right]^{-1}.$$

Since expected output is affine in beliefs, $\Pi_2'' < 0$ implies $C'' > 0$. Similarly, the first-period wage cost is

$$\frac{(U + c_1)^2}{2} + \frac{c_1^2}{2\chi^2 \mathbb{V}_m(\mu)}.$$

Dropping constants and dividing by $c_1^2/2$, the evaluation problem therefore becomes

$$\min_{m \leq_{MPS} \bar{m}} \left\{ \frac{1}{\chi^2 \mathbb{V}_m(\mu)} + \delta \left(\frac{c_2}{c_1} \right)^2 \mathbb{E}_m[C(\mu)] \right\}.$$

The objective depends on δ, c_1, c_2 only through $\delta \left(\frac{c_2}{c_1} \right)^2$. Let m, m' be optimal at $\delta \left(\frac{c_2}{c_1} \right)^2 < \delta' \left(\frac{c_2}{c_1} \right)^2$, respectively. Adding the inequalities stating that they are better than the other at

their respective objective yields

$$\left[\delta' \left(\frac{c'_2}{c'_1} \right)^2 - \delta \left(\frac{c_2}{c_1} \right)^2 \right] \left[\int C(\mu) dm'(\mu) - \int C(\mu) dm(\mu) \right] \leq 0.$$

Thus $\int C dm' \leq \int C dm$. Since $C'' > 0$, and since the censorship families are one-parameter families indexed by the pooling threshold and greater pooling within the optimal censorship family strictly reduces $\int C dm$, the optimal degree of pooling is weakly increasing in $\delta(c_2/c_1)^2$.

For the binary comparison, recall that we write $P_\mu = a + b + (\Delta_a + \Delta_b)\mu$ for the probability of high output. Under a tough evaluation, the good rating occurs with probability $(1 - \sigma)P_{\mu_0}$, and denote the posterior beliefs by

$$\mu_G = \mu_0 + \frac{\mu_0(1 - \mu_0)(\Delta_a + \Delta_b)}{P_{\mu_0}}, \quad \mu_B(\sigma) = \mu_0 - \frac{(1 - \sigma)P_{\mu_0}}{1 - (1 - \sigma)P_{\mu_0}}(\mu_G - \mu_0).$$

Thus μ_G is independent of σ , while μ_B increases toward μ_0 as the evaluation becomes tougher.

The ratio of the marginal continuation gain to the marginal current wage cost of increasing σ is

$$\mathcal{R}(\sigma) = \frac{2\delta(b + \Delta_b\mu_0)^2(1 - \sigma)^2}{c_1^2} [\Pi_2(\mu_B) + \Pi'_2(\mu_B)(\mu_G - \mu_B) - \Pi_2(\mu_G)].$$

Concavity of Π_2 makes the bracket positive and decreasing in σ . Hence $\mathcal{R}' < 0$ and $\mathcal{R}(\sigma) \rightarrow 0$ as $\sigma \rightarrow 1$. The optimum is therefore uniquely characterized by

$$\sigma^* = 0 \iff \mathcal{R}(0) \leq 1, \quad \mathcal{R}(\sigma^*) = 1 \text{ otherwise.}$$

In the binary case,

$$\Pi_2(\mu) = \int_Y y dF(y|1, \mu) - \frac{(U + c_2)^2}{2} - \frac{c_2^2 P_\mu (1 - P_\mu)}{2(b + \Delta_b \mu)^2}.$$

Substituting this expression and the posterior formulas into $\mathcal{R}$, we obtain $\mathcal{R}$ in the binary case.

Lemma 4. *In the binary case, suppose that $a\Delta_b > b\Delta_a$. For every fixed $\sigma \in [0, 1)$,*

$$\frac{\partial \mathcal{R}(\sigma)}{\partial \Delta_a} > 0, \quad \frac{\partial \mathcal{R}(\sigma)}{\partial \Delta_b} > 0.$$

Proof of Lemma: By lengthy computation; deferred to Online Appendix F. $\square$

By Lemma 4, strong complementarity implies that $\mathcal{R}(\sigma)$ is strictly increasing in both Δ_a and Δ_b at every fixed σ . Since $\mathcal{R}$ is strictly decreasing in σ , either parameter increase raises the unique solution to $\mathcal{R}(\sigma^*) = 1$. The comparison is therefore strict whenever $\sigma^* > 0$; at full revelation, the condition $\mathcal{R}(0) \leq 1$ gives the corresponding weak comparison. $\square$

B Private Information of the Principal (Online Appendix)

This appendix formalizes the acquisition game of Section 5.1 and proves the results stated there. Throughout, limited liability is slack at every history, and the principal implements high effort

in both periods. For a belief μ , $w_2(\cdot; \mu)$ denotes the optimal wage schedule of (5). It is unique up to null sets, and both constraints bind (Proposition 1). For a principal with belief μ_P , her terminal profit is

$$\Pi_2(\mu_P; \mu) = \int_Y (y - w_2(y; \mu)) dF(y|1, \mu_P), \quad (38)$$

which is affine in μ_P , with $\Pi_2(\mu; \mu) = \Pi_2(\mu)$.

B.1 The Acquisition Game

This subsection defines the observable- and covert-acquisition games and their perfect Bayesian equilibria. Let $\mathcal{C}$ denote the set of contracts $C = (S, p)$ of Section 2. For a first-period contract $C_1 = (S_1, p_1)$, let $\mathcal{K}(C_1)$ denote the set of experiments $K = (S_P, p_P)$, where S_P is a signal space and $p_P : Y \times S_1 \times \mathbb{R}_+ \rightarrow \Delta(S_P)$. The null experiment $K^\emptyset$ has a singleton signal space. For a terminal contract $C_2 = (S_2, p_2)$, let

$$\bar{u}_2(y) = \int u(w) dp_2(s, w|y), \quad \bar{w}_2(y) = \int w dp_2(s, w|y)$$

denote the expected utility and the expected wage it pays at output y . We suppress their dependence on C_2 .

Timing and Information Ability $\theta \in \{\theta_L, \theta_H\}$ is drawn according to the common prior μ_0 and is observed by neither party.

1. The principal chooses a contract $C_1 \in \mathcal{C}$ and an experiment $K \in \mathcal{K}(C_1)$.
2. The agent observes C_1 and, in the observable-acquisition game, K . He exits, which ends the relationship, or accepts and chooses effort $e_1 \in \{0, 1\}$.
3. Output y_1 is drawn from $F(\cdot|e_1, \theta)$, the public signal and wage (s_1, w_1) from $p_1(\cdot|y_1)$, and the private signal s_P from $p_P(\cdot|y_1, s_1, w_1)$. Both parties observe e_1 and (s_1, w_1) , only the principal observes s_P , and neither observes y_1 .
4. The principal offers a terminal contract $C_2 = (S_2, p_2) \in \mathcal{C}$.
5. The agent observes C_2 . He exits, or accepts and chooses effort $e_2 \in \{0, 1\}$.
6. Output y_2 is drawn from $F(\cdot|e_2, \theta)$, independently of (y_1, s_1, w_1, s_P) given θ and the agent's effort choice, and (s_2, w_2) is drawn from $p_2(\cdot|y_2)$.

In each period in which the agent works, the principal receives $y_t - w_t$ and the agent receives $u(w_t) - ce_t$. In each period after the agent exits, the principal receives 0 and the agent receives U . Second-period payoffs are discounted by δ .

If the agent works in the first period, the public history at the start of the second period is $h = (S_1, p_1, s_1, w_1, e_1)$. In the observable-acquisition game, h also contains K . The principal's private history is (h, K, s_P) , and the agent's terminal information set is (h, C_2) .

Strategies A strategy of the principal consists of a distribution σ_1 over first-period choices (C_1, K) with $K \in \mathcal{K}(C_1)$, and a kernel $\sigma_2(C_2|h, K, s_P)$ that assigns a distribution over terminal offers to every private history. In the covert-acquisition game, we take σ_1 to select the experiment as a function of the contract. This is without loss: a randomization over experiments that the agent does not observe is itself an experiment whose signal records the draw. A strategy of the agent consists of kernels $\gamma_1(\cdot|C_1, K)$ and $\gamma_2(\cdot|h, C_2)$ with values in $\Delta(\{\text{exit}, 0, 1\})$, where $e \in \{0, 1\}$ denotes acceptance with effort e . In the covert-acquisition game, γ_1 does not depend on K .

Beliefs The public belief $\mu(h)$ is given by (2) with $(p, s, w, e) = (p_1, s_1, w_1, e_1)$. The principal knows her experiment. Her belief $\mu_P(h, s_P)$ is given by (23), computed with the experiment K she actually chose. These updating formulas apply after every choice of contract, experiment, and effort, including deviations, almost surely under the distribution generated by those choices. At null realizations, we fix versions of the conditional distributions and use them throughout. As in (2), ratios of differentials denote Radon–Nikodym derivatives, and we fix a version of each. Let $\underline{q}(h)$ and $\bar{q}(h)$ denote the essential infimum and supremum, with respect to the conditional distribution of first-period output given h , of the belief

$$\frac{\mu_0 dF(y|e_1, \theta_H)}{dF(y|e_1, \mu_0)}$$

that the principal would hold after observing output y . Since $\mu_P(h, s_P)$ averages this belief over the conditional distribution of output given (h, s_P) , every belief the principal can hold at h lies in $[\underline{q}(h), \bar{q}(h)]$.

The agent's beliefs have two components. First, he attributes to the principal an experiment $\hat{K} = (\hat{S}_P, \hat{p}_P)$. In the observable-acquisition game, $\hat{K} = K$ is the announced experiment. In the covert-acquisition game, $\hat{K} = \hat{K}(C_1) \in \mathcal{K}(C_1)$ depends only on the first-period contract, since the experiment does not affect the distribution of (s_1, w_1) . Let $\hat{\mu}_P(h, s_P)$ denote (23) computed with $\hat{p}_P$, and let

$$d\eta(s_P|h) = \frac{\int_Y d\hat{p}_P(s_P|y, s_1, w_1) dp_1(s_1, w_1|y) dF(y|e_1, \mu_0)}{\int_Y dp_1(s_1, w_1|y) dF(y|e_1, \mu_0)}$$

denote the distribution of the principal's signal given h that the agent expects. Second, at every terminal information set (h, C_2) , he holds a belief $\mu_A(h, C_2) \in [0, 1]$ that his ability is high. The game ends after his terminal choice, so this is the only payoff-relevant part of his belief about ability and the principal's private history. An *assessment* is a tuple $(\sigma_1, \sigma_2, \gamma_1, \gamma_2, \hat{K}, \mu_A)$.

Continuation Payoffs Given an assessment, the agent's expected second-period utility at the public history h is

$$U_2(h) = \iint_{\hat{S}_P \times \mathcal{C}} \left[\gamma_2(\text{exit}|h, C_2) U + \sum_{e \in \{0, 1\}} \gamma_2(e|h, C_2) \left(\int_Y \bar{u}_2(y) dF(y|e, \hat{\mu}_P(h, s_P)) - ce \right) \right] d\sigma_2(C_2|h, \hat{K}, s_P) d\eta(s_P|h). \quad (39)$$

It generalizes U_2 in (4). The principal's expected second-period profit at her private history (h, K, s_P) is

$$\mathbb{E}[y_2 - w_2|h, K, s_P] = \int_{\mathcal{C}} \sum_{e \in \{0,1\}} \gamma_2(e|h, C_2) \int_Y (y - \bar{w}_2(y)) dF(y|e, \mu_P(h, s_P)) d\sigma_2(C_2|h, K, s_P). \quad (40)$$

The agent evaluates his continuation with the experiment $\hat{K}$ that he attributes to the principal. The principal evaluates hers with the experiment that she chose.

Definition 3 (Perfect Bayesian equilibrium). An assessment $(\sigma_1, \sigma_2, \gamma_1, \gamma_2, \hat{K}, \mu_A)$ is a *perfect Bayesian equilibrium* of the observable- or covert-acquisition game if it satisfies the following conditions.

- (i) *Terminal choice of the agent.* At every terminal information set (h, C_2) , $\gamma_2(\cdot|h, C_2)$ assigns probability one to actions that maximize the agent's expected utility at belief $\mu_A(h, C_2)$. Exit yields U , and acceptance with effort $e \in \{0, 1\}$ yields

$$\int_Y \bar{u}_2(y) dF(y|e, \mu_A(h, C_2)) - ce.$$

- (ii) *Terminal offer of the principal.* At every private history (h, K, s_P) , $\sigma_2(\cdot|h, K, s_P)$ assigns probability one to offers $C_2 \in \mathcal{C}$ that maximize

$$\sum_{e \in \{0,1\}} \gamma_2(e|h, C_2) \int_Y (y - \bar{w}_2(y)) dF(y|e, \mu_P(h, s_P)).$$

- (iii) *First-period choice of the agent.* For every first-period choice (C_1, K) , $\gamma_1(\cdot|C_1, K)$ assigns probability one to actions that maximize the agent's expected utility. Exit yields $(1 + \delta)U$, and acceptance with effort $e_1 \in \{0, 1\}$ yields

$$\iint_{S_1 \times \mathbb{R}_+ \times Y} (u(w_1) + \delta U_2(h)) dp_1(s_1, w_1|y) dF(y|e_1, \mu_0) - ce_1,$$

where h is the public history that (C_1, K, s_1, w_1, e_1) generates.

- (iv) *First-period choice of the principal.* σ_1 assigns probability one to first-period choices (C_1, K) that maximize

$$\sum_{e_1 \in \{0,1\}} \gamma_1(e_1|C_1, K) \iiint_{S_P \times S_1 \times \mathbb{R}_+ \times Y} (y - w_1 + \delta \mathbb{E}[y_2 - w_2|h, K, s_P]) dp_P(s_P|y, s_1, w_1) dp_1(s_1, w_1|y) dF(y|e_1, \mu_0).$$

- (v) *Consistency.* In the covert-acquisition game, $\hat{K}(C_1) = K$ for σ_1 -almost every (C_1, K) . At every public history h , on or off the equilibrium path, the agent's terminal belief follows from σ_2 and $\hat{K}$ by Bayes' rule:

$$\mu_A(h, C_2) = \frac{\int \hat{\mu}_P(h, s_P) d\sigma_2(C_2|h, \hat{K}, s_P) d\eta(s_P|h)}{\int d\sigma_2(C_2|h, \hat{K}, s_P) d\eta(s_P|h)}. \quad (41)$$

As in (2), the ratio in (41) is a Radon–Nikodym derivative. Condition (v) thus requires

$$\iint_{\hat{S}_P \times B} \mu_A(h, C_2) d\sigma_2(C_2|h, \hat{K}, s_P) d\eta(s_P|h) = \int_{\hat{S}_P} \hat{\mu}_P(h, s_P) \sigma_2(B|h, \hat{K}, s_P) d\eta(s_P|h)$$

for every measurable set $B \subseteq \mathcal{C}$ of offers. It pins down $\mu_A(h, \cdot)$ almost everywhere with respect to the distribution of offers that the agent expects at h . At an offer made with positive probability, such as a pooling offer, it is an ordinary ratio of probabilities. On a set of offers having zero probability, Definition 3 leaves $\mu_A(h, C_2) \in [0, 1]$ unrestricted. The refinements of Section 5.1 restrict it further. Condition (v) applies even after a covert change of experiment, which the agent cannot detect. He then updates with $\hat{K}$, while the principal updates with the experiment she chose. Condition (v) imposes Bayes' rule at every public history, not only on the equilibrium path, which is a version of no-signaling-what-you-don't-know for the first period offer which we baked into the definition of PBE.

Two consequences are used repeatedly. First, the agent's payoff from a given terminal action is affine in his belief. By condition (v), $\hat{\mu}_P(h, s_P)$ may therefore be replaced by $\mu_A(h, C_2)$ in (39), and condition (i) implies $U_2(h) \geq U$ at every h . Condition (iii) thus compares complete continuation plans, including rejecting every terminal offer. Second, the principal observes (s_1, w_1) in addition to s_P , so $\mathbb{E}[\mu_P(h, s_P)|h] = \mu(h)$ for the experiment she chose. On the equilibrium path, $\hat{K} = K$, and (41) yields $\mathbb{E}[\mu_P(h, s_P)|h, C_2] = \mu_A(h, C_2)$ at almost every terminal information set.

Terminology The prior, the technology, and the strategies induce a distribution over plays of the game. Statements about almost every on-path terminal information set, as in Definition 2, refer to the distribution of (h, C_2) that it induces. An equilibrium *implements high effort in both periods* if, almost surely under this distribution, the agent accepts both contracts and chooses $e_1 = e_2 = 1$. The principal's *expected profit* is the objective in condition (iv), averaged over σ_1 .

B.2 Proof of Proposition 6

Proof of Proposition 6. Fix a no-holdup perfect Bayesian equilibrium that implements high effort in both periods, and condition on its first-period choice (C_1, K) . Write $\mathbb{E}[\cdot|e_1]$ for expectations when the agent exerts effort e_1 in the first period, and Π^{eq} for the principal's equilibrium profit.

By no-holdup, the agent's equilibrium utility is $\mathbb{E}[u(w_1)|1] - c + \delta U$. He can instead exert low effort and reject every terminal offer, which yields $\mathbb{E}[u(w_1)|0] + \delta U$ whatever he believes after the deviation. Participation and this deviation imply

$$\mathbb{E}[u(w_1)|1] - c \geq U, \quad \mathbb{E}[u(w_1)|1] - \mathbb{E}[u(w_1)|0] \geq c. \quad (42)$$

Let $\sigma_2^1(C_2|s_1, w_1, s_P)$ denote the equilibrium offer rule at the high-effort history $h = (S_1, p_1, s_1, w_1, 1)$. Consider the public contract with signal (s_1, C_2) , wage w_1 , and

$$d\hat{p}_1(s_1, C_2, w_1|y) = \int_{S_P} d\sigma_2^1(C_2|s_1, w_1, s_P) dp_P(s_P|y, s_1, w_1) dp_1(s_1, w_1|y). \quad (43)$$

This contract garbles first-period output with the same kernel whatever effort the agent exerts. Its marginal on (s_1, w_1) is p_1 . The wage therefore has the same distribution as under the original contract, for each effort, and (42) holds for this contract. Under high effort, its posterior after (s_1, C_2, w_1) is the agent's equilibrium belief $\mu_A(h, C_2)$.

On the equilibrium path, the agent accepts C_2 and exerts high effort at belief $\mu_A = \mu_A(h, C_2)$. Hence $\bar{u}_2$ satisfies the constraints of (5) at μ_A . Three facts hold conditional on (h, C_2) : the principal's belief averages to μ_A ; second-period output is independent of first-period signals given ability; and her profit from a given contract is affine in her belief. Hence

$$\mathbb{E}[y_2 - w_2 | h, C_2] = \int_Y (y - \bar{w}_2(y)) dF(y|1, \mu_A) \leq \int_Y (y - u^{-1}(\bar{u}_2(y))) dF(y|1, \mu_A) \leq \Pi_2(\mu_A). \quad (44)$$

The first inequality is Jensen's inequality. The second holds because $u^{-1} \circ \bar{u}_2$ is feasible in (5).

Let m be the distribution of $\mu_A(h, C_2)$ under high effort, and let $w(\mu) = u^{-1}(\mathbb{E}[u(w_1) | \mu_A = \mu])$. The contract (43) garbles output, so $m \leq_{MPS} \bar{m}$. As in the derivation of (11), (42) implies the participation and incentive constraints of (11) for (m, w) . By Jensen's inequality, $\mathbb{E}_m[w(\mu)] \leq \mathbb{E}[w_1 | 1]$. Using (44),

$$\Pi^{\text{eq}} \leq \mathbb{E}[y] - \mathbb{E}[w_1 | 1] + \delta \mathbb{E}_m[\Pi_2(\mu)] \quad (45)$$

$$\leq \mathbb{E}[y] + \mathbb{E}_m[\delta \Pi_2(\mu) - w(\mu)] \quad (46)$$

$$\leq \Pi(\mu_0). \quad (47)$$

Averaging over mixed first-period choices proves the bound.

It remains to show that an equilibrium attaining the bound replicates the optimal contract. Suppose $\Pi^{\text{eq}} = \Pi(\mu_0)$. Then (45)–(47) hold with equality for almost every first-period choice. The first two integrate pointwise inequalities, so these hold with equality almost surely.

By (47), (m, w) solves (11). In particular, w minimizes the wage cost given m . By (17) and slack limited liability, $w(\mu) = \frac{1}{2}(\lambda_P + \lambda_{IC}\chi(\mu - \mu_0))^2$ m -almost surely, for some multipliers. Moreover, $\lambda_{IC} > 0$: since $\mathbb{E}_m[\mu] = \mu_0$, a constant wage violates the incentive constraint. Hence w is strictly increasing on the support of m .

By (46) and strict concavity of u , $w_1 = w(\mu_A(h, C_2))$ almost surely. So $\mu_A(h, C_2)$ is a function of the public outcome (s_1, w_1) , and $\mu(h) = \mathbb{E}[\mu_A(h, C_2) | h] = \mu_A(h, C_2)$, establishing point 2. The first-period contract therefore induces the distribution m of $\mu(h)$ and the wage $w(\mu(h))$. It solves (11), and hence (4). In the cases covered by Theorem 1, this solution is unique up to informationally equivalent evaluations, establishing point 1.

By (44), the terminal wage is deterministic almost surely, and $u^{-1} \circ \bar{u}_2$ solves (5) at $\mu_A = \mu(h)$. That solution is unique, so the terminal contract pays $w_2(\cdot; \mu(h))$, establishing point 3. $\square$

B.3 Attaining the Bound

Observable acquisition

Proof of Remark 1. Let C_1^* solve (4), and let $K^\emptyset$ denote the null experiment. After $(C_1^*, K^\emptyset)$, no-signaling-what-you-don't-know gives $\mu_A(h, C_2) = \mu(h)$ after every terminal offer. Every terminal subgame is therefore the common-information problem (5). In any equilibrium, the

principal offers $w_2(\cdot; \mu(h))$, and the agent accepts and exerts high effort; otherwise a slightly more generous and steeper offer would be strictly accepted with high effort, and the principal would have no best response. The agent therefore obtains U in the second period after either first-period effort, and his first-period choices are those of (4). The principal can choose $(C_1^*, K^\varnothing)$ and secure a profit of $\Pi(\mu_0)$ (respectively, offer a slightly more generous and steeper contract, to ensure the participation and incentive constraint are satisfied strictly, obtaining a profit arbitrarily close to $\Pi(\mu_0)$). Every such equilibrium thus gives her at least $\Pi(\mu_0)$. Proposition 6 gives the reverse inequality and the characterization. $\square$

Existence is a separate question. A no-holdup equilibrium of this kind, satisfying no-signaling-what-you-don't-know and attaining $\Pi(\mu_0)$, exists if and only if one can select, after every public first-period choice (C_1, K) , a continuation equilibrium that satisfies no-signaling-what-you-don't-know and gives the principal at most $\Pi(\mu_0)$, with the continuation of the preceding proof selected after $(C_1^*, K^\varnothing)$. Necessity follows because the principal must not gain by deviating to another choice. Sufficiency follows because, with such a selection, choosing $(C_1^*, K^\varnothing)$ is optimal for her.

It suffices that, after every public first-period choice (C_1, K) , the continuation game has an equilibrium that satisfies no-signaling-what-you-don't-know and in which, after every first-period action the agent takes with positive probability, the terminal offer is accepted with high effort and leaves the agent utility U , as in (24). In any such continuation, the principal obtains at most $\Pi(\mu_0)$. If the agent exits, she obtains zero. If he works, the proof of Proposition 6 applies without change: it uses the first-period choice only through the agent's best response to it, so (42)–(47) hold conditional on (C_1, K) . If he shirks, the same argument applies with expectations taken under $e_1 = 0$. Participation gives $\mathbb{E}[u(w_1)|0] \geq U$, the contract (43) garbles first-period output, and (44) bounds terminal profit by $\Pi_2(\mu_A(h, C_2))$. Her payoff is therefore at most the value of a public-information contract under which the agent shirks in the first period, which is at most $\Pi(\mu_0)$ because implementing high effort is optimal (Appendix A.1). If the agent mixes, her payoff is an average of these cases. With binary output and $\Delta_b \geq 0$, the construction in the proof of Proposition 7 provides such a continuation after every (C_1, K) .

Covert acquisition

Let output be binary. As in the proof of Proposition 2, $P_\mu = a + b + (\Delta_a + \Delta_b)\mu$ denotes the probability of high output under high effort at belief μ ; under low effort, this probability is $a + \Delta_a\mu$. Both are nondecreasing in μ . The optimal terminal contract $w_2(\cdot; \mu)$ pays the utilities u_L and u_H given there. Thus $(1 - P_\mu)u_L + P_\mu u_H = U + c$, $u_H - u_L = c/(b + \Delta_b\mu)$, and

$$u_H = U + \frac{1 - a - \Delta_a\mu}{b + \Delta_b\mu}c.$$

This is at most $U + c/b$ when $\Delta_b \geq 0$.

The agent's beliefs are *skeptical* if, after every terminal offer other than $w_2(\cdot; \mu(h))$, $\mu_A(h, C_2) = \underline{q}(h)$ when $\bar{u}_2(y_H) \geq \bar{u}_2(y_L)$, and $\mu_A(h, C_2) = \bar{q}(h)$ otherwise. The agent thus evaluates an unexpected offer at the attainable belief at which accepting it is least attractive to him.

With binary nonnegative output and $\Delta_b \geq 0$, the maintained condition (31) reads

$$b(y_H - y_L) > u^{-1}\left(U + \frac{c}{b}\right). \quad (48)$$

The effort of a low-ability agent raises expected output by more than the highest wage that any $w_2(\cdot; \mu)$ pays.

Lemma 5. *Let output be binary and $\Delta_b \geq 0$, fix a public history h , and let the agent's beliefs be skeptical. Consider any principal belief $\mu_P \in [\underline{q}(h), \bar{q}(h)]$ and any terminal offer C_2 other than $w_2(\cdot; \mu(h))$.*

(i) *If accepting C_2 with high effort is optimal for the agent,*

$$\text{then } \int \bar{w}_2 dF(y|1, \mu_P) \geq \int w_2(y; \mu(h)) dF(y|1, \mu_P).$$

(ii) *If accepting C_2 with low effort is optimal for the agent, then $\int \bar{w}_2 dF(y|0, \mu_P) \geq u^{-1}(U)$.*

Proof of Lemma: By Jensen's inequality, $\bar{w}_2 \geq u^{-1} \circ \bar{u}_2$, so it suffices to prove both claims with $u^{-1} \circ \bar{u}_2$ in place of $\bar{w}_2$. Write $z_L = \bar{u}_2(y_L)$ and $z_H = \bar{u}_2(y_H)$. Abbreviate $\underline{q} = \underline{q}(h)$, and let u_L and u_H be the utilities that $w_2(\cdot; \mu(h))$ pays.

Low effort. Suppose first that $z_H \geq z_L$. The agent's belief is then $\underline{q} \leq \mu_P$, and participation requires $(1 - a - \Delta_a \underline{q})z_L + (a + \Delta_a \underline{q})z_H \geq U$. The left side is nondecreasing in the belief, so the inequality also holds at μ_P . Since u^{-1} is convex and increasing, Jensen's inequality gives the claim. If $z_H < z_L$, the agent's belief is $\bar{q}(h) \geq \mu_P$, the left side is nonincreasing in the belief, and the same argument applies.

High effort. Incentive compatibility requires $(b + \Delta_b \mu_A(h, C_2))(z_H - z_L) \geq c > 0$. Hence $z_H > z_L$ and $\mu_A(h, C_2) = \underline{q}$. Acceptance with high effort then requires

$$(1 - P_{\underline{q}})z_L + P_{\underline{q}}z_H \geq U + c, \quad z_H - z_L \geq \frac{c}{b + \Delta_b \underline{q}}. \quad (49)$$

Let $d_L = z_L - u_L$ and $d_H = z_H - u_H$ denote the changes in utility relative to $w_2(\cdot; \mu(h))$. Since $(1 - P_{\mu(h)})u_L + P_{\mu(h)}u_H = U + c$ and $u_H - u_L = c/(b + \Delta_b \mu(h))$, (49) implies

$$(1 - P_{\underline{q}})d_L + P_{\underline{q}}d_H \geq (P_{\mu(h)} - P_{\underline{q}})(u_H - u_L) \geq 0, \quad d_H - d_L \geq \frac{c}{b + \Delta_b \underline{q}} - \frac{c}{b + \Delta_b \mu(h)} \geq 0,$$

because $\mu(h) \geq \underline{q}$ and $\Delta_b \geq 0$. In particular, $d_H \geq 0$. Let $\mu_P \geq \underline{q}$. Since u^{-1} is convex, $u^{-1}(z) - u^{-1}(u) \geq (u^{-1})'(u)(z - u)$, so the principal's cost difference satisfies

$$\begin{aligned} (1 - P_{\mu_P})[u^{-1}(z_L) - u^{-1}(u_L)] + P_{\mu_P}[u^{-1}(z_H) - u^{-1}(u_H)] \\ \geq (1 - P_{\mu_P})(u^{-1})'(u_L)d_L + P_{\mu_P}(u^{-1})'(u_H)d_H \\ \geq (u^{-1})'(u_L)[(1 - P_{\mu_P})d_L + P_{\mu_P}d_H] \\ \geq (u^{-1})'(u_L)[(1 - P_{\underline{q}})d_L + P_{\underline{q}}d_H] \geq 0. \end{aligned}$$

The second inequality uses $d_H \geq 0$ and $(u^{-1})'(u_H) \geq (u^{-1})'(u_L)$. The third uses $d_H \geq d_L$ and $P_{\mu_P} \geq P_{\underline{q}}$. The last uses $(u^{-1})'(u_L) = u_L \geq 0$. This proves the first claim. $\square$

Proof of Proposition 7. Let C_1^* solve (4), and consider the following assessment.

- The principal offers C_1^* and observes first-period output. At every public history h , she offers $w_2(\cdot; \mu(h))$, whatever her private information.
- The agent attributes this experiment to her. After $w_2(\cdot; \mu(h))$, he holds belief $\mu(h)$; after other terminal offers, his beliefs are skeptical.
- The agent accepts C_1^* and exerts high effort, and he accepts $w_2(\cdot; \mu(h))$ and exerts high effort. He best-responds to all other offers.

By (41), the agent's belief after the pooling offer $w_2(\cdot; \mu(h))$ is $\mu(h)$. Both constraints bind at this belief, so accepting and working is optimal. Consider a principal with belief $\mu_P \in [\underline{q}(h), \bar{q}(h)]$. The offer $w_2(\cdot; \mu(h))$ yields $\Pi_2(\mu_P; \mu(h))$. By Lemma 5, an offer accepted with high effort yields no more. An offer accepted with low effort yields at most $\int y dF(y|0, \mu_P) - u^{-1}(U)$, and rejection yields zero. Moreover, $w_2(\cdot; \mu(h)) \leq u^{-1}(u_H) \leq u^{-1}(U + c/b)$ and $\int y dF(y|1, \mu_P) - \int y dF(y|0, \mu_P) = (b + \Delta_b \mu_P)(y_H - y_L) \geq b(y_H - y_L)$. Condition (48) therefore gives

$$\Pi_2(\mu_P; \mu(h)) \geq \int y dF(y|0, \mu_P) + b(y_H - y_L) - u^{-1}\left(U + \frac{c}{b}\right) > \int y dF(y|0, \mu_P) \geq 0,$$

where the last inequality uses $y_L \geq 0$. Hence $w_2(\cdot; \mu(h))$ is optimal for every belief the principal can hold, including after a covert change of experiment.

Since $\Pi_2(\mu_P; \mu(h))$ is affine in μ_P and $\mathbb{E}[\mu_P(h, s_P)|h] = \mu(h)$, the principal's expected terminal profit at h is $\Pi_2(\mu(h))$, whatever experiment she chooses. In particular, observing output is optimal.

At every public history, including those after other first-period contracts and after low effort, the agent obtains U in the second period. His first-period constraints are therefore those of (4). The principal's first-period problem is also (4), and C_1^* solves it. The agent does not observe the experiment, so a change of experiment affects neither his first-period behavior nor, by the acquisition step, her payoff.

The assessment is therefore a no-holdup perfect Bayesian equilibrium with profit $\Pi(\mu_0)$. By Proposition 6, it is principal-preferred among no-holdup equilibria that implement high effort.

In the observable-acquisition game, the same construction applies with a different $\underline{q}(h)$ and $\bar{q}(h)$: the smallest and largest points of the closed convex hull of the support of $\mu_P(h, \cdot)$ under the announced experiment. The agent's beliefs then satisfy no-signaling-what-you-don't-know. Lemma 5 and the argument above still apply, because $\mu(h)$ and every belief the principal can hold lie between the bounds of the Lemma. $\square$

C Unobservable Effort (Online Appendix)

In this section, all notation pertains to the hidden action problem; we do not use decorators to distinguish contracts, profits, continuation values and the like from the baseline model to keep the notation light.

The principal's problem with hidden effort is

$$\Pi(\mu_0) = \max_{S,p} \iint_{S \times \mathbb{R}_+ \times Y} (y - w_1 + \delta \Pi_2(s_1, w_1)) \, dp(s_1, w_1|y) \, dF(y|1, \mu_0) \quad (50)$$

$$\text{s.t.} \quad \iint_{S \times \mathbb{R}_+ \times Y} (u(w_1) + \delta U_2(s_1, w_1, 1)) \, dp(s_1, w_1|y) \, dF(y|1, \mu_0) - c \geq (1 + \delta)U \quad (51)$$

$$\begin{aligned} & \iint_{S \times \mathbb{R}_+ \times Y} (u(w_1) + \delta U_2(s_1, w_1, 1)) \, dp(s_1, w_1|y) \, dF(y|1, \mu_0) - c \\ & \geq \iint_{S \times \mathbb{R}_+ \times Y} (u(w_1) + \delta U_2(s_1, w_1, 0)) \, dp(s_1, w_1|y) \, dF(y|0, \mu_0) \end{aligned} \quad (52)$$

where

$$\Pi_2(s_1, w_1) = \max_{S,p} \iint_{S \times \mathbb{R}_+ \times Y} (y - w_2) \, dp(s_2, w_2|y) \, dF(y|1, \mu(s_1, w_1, 1)) \quad (53)$$

$$\text{s.t.} \quad \iint_{S \times \mathbb{R}_+ \times Y} u(w_2) \, dp(s_2, w_2|y) \, dF(y|1, \mu(s_1, w_1, 1)) - c \geq U \quad (54)$$

$$\begin{aligned} & \iint_{S \times \mathbb{R}_+ \times Y} u(w_2) \, dp(s_2, w_2|y) \, dF(y|1, \mu(s_1, w_1, 1)) - c \\ & \geq \iint_{S \times \mathbb{R}_+ \times Y} u(w_2) \, dp(s_2, w_2|y) \, dF(y|0, \mu(s_1, w_1, 1)) \end{aligned} \quad (55)$$

and

$$U_2(s_1, w_1, e) := \max \left\{ U, \right. \quad (56)$$

$$\left. \iint_{S \times \mathbb{R}_+ \times Y} u(w_2) \, dp(s_2, w_2|y) \, dF(y|1, \mu(s_1, w_1, e)) - c, \right. \quad (57)$$

$$\left. \iint_{S \times \mathbb{R}_+ \times Y} u(w_2) \, dp(s_2, w_2|y) \, dF(y|0, \mu(s_1, w_1, e)) \right\} \quad (58)$$

where (S, p) are the optimal contract under $\Pi_2(s_1, w_1)$.

Deviation Posterior Formula Consider an arbitrary evaluation structure (S, p) . An evaluation s is drawn with probability $p(s, w|y)$ conditional on output. Let $\mu(s, w)$ and $\tilde{\mu}(s, w)$ be the posteriors after observing s, w when effort was high and low, respectively.

Bayes' rule gives

$$\mu(s, w) - \mu_0 = \mu_0(1 - \mu_0)(\Delta_a + \Delta_b) \frac{\int dp(s, w|y) \, d\Delta_F(y)}{\int dp(s, w|y) \, dF(y|1, \mu_0)}, \quad (59)$$

$$\tilde{\mu}(s, w) - \mu_0 = \mu_0(1 - \mu_0)\Delta_a \frac{\int dp(s, w|y) \, d\Delta_F(y)}{\int dp(s, w|y) \, dF(y|0, \mu_0)}. \quad (60)$$

Moreover, $\int p(s, w|y) \, dF(y|1, \mu_0) - \int p(s, w|y) \, dF(y|0, \mu_0) = (b + \Delta_b \mu_0) \int p(s, w|y) \, d\Delta_F(y)$. Using (59),

$$\frac{\int p(s, w|y) \, dF(y|0, \mu_0)}{\int p(s, w|y) \, dF(y|1, \mu_0)} = 1 - \frac{b + \Delta_b \mu_0}{\mu_0(1 - \mu_0)(\Delta_a + \Delta_b)} [\mu(s, w) - \mu_0] = 1 - x(\mu(s, w)).$$

Hence, we see that if m is the distribution over posteriors under high effort, the distribution under low effort is $(1 - x(\cdot))m(\cdot)$. Substituting into (60) gives

$$\tilde{\mu}(s, w) = \mu_0 + \frac{\Delta_a}{\Delta_a + \Delta_b} \frac{\mu(s, w) - \mu_0}{1 - x(\mu(s, w))}.$$

Hence the private posterior depends on the evaluation only through the public posterior, and we obtain (25).

Second-Period Problem The terminal contract $w_2(y; \mu)$ is unaffected by hidden effort on path. Therefore, the continuation value of the principal is $\Pi_2(\mu(s_1, w_1, 1))$ from the baseline problem. Its binding participation and incentive constraints imply

$$\int_Y u(w_2(y; \mu)) dF(y|1, \mu) = U + c, \quad \int_Y u(w_2(y; \mu)) d\Delta_F(y) = \frac{c}{b + \Delta_b \mu}.$$

Consequently, an agent with private belief $\tilde{\mu}$ obtains

Terminal choice	Expected utility net of effort cost
Leave	U
Accept and shirk	$U + \frac{c\Delta_a}{b + \Delta_b \mu}(\tilde{\mu} - \mu)$
Accept and work	$U + \frac{c(\Delta_a + \Delta_b)}{b + \Delta_b \mu}(\tilde{\mu} - \mu).$

We denote the rent by

$$R(\mu) := \frac{c \max\{\Delta_a, \Delta_a + \Delta_b\}}{b + \Delta_b \mu} \max\{\tilde{\mu}(\mu) - \mu, 0\}. \quad (61)$$

First-Period Problem in Posterior Space From the previous paragraph, it is immediate that both the principal's and the agent's continuation value, for the principal on-path ($e_1 = 1$), for the agent both on and off path, can be written as a function of the first-period public belief. We can therefore write the first period problem in belief space.

$$\Pi(\mu_0) = \max_{m \in \Delta([0,1]), w: [0,1] \rightarrow \mathbb{R}_+} \int_Y y dF(y|1, \mu_0) + \int_{[0,1]} (\delta \Pi_2(\mu) - w(\mu)) dm(\mu) \quad (62)$$

$$\text{s.t.} \quad \int_{[0,1]} u(w(\mu)) dm(\mu) - c \geq U \quad (63)$$

$$\int_{[0,1]} u(w(\mu)) x(\mu) dm(\mu) - c \geq \int_{[0,1]} \delta R(\mu)(1 - x(\mu)) dm(\mu) \quad (64)$$

C.1 Binary Output Analysis

We now specialize to binary output. Recall that we write $P_\mu = a + b + (\Delta_a + \Delta_b)\mu$ for the probability that high output realizes under high effort. Recall further that the posterior after any evaluation is an average of μ_L and μ_H , the posteriors after observing y_L and y_H . It therefore lies in $[\mu_L, \mu_H]$. Averaging over evaluations gives the prior μ_0 . Conversely, any distribution m on this interval with mean μ_0 can be induced by an evaluation. Recall that the contract in the

second period in the binary case is

$$u(w_2(y_L; \mu)) = U - \frac{c(a + \Delta_a \mu)}{b + \Delta_b \mu}, \quad u(w_2(y_H; \mu)) = U + \frac{c(1 - a - \Delta_a \mu)}{b + \Delta_b \mu}. \quad (65)$$

Proof of Proposition 8. The observed-effort Lagrangian integrand is, up to terms affine in μ ,

$$\ell^*(\mu; \lambda_{IC}) = \frac{\lambda_{IC}^2}{2} x(\mu)^2 + \delta \Pi_2(\mu).$$

With hidden effort it becomes

$$\tilde{\ell}(\mu; \lambda_{IC}) = \ell^*(\mu; \lambda_{IC}) - \lambda_{IC} \delta(1 - x(\mu)) R(\mu). \quad (66)$$

and m^* maximizes the expected value of $\tilde{\ell}$. Maximizing this expected value can be done via concavification (Aumann and Maschler, 1995; Kamenica and Gentzkow, 2011). In particular, there is an affine function A above $\tilde{\ell}$ that coincides with it on the support of m^* , i.e.

$$A(\mu) \geq \tilde{\ell}(\mu; \lambda_{IC}), \quad A(\mu) = \tilde{\ell}(\mu; \lambda_{IC}) \quad \text{on the support of } m^*. \quad (67)$$

Such a line proves optimality because its expectation is $A(\mu_0)$ under every feasible distribution. We now determine the shape of $\tilde{\ell}$ to determine where the line can meet its graph.

Equation (25) implies that

$$(1 - x(\mu))[\tilde{\mu}(\mu) - \mu] = x(\mu) \left[\mu - \mu_0 - \frac{\mu_0(1 - \mu_0)\Delta_b}{b + \Delta_b \mu_0} \right]. \quad (68)$$

The last parenthesis is nonzero on $[\mu_L, \mu_H]$: The second crossing between μ and $\tilde{\mu}$ lies above μ_H with strong complements and below μ_L with strong substitutes, since

$$\begin{aligned} \mu_0 + \frac{\mu_0(1 - \mu_0)\Delta_b}{b + \Delta_b \mu_0} - \mu_H &= \frac{\mu_0(1 - \mu_0)(a\Delta_b - b\Delta_a)}{(b + \Delta_b \mu_0)P_{\mu_0}} > 0, \\ \mu_0 + \frac{\mu_0(1 - \mu_0)\Delta_b}{b + \Delta_b \mu_0} - \mu_L &= \frac{\mu_0(1 - \mu_0)[b\Delta_a + (1 - a)\Delta_b]}{(b + \Delta_b \mu_0)(1 - P_{\mu_0})} < 0. \end{aligned}$$

Hence R is positive precisely below the prior with complements and above it with substitutes.

Shape of the Lagrangian: Under either condition, $\Pi_2'' < 0$ and Π_2''' has the sign of Δ_b . Where $R(\mu) > 0$,

$$\frac{\partial^2}{\partial^2 \mu} (1 - x(\mu)) R(\mu) = \frac{2c \max\{\Delta_a, \Delta_a + \Delta_b\}}{\Delta_a + \Delta_b} \frac{(b + \Delta_b \mu_0) \left[(\Delta_b)^2 + \frac{(b + \Delta_b \mu_0)^2}{\mu_0(1 - \mu_0)} \right]}{(b + \Delta_b \mu)^3} > 0, \quad (69)$$

$$\frac{\partial^3}{\partial^3 \mu} (1 - x(\mu)) R(\mu) = -\frac{3\Delta_b}{b + \Delta_b \mu} \left[\frac{\partial^2}{\partial^2 \mu} (1 - x(\mu)) R(\mu) \right]. \quad (70)$$

It follows that $\tilde{\ell}''' > 0$ with complements and $\tilde{\ell}''' < 0$ with substitutes on either side of the prior. At the prior, the function is continuous but its slope jumps downward by

$$\tilde{\ell}'(\mu_0+) - \tilde{\ell}'(\mu_0-) = -\lambda_{IC} \delta c \frac{\max\{\Delta_a, \Delta_a + \Delta_b\}}{\Delta_a + \Delta_b} \frac{|\Delta_b|}{b + \Delta_b \mu_0} < 0. \quad (71)$$

The second derivative jumps up with complements and down with substitutes. This follows from (69): the rent term disappears at the prior with complements and appears there with substitutes. The second derivative is therefore increasing across the two intervals with complements and decreasing with substitutes.

Concavification

Complements: Since the distribution is informative and has mean μ_0 , it must assign positive probability to a posterior below μ_0 and to a posterior above μ_0 . We show that the upper posterior must be μ_H .

Suppose instead that an upper posterior lies strictly below μ_H . At this posterior, A touches the graph of $\tilde{\ell}$ from above. Hence A is tangent to the graph and the second derivative is nonpositive. Since the second derivative is strictly increasing, it is strictly negative at every lower posterior. The downward slope jump at μ_0 preserves this strict concavity. A tangent to a strictly concave function lies strictly above its graph at every other point. Thus A cannot touch the graph at any lower posterior. But the distribution must use a posterior below μ_0 , and every posterior it uses must be a point where A touches the graph. This is a contradiction. The upper posterior must therefore be μ_H .

There can also be at most one posterior below μ_0 . If there were two, the larger would be an interior point where A is tangent and the second derivative is nonpositive. The same strict-concavity argument would imply that A lies strictly above the graph at the smaller posterior, again a contradiction.

The prior itself may also receive positive probability. Thus the support is contained in

$$\{\mu_0 - d_B, \mu_0, \mu_H\}, \quad 0 < d_B \leq \mu_0 - \mu_L.$$

For substitutes, the second derivative is strictly decreasing. Applying the same argument from right to left shows that the lower posterior must be μ_L and that there can be at most one posterior above μ_0 . The support is therefore contained in

$$\{\mu_L, \mu_0, \mu_0 + d_G\}, \quad 0 < d_G \leq \mu_H - \mu_0.$$

In either case, both posteriors different from the prior receive positive probability.

This establishes Proposition 8 for the log-submodular case. It remains to show that the prior is never part of the support under log-supermodularity.

The probability of a neutral evaluation

Write the two informative posteriors as $\mu_0 - d_B$ and $\mu_0 + d_G$. Conditional on an informative evaluation, their probabilities are $d_G/(d_B + d_G)$ and $d_B/(d_B + d_G)$. Define

$$\begin{aligned} Q &= x'(\mu_0)^2 d_B d_G, \\ D &= \frac{d_G(1 - x(\mu_0 - d_B))R(\mu_0 - d_B) + d_B(1 - x(\mu_0 + d_G))R(\mu_0 + d_G)}{d_B + d_G}, \\ J &= \Pi_2(\mu_0) - \frac{d_G \Pi_2(\mu_0 - d_B) + d_B \Pi_2(\mu_0 + d_G)}{d_B + d_G}. \end{aligned}$$

These are the expected squared score, expected deviation rent, and loss of continuation profit, conditional on being informative. If the neutral evaluation has probability p_N , the expected first-period wage is

$$\frac{(U + c)^2}{2} + \frac{[c + \delta(1 - p_N)D]^2}{2(1 - p_N)Q}.$$

The expected second-period wage is

$$\frac{(U + c)^2}{2} + \frac{c^2 P_{\mu_0}(1 - P_{\mu_0})}{2(b + \Delta_b \mu_0)^2} + (1 - p_N)J.$$

The terms independent of the evaluation can be omitted from the minimization. The principal therefore chooses d_B , d_G , and p_N to minimize

$$\frac{[c + \delta(1 - p_N)D]^2}{2(1 - p_N)Q} + \delta(1 - p_N)J.$$

The first term is the additional first-period wage payment needed to provide incentives. The second is the discounted increase in second-period wages caused by learning. Differentiating gives the unique minimizer

$$p_N = \max\{0, 1 - \Omega^{-1/2}\}, \quad \Omega = \frac{\delta^2 D^2 + 2\delta QJ}{c^2}. \quad (72)$$

First suppose $\Delta_a = 0$. Fix $d_G = \mu_H - \mu_0$ and let primes denote derivatives in d_B . We first establish two inequalities, then use them to rule out a positive neutral probability. The idea will be that if the neutral evaluation is optimal, you also want to shift up the bad posterior towards the prior, but that cannot be in the optimum.

First, note J is a difference of slopes, and so really is a curvature measure and that $\frac{J}{Q}$ measures the extra second-period wage payments per unit of first-period informativeness. Twice applying the fundamental theorem of calculus and differentiating on both sides gives

$$\left(\frac{J}{Q}\right)' = \frac{1}{x'(\mu_0)^2} \int_0^1 \int_0^{1-t} t \Pi_2'''(t(\mu_0 - d_B) + s\mu_0 + (1 - s - t)\mu_H) ds dt > 0, \quad (73)$$

where the inequality follows from $\Pi_2''' > 0$. Shifting down the low posterior increases the cost of doing so as the value becomes more concave.

Next, we prove

$$\left(\frac{D}{Q}\right)' \leq 0 \implies \Omega \leq \frac{13}{16}. \quad (74)$$

Suppose $(D/Q)' \leq 0$. For this calculation, write

$$s = \frac{b + \Delta_b \mu_0}{P_{\mu_0}}, \quad G = \frac{\mu_0(1 - \mu_0)(\Delta_b)^2}{(b + \Delta_b \mu_0)P_{\mu_0}}, \quad X = \frac{\Delta_b d_B}{b + \Delta_b \mu_0}.$$

Here $0 < s < 1$, $G > 0$, $0 < X < 1$, and feasibility implies $P_{\mu_0} \geq X/(X + G)$. Substitution gives

$$Q = \frac{s^2 X}{G}, \quad \frac{D}{c} = \frac{X(G + sX)}{(G + X)(1 - X)}.$$

Differentiating D/Q , the assumed inequality becomes

$$G^2 + 2GX + sX^2 \leq (1-s)G.$$

It follows that

$$G + 2X \leq 1 - s, \quad (G + sX)(G + X) \leq (1-s)G(1-X).$$

Consequently, $\frac{D}{c} \leq \frac{(1-s)GX}{(G+X)^2} \leq \frac{1}{4}$. Substituting in J and using $P_{\mu_0} \geq X/(X+G)$ also gives

$$\frac{2QJ}{c^2} \leq \frac{D}{c} \frac{G+X}{1-X} \frac{(1-s)(3-2X+2G-XG) + s(1-X)(1+G)}{(1+G)^2} \leq 3 \frac{D}{c}.$$

For the last inequality, use $G + 2X \leq 1$, $3 - 2X + 2G - XG \leq 3(1+G)^2$, and $(1-X)(1+G) \leq (1+G)^2$. Since $\delta \leq 1$,

$$\Omega \leq \left(\frac{D}{c}\right)^2 + \frac{2QJ}{c^2} \leq \left(\frac{D}{c}\right)^2 + 3 \frac{D}{c} \leq \frac{13}{16}.$$

This proves (74).

We can now rule out a neutral evaluation. Suppose an optimum has $p_N > 0$. By (72), this requires $\Omega > 1$, so (74) implies $(D/Q)' > 0$. After choosing the optimal neutral probability, the part of expected discounted wage payments that depends on the evaluation is

$$c\delta \frac{D}{Q} + c\sqrt{\delta^2 \left(\frac{D}{Q}\right)^2 + 2\delta \frac{J}{Q}}.$$

Its derivative in d_B is strictly positive because $(D/Q)' > 0$ and, by (73), $(J/Q)' > 0$. Since $d_B > 0$, reducing d_B slightly is feasible and lowers expected wage payments. This contradicts optimality. Hence $p_N = 0$ when $\Delta_a = 0$.

Finally, consider $\Delta_a > 0$. Apply the change of parameters

$$\begin{aligned} \hat{a} &= a - \frac{b\Delta_a}{\Delta_b}, & \hat{b} &= \frac{\Delta_a + \Delta_b}{\Delta_b}b, & \hat{\Delta}_a &= 0, & \hat{\Delta}_b &= \Delta_a + \Delta_b, \\ \hat{c} &= \frac{\Delta_a + \Delta_b}{\Delta_b}c, & \hat{U} &= U - \frac{\Delta_a}{\Delta_b}c. \end{aligned}$$

Strong complementarity gives $\hat{a} > 0$. This change preserves high-effort output probabilities and terminal payments. It also preserves $U + c$ and multiplies x , $(1-x)R$, and c by the same factor $(\Delta_a + \Delta_b)/\Delta_b$. These factors cancel from the first-period utility schedule and expected wage payment. Thus every evaluation has the same expected discounted wage payments under the two sets of parameters. The result already proved for $\hat{\Delta}_a = 0$ therefore gives $p_N = 0$ in the original problem as well. $\square$

C.2 Substitutes: An Example with a Neutral Evaluation

Example 2. Take

$$a = .005, \quad b = .7001, \quad \Delta_a = .82, \quad \Delta_b = -.70, \quad \mu_0 = .96,$$

with $\delta = 1$, $c = .01$, $U = 1$, and $(y_L, y_H) = (0, 1)$. Numerically, the optimal evaluation satisfies

$$p_N \simeq .004902, \quad p(G|y_L) \simeq .002077.$$

Thus a neutral evaluation occurs with probability about 0.49%, and low output receives a good evaluation with probability about 0.21%. The corresponding posteriors are

$$(\mu(B), \mu(N), \mu(G)) \simeq (.934357, .96, .965603).$$

All current and terminal payments are positive, and high effort is optimal at every terminal posterior.

D Commitment to Continuation Values (Online Appendix)

The first-period contract is (S, p) , where $p : Y \rightarrow \Delta(S \times \mathbb{R}_+ \times \mathbb{R}_+)$ assigns a signal s , a current wage w , and a continuation value U_2 . The continuation value is an enforceable right to a flat second-period wage $U_2^2/2$, which gives the agent utility U_2 without effort. After accepting the initial contract, the agent cannot exit to obtain the exogenous outside utility U in the second period. Initial participation instead requires total expected utility of at least $(1 + \delta)U$. At the beginning of the second period, the principal observes first-period effort and offers a replacement contract, which the agent may reject in favor of the flat wage. Both parties observe (s, w, U_2, e) and share the posterior $\mu(s, w, U_2, e)$. As in the baseline model, we maintain that the principal implements high effort after every history. We initially suppose that nonnegativity constraints on wages and continuation values are slack and give sufficient conditions below.

Write $\Pi_2(\mu, U_2)$ for terminal profit at belief μ when the agent holds continuation value U_2 . Recall that

$$x(y; \mu) = 1 - \frac{dF(y|0, \mu)}{dF(y|1, \mu)}, \quad I(\mu) = \int_Y x(y; \mu)^2 dF(y|1, \mu).$$

The terminal contract satisfies

$$\int_Y u(w_2(y)) dF(y|1, \mu) - c \geq U_2, \tag{75}$$

$$\int_Y x(y; \mu) u(w_2(y)) dF(y|1, \mu) \geq c. \tag{76}$$

By the argument of Proposition 1, both constraints bind. Consequently, the agent obtains U_2 after either first-period effort choice, although the replacement contract adjusts to the resulting posterior.

The transformation to posterior space therefore continues to apply. Including the wage and continuation value in the observed outcome, and averaging utility payments conditional on the posterior, allows us to restrict attention to deterministic schedules $w(\mu)$ and $U_2(\mu)$. The

principal's problem is

$$\begin{aligned}
& \max_{m \leq_{MPS} \bar{m}, w, U_2} \int_Y y \, dF(y|1, \mu_0) + \int_{[0,1]} (\delta \Pi_2(\mu, U_2(\mu)) - w(\mu)) \, dm(\mu) \\
& \text{s.t.} \quad \int_M (u(w(\mu)) + \delta U_2(\mu)) \, dm(\mu) - c \geq (1 + \delta)U, \\
& \quad \int_M \chi(\mu - \mu_0)[u(w(\mu)) + \delta U_2(\mu)] \, dm(\mu) \geq c,
\end{aligned} \tag{77}$$

Proof of Proposition 9. Since $u(w) = \sqrt{2w}$, the terminal problem minimizes a quadratic cost subject to (75)–(76). Using $\int_Y x(y; \mu) \, dF(y|1, \mu) = 0$, its solution is

$$u(w_2(y; \mu, U_2)) = U_2 + c + \frac{c}{I(\mu)} x(y; \mu).$$

Hence

$$\Pi_2(\mu, U_2) = \int_Y y \, dF(y|1, \mu) - \frac{(U_2 + c)^2}{2} - \frac{c^2}{2I(\mu)}.$$

The cost of delivering the promise is thus separable from the cost of terminal incentives. This expression also justifies conditional averaging in (77): averaging $u(w)$ and U_2 at a given posterior preserves both constraints and weakly reduces compensation costs.

Fix a posterior and total promised utility $u(w) + \delta U_2$. The principal minimizes

$$\frac{u(w)^2}{2} + \delta \frac{(U_2 + c)^2}{2}.$$

Equating marginal costs gives

$$u(w(\mu)) = U_2(\mu) + c.$$

Substituting into the first-period constraints yields

$$\int_M u(w(\mu)) \, dm(\mu) - c \geq U, \quad \int_M \chi(\mu - \mu_0) u(w(\mu)) \, dm(\mu) \geq \frac{c}{1 + \delta}.$$

For a fixed nondegenerate distribution m , the least-cost schedule therefore satisfies

$$\begin{aligned}
u(w(\mu)) &= U + c + \frac{c(\mu - \mu_0)}{(1 + \delta)\chi \mathbb{V}_m(\mu)}, \\
U_2(\mu) &= U + \frac{c(\mu - \mu_0)}{(1 + \delta)\chi \mathbb{V}_m(\mu)}.
\end{aligned}$$

Indeed, participation fixes the mean, and the Cauchy–Schwarz inequality gives the minimum variance required by incentive compatibility, which holds with equality at this schedule.

Expected compensation costs are consequently

$$\frac{1 + \delta}{2} (U + c)^2 + \frac{c^2}{2(1 + \delta)\chi^2 \mathbb{V}_m(\mu)} + \frac{\delta c^2}{2} \int_M \frac{1}{I(\mu)} \, dm(\mu).$$

Expected output is independent of the evaluation because it is affine in beliefs. Dropping terms independent of m , the evaluation problem becomes

$$\min_{m \leq MPS\bar{m}} \left\{ \frac{c^2}{2(1+\delta)\chi^2 \mathbb{V}_m(\mu)} + \frac{\delta c^2}{2} \int_M \frac{1}{I(\mu)} dm(\mu) \right\}.$$

This is precisely the interior period-by-period evaluation problem with first-period effort cost $c/\sqrt{1+\delta}$ and unchanged second-period effort cost c . Theorem 1 therefore applies. Moreover, the relative weight on terminal incentive costs increases by the factor $1+\delta$. Under the conditions of Proposition 5, the optimal evaluation consequently pools weakly more. $\square$

E Full Commitment (Online Appendix)

The principal offers a contract $C = (S_1, S_2, p_1, p_2)$ comprising signal spaces S_1, S_2 for the first and second period evaluation, a first period evaluation and wage $p_1 : Y \rightarrow \Delta(S_1 \times \mathbb{R}_+)$, and a second period evaluation and wage $p_2 : S_1 \times \mathbb{R}_+ \times Y \times Y \rightarrow \Delta(S_2 \times \mathbb{R}_+)$.

Write $h_1 = (s_1, w_1)$ for the first-period public realization. Two objects describe what is known about first-period output. First, let

$$\hat{\mu}(y_1, e_1) = \frac{\mu_0 dF(y_1|e_1, 1)}{dF(y_1|e_1, \mu_0)}$$

denote the belief about ability given output y_1 and effort e_1 . Conditional on y_1 , second-period output is distributed according to $F(\cdot|e_2, \hat{\mu}(y_1, e_1))$, which captures the correlation of y_1 and y_2 through the agent's ability. Second, the agent does not observe y_1 . Let $v(\cdot|h_1, e_1)$ denote his posterior distribution over first-period output after h_1 and effort e_1 ,

$$dv(y_1|h_1, e_1) = \frac{dp_1(h_1|y_1) dF(y_1|e_1, \mu_0)}{\int_Y dp_1(h_1|y) dF(y|e_1, \mu_0)}.$$

To ensure the contracting problem fits on the page, we omit the spaces integration is taken over; outer integrals are over $S_1 \times \mathbb{R}_+ \times Y$, inner integrals in second period constraints are over $S_2 \times \mathbb{R}_+ \times Y$ and integrals in second period agent values are over $S_2 \times \mathbb{R}_+ \times Y \times Y$.

The full commitment problem is

$$\Pi(\mu_0) = \max_C \iint \left(y_1 - w_1 + \delta \iint (y_2 - w_2) dp_2(s_2, w_2 | h_1, y_1, y_2) dF(y_2 | 1, \hat{\mu}(y_1, 1)) \right) dp_1(h_1 | y_1) dF(y_1 | 1, \mu_0) \quad (78)$$

$$\text{s.t.} \quad \iint (u(w_1) + \delta V_2(h_1, 1, 1)) dp_1(h_1 | y_1) dF(y_1 | 1, \mu_0) - c \geq (1 + \delta)U, \quad (79)$$

$$V_2(h_1, 1, 1) \geq U, \quad V_2(h_1, 1, 1) \geq V_2(h_1, 1, 0) \quad \text{for almost every } h_1, \quad (80)$$

$$\iint (u(w_1) + \delta V_2(h_1, 1, 1)) dp_1(h_1 | y_1) dF(y_1 | 1, \mu_0) - c \quad (81)$$

$$\geq \iint (u(w_1) + \delta U_2(h_1, 0)) dp_1(h_1 | y_1) dF(y_1 | 0, \mu_0) \quad (82)$$

where

$$V_2(h_1, e_1, e_2) := \iint u(w_2) dp_2(s_2, w_2 | h_1, y_1, y_2) dF(y_2 | e_2, \hat{\mu}(y_1, e_1)) dv(y_1 | h_1, e_1) - ce_2, \quad (83)$$

$$U_2(h_1, 0) := \max\{U, V_2(h_1, 0, 1), V_2(h_1, 0, 0)\}, \quad (84)$$

Proof of Proposition 10. As $\theta_H = \theta_L$, we omit the belief as an argument throughout the proof. Recall that

$$x(y) = 1 - \frac{dF(y|0)}{dF(y|1)}, \quad \mathbb{E}[x(y)] = 0, \quad \mathbb{E}[x(y)^2] = I > 0,$$

where expectations are under high effort.

Take any feasible contract, write $h_1 = (s_1, w_1)$, and let

$$u_1 = u(w_1), \quad u_2 = u(w_2), \quad z(y_1, h_1) = \mathbb{E}[u_2 | y_1, h_1], \quad \bar{z}(h_1) = \mathbb{E}[z(y_1, h_1) | h_1].$$

Reveal y_1 , retain h_1 as part of the public signal, and replace the utility payments by

$$\hat{u}_1 = u_1 + \delta(z - \bar{z}), \quad \hat{u}_2 = \bar{z} + \frac{c}{I}x(y_2). \quad (85)$$

The first change moves the variation in expected terminal pay into first-period pay. The second gives every revealed history the same continuation promise as the original public history, with a bonus that provides second-period incentives. These payments may be negative.

The modification preserves expected total utility conditional on (y_1, h_1) :

$$\hat{u}_1 + \delta \mathbb{E}[\hat{u}_2 | y_1, h_1] = u_1 + \delta z(y_1, h_1).$$

It therefore preserves initial participation and the payoff comparison between working in both periods and shirking in the first period followed by work in the second.

At every revealed history,

$$\mathbb{E}[\hat{u}_2 | y_1, h_1] = \bar{z}(h_1) \geq U + c, \quad \mathbb{E}[x(y_2)\hat{u}_2 | y_1, h_1] = c.$$

The first inequality follows from the original contract's second-period participation constraint. Hence the modified schedule satisfies second-period participation and incentive compatibility

after every revealed history.

Under the modification, the agent faces the same second-period wage schedule and output distribution whether he worked or shirked in the first period. Second-period participation and incentive compatibility therefore make working at least as good as quitting or shirking also after a deviation. To verify first-period incentive compatibility, it thus suffices to compare working in both periods with shirking in the first period and working in the second, which we have done.

The original second-period incentive constraint gives

$$c \leq \mathbb{E}[x(y_2)u_2|h_1] = \mathbb{E}[x(y_2)(u_2 - z)|h_1].$$

The equality uses independence of y_2 from (y_1, h_1) and $\mathbb{E}[x(y_2)] = 0$. Cauchy–Schwarz and $z = \mathbb{E}[u_2|y_1, h_1]$ imply

$$\mathbb{E}[u_2^2|h_1] = \mathbb{E}[z^2|h_1] + \mathbb{E}[(u_2 - z)^2|h_1] \geq \mathbb{E}[z^2|h_1] + \frac{c^2}{I}.$$

For the modified payments,

$$\mathbb{E}[\hat{u}_1^2 + \delta \hat{u}_2^2|h_1] = u_1^2 + \delta^2 \mathbb{V}(z|h_1) + \delta \bar{z}(h_1)^2 + \frac{\delta c^2}{I}.$$

Since $\mathbb{E}[z^2|h_1] = \bar{z}(h_1)^2 + \mathbb{V}(z|h_1)$, averaging yields

$$\frac{1}{2} \mathbb{E}[\hat{u}_1^2 + \delta \hat{u}_2^2] \leq \mathbb{E}[w_1 + \delta w_2] - \frac{\delta(1-\delta)}{2} \mathbb{E}[\mathbb{V}(z|h_1)] \leq \mathbb{E}[w_1 + \delta w_2].$$

Thus the modification satisfies all participation and incentive constraints and weakly reduces quadratic wage costs.

If its payments are nonnegative, we are done. If not, let C^F be an optimal fully revealing contract. The full-revelation problem has a concave objective and constraints that are linear in utility payments, so the Lagrangian is separable across outcomes and strictly concave in each payment. Since the payments of C^F are strictly positive, it maximizes the Lagrangian over all schedules and therefore solves the full-revelation problem without nonnegativity constraints. Therefore, C^F costs weakly less than the modified contract, which costs weakly less than the original contract. Since the original contract was arbitrary, full revelation is optimal. $\square$

E.1 Examples

Example 3. Output is binary with $y_L = 0$ and $y_H = 15$, and the technology is log-supermodular with

$$(a, \Delta_a, b, \Delta_b, \mu_0) = \left(\frac{1}{5}, \frac{1}{10}, \frac{1}{5}, \frac{1}{5}, \frac{1}{100} \right), \quad c = \frac{1}{10}, \quad \delta = \frac{1}{2}, \quad U = 1.$$

Under full commitment, allowing second-period wages to depend on a retained record of first-period output, the optimal contract implementing high effort in both periods fully reveals first-period output. All wages are strictly positive, and the expected discounted wage is approximately 0.946.

Example 4. Output is binary with $y_L = 0$ and $y_H = 15$, and the technology is log-supermodular with

$$(a, \Delta_a, b, \Delta_b, \mu_0) = (.30, .65, .01, .03, .90), \quad c = .01, \quad \delta = 1, \quad U = 1.$$

Under full commitment with a retained record of first-period output, the optimal contract implementing high effort in both periods reveals no information in the first period. The first-period wage is constant at $w_1 = 1/2$, and incentives are provided through the second-period wage, which depends on both outputs. All wages are strictly positive, and the expected discounted wage is approximately 1.023.

F Additional Lemmata (Online Appendix)

Proof of Lemma 4. Write $\tau = 1 - \sigma$ and introduce the two dimensionless quantities

$$\phi = \frac{\mu_0(1 - \mu_0)(\Delta_a + \Delta_b)\Delta_b}{b + \Delta_b\mu_0}, \quad r = \frac{(\Delta_a + \Delta_b)(b + \Delta_b\mu_0)}{\Delta_b P_{\mu_0}}.$$

Strong complementarity gives

$$1 - r = \frac{a\Delta_b - b\Delta_a}{\Delta_b P_{\mu_0}} > 0.$$

Moreover, $0 < P_{\mu_0} < P_{\mu_0} + \phi < 1$. Thus $r \in (0, 1)$ and $\zeta := 1 - \tau(P_{\mu_0} + \phi) \in (0, 1)$.

Substituting the binary continuation value and the posterior formulas gives

$$\mathcal{R}(\sigma) = \delta \left(\frac{c_2}{c_1} \right)^2 \mathcal{H},$$

where

$$\begin{aligned} \mathcal{H} = & \frac{\tau^2 \phi^2}{(P_{\mu_0} + \phi)^2 \zeta^3} \left\{ 2\phi(1 - r)(1 - P_{\mu_0} + P_{\mu_0}r) \right. \\ & + \zeta \left[3P_{\mu_0}(1 - P_{\mu_0})(1 - r)^2 \right. \\ & \left. \left. + (\phi(1 - 2P_{\mu_0}) + 2P_{\mu_0}(2 - P_{\mu_0}))r(1 - r) + (P_{\mu_0} + \phi)r^2 \right] \right\}. \end{aligned}$$

Regard $\mathcal{H}$ as a function of (P_{μ_0}, ϕ, r) , holding τ fixed. The relevant derivative combinations are

$$\begin{aligned} D_0 &= \phi \mathcal{H}_\phi - r \mathcal{H}_r, \\ D_1 &= \phi \mathcal{H}_\phi + r \mathcal{H}_r, \\ D_2 &= \phi \mathcal{H}_\phi + P_{\mu_0} r \mathcal{H}_{P_{\mu_0}} + r(1 - r) \mathcal{H}_r. \end{aligned}$$

We show below that all three are positive. The chain rule then gives

$$(\Delta_a + \Delta_b) \frac{\partial \mathcal{H}}{\partial \Delta_a} = \frac{b}{b + \Delta_b \mu_0} D_1 + \frac{\mu_0 \Delta_b}{b + \Delta_b \mu_0} D_2 > 0,$$

and

$$\frac{\partial \mathcal{H}}{\partial \Delta_b} = \frac{\partial \mathcal{H}}{\partial \Delta_a} + \frac{b}{\Delta_b(b + \Delta_b \mu_0)} D_0 > 0.$$

These derivatives include the changes in P_{μ_0} and both posteriors.

It remains to verify $D_0, D_1, D_2 > 0$. After differentiating and dividing by the positive factor $\frac{\tau^2 \phi^2}{(P_{\mu_0} + \phi)^3 \zeta^4}$, each D_i is a polynomial in r , which can be written in Bernstein form with the coefficients:

	degree	coefficients			
D_0	2	$2(1 - P_{\mu_0})A_0$	A_1	$2\zeta(P_{\mu_0} + \phi)A_2$	
D_1	2	$2(1 - P_{\mu_0})A_0$	A_0	$2P_{\mu_0}\zeta^2(P_{\mu_0} + \phi)^2$	
D_2	3	$2(1 - P_{\mu_0})A_0$	$B_1/3$	$(P_{\mu_0} + \phi)B_2/3$	$\zeta(P_{\mu_0} + \phi)^2(2 - \zeta)$

Here coefficients $(z_0, \dots, z_n)$ represent $\sum_{j=0}^n \binom{n}{j} z_j r^j (1 - r)^{n-j}$, and

$$\begin{aligned}
A_0 &= \phi^2(3 - 2\zeta) + 3\phi P_{\mu_0}\zeta(2 - \zeta) + 3P_{\mu_0}^2\zeta^2, \\
A_1 &= \phi^2[(1 - P_{\mu_0})(3 - \zeta^2) + P_{\mu_0}(1 - \zeta)(3 - \zeta)] \\
&\quad + 2\phi P_{\mu_0}\zeta[3(1 - P_{\mu_0}) + (1 - \zeta)(P_{\mu_0} + 1)] + \zeta^2 P_{\mu_0}^2(5 - 4P_{\mu_0}), \\
A_2 &= \phi[2 - \zeta(1 + P_{\mu_0})] + \zeta P_{\mu_0}(2 - P_{\mu_0}), \\
B_1 &= \phi^2[(1 - P_{\mu_0})(12 - 8\zeta) + 6P_{\mu_0}(1 - \zeta)] \\
&\quad + \phi P_{\mu_0}[(1 - P_{\mu_0})(6 + 14\zeta - 9\zeta^2) + P_{\mu_0}\zeta(10 - 9\zeta)] \\
&\quad + P_{\mu_0}^2[3\zeta^2 + 6\zeta(1 - P_{\mu_0})], \\
B_2 &= \phi[(1 - P_{\mu_0})(6 - 2\zeta - \zeta^2) + P_{\mu_0}(6 - 6\zeta + \zeta^2)] \\
&\quad + 2P_{\mu_0}\zeta(2 - \zeta)(2 - P_{\mu_0}).
\end{aligned}$$

Every displayed coefficient is strictly positive for $0 < P_{\mu_0}, \zeta < 1$. Since the Bernstein weights are nonnegative, $D_0, D_1, D_2 > 0$, completing the proof. $\square$